

# Learning Dexterous Manipulation

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Hands, simulators, training, and evaluation

*Compiled 2026-10-05. Every claim traces to a note in `papers/notes/`, every note to a parsed source in `papers/md/` or `code/md/`, and every source to a hash or commit in `corpus/manifest.json`. The method is in Appendix A.*

## Abstract

A dexterous hand can change an object's pose without setting it down. This survey covers 112 learning-based methods that attempt it, the hands they run on, the simulators they train in, and how they report results. The field has largely converged on one recipe: reinforcement learning on privileged state in a GPU simulator, distilled into a vision-only student for transfer to hardware. Almost half the corpus trains that way, nearly all of it in one simulator, whose own successors are barely used. A second and faster-growing branch learns from teleoperated demonstration instead, through vision-language-action models, diffusion policies, and systems whose contribution is the collection rig itself. Published work has tripled since 2022. Grasping remains the largest task family and bimanual coordination the fastest-rising, and almost every experiment runs on one of four hands. What the survey adds comes from reading released code and vendor specifications beside the papers. Eight hands that can be bought or built from published designs appear in no method paper here. No closed-loop policy in the corpus reports how far its hand passes into the object it holds, which a position-only success criterion cannot see. Section VII proposes a protocol for the unreported quantities; the body gives the counts behind every claim above.

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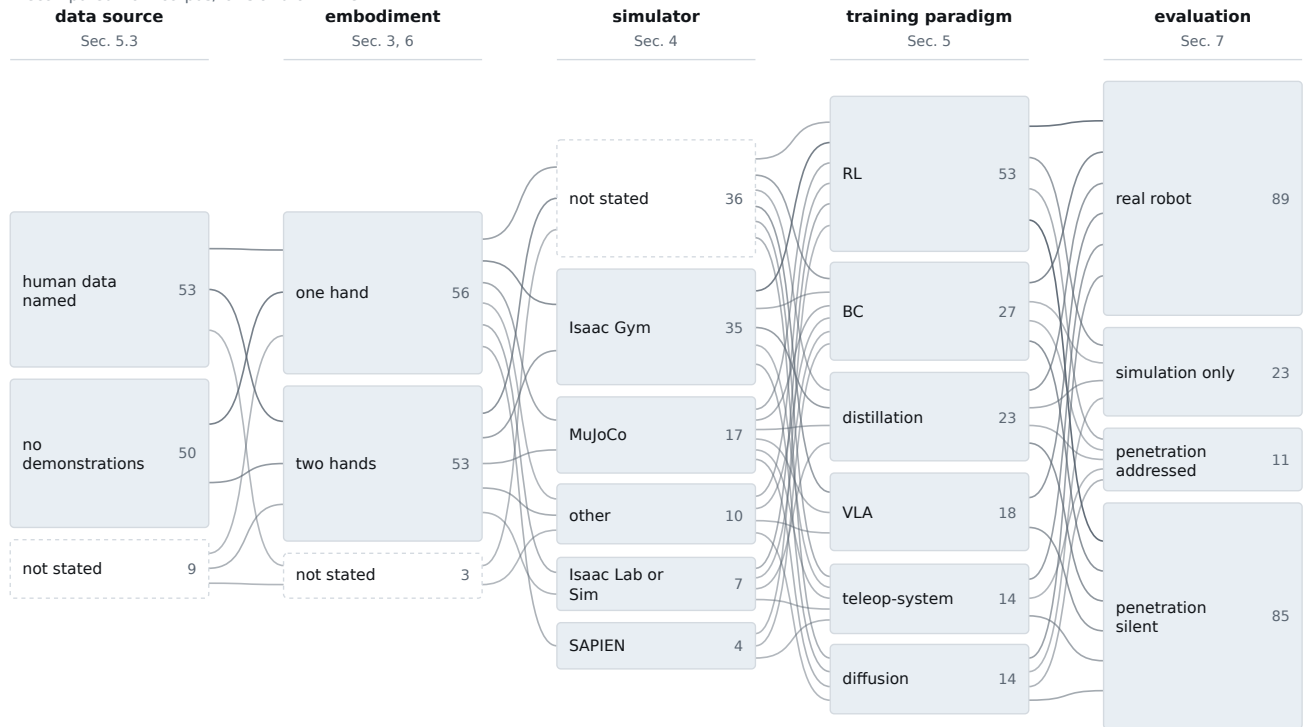
Luai Abuelsamen, [luai\\_abuelsamen@berkeley.edu](mailto:luai_abuelsamen@berkeley.edu). Corrections go to that address, which is the route section 5.6 names; the corpus is available from the author and is not yet deposited.

A survey of 221 bibliography entries, 218 of which carry a structured row read from a note: 112 method papers, 33 hands, 15 simulators, 15 datasets, 14 benchmarks, 14 surveys, 8 tactile sensors and 7 evaluation protocols. "Method row" throughout means one of the 112, and every headline count here has one of those eight classes as its denominator.

# 1. Introduction

**Figure 1. The field on one page, and the map of this survey**

Every node is a count over the 112 method papers in the corpus, and every column head names the section that covers it.  
Dashed nodes are reporting gaps, not choices. A paradigm count may include a paper twice, because a method may use several.  
Recomputed from corpus/rows at draw time.



The routes converge. The heaviest single path is no demonstrations, one hand, Isaac Gym, reinforcement learning, real robot.  
Whatever enters on the left, almost everything leaves through a GPU simulator and arrives at a real robot without contact quality ever being measured.

A hand is dexterous when it can change an object's pose without putting the object down. That property, and not the finger count, is what separates a hand from a gripper. [bicchi\\_grasping\\_chapter\\_2001](#) draws the same line in Sec. 1.1 between restraining an object and manipulating it with the fingers: restraint is a static question about whether the contacts prevent motion, while in-hand manipulation is a dynamic one about whether contacts can be broken and remade while the object stays held. A parallel-jaw gripper can hold almost anything and reorient almost nothing.

The classical theory of grasping is about holding, and it is good at it. Given a set of contact points and an assumption about friction, it can decide whether an object is immobilized, and what force each finger has to apply to keep it that way. What it cannot do is say what the fingers should do next. Contact is what makes that hard: a finger is either touching or it is not, and it can push but not pull, so the equations of motion change form every time a contact is made or broken. The standard treatment says as much about its own results, that immobilizing an object "does not guarantee stability", and that "the nonsmooth nature of grasp dynamics, because of the unilateral constraints on displacements and forces, has made a thorough analysis very difficult" ([bicchi\\_grasping\\_chapter\\_2001](#), Sec. 1.5 and 1.8). The tradition produced tests. You can check a candidate grasp against them; you cannot run them on a robot.

Learning changed what a practitioner has to produce. The old deliverable was a proof that a configuration holds. The new one is a policy that scores well over many simulated attempts, which means the simulator now stands in for the analysis and the score stands in for the proof. That is why the field moved, and it is also why this survey is organized as it is. It shifts two burdens onto the author. The simulator has to be faithful where fidelity matters. The measurement has to mean something. Much of what follows asks how well this literature carries those two.

One recipe now carries most of the field. Train a policy with reinforcement learning on privileged state, the object pose, velocities and physical parameters the robot cannot sense, in a GPU simulator running thousands of environments in parallel;

distill that teacher into a student that sees only what a camera and joint encoders provide; transfer the student. Fifty-three of the 112 method papers here train this way, 35 of them in Isaac Gym, and 23 distill a student from the teacher. In-hand cube reorientation is the recipe's benchmark, inherited from [openai\\_dexterity\\_2018](#) and still the task a new method is expected to show.

A second branch is growing underneath it and inverts the dependency. Instead of a simulator and a reward it takes human demonstrations through a teleoperation rig and fits a policy directly: 18 vision-language-action models, 14 diffusion policies, and 14 papers whose contribution is the collection apparatus itself. It is the newer literature, most of it from the last two years, and it trades the reward-design problem for a data-collection problem. Whether it displaces the simulator pipeline or ends up feeding it is the question this corpus cannot settle, but the growth is the most visible trend in it.

Both branches run on remarkably little hardware. Of the 33 hands with a documented specification here, four carry almost every experiment: the Allegro at 35 method papers, a Shadow or Adroit model at 21, the Inspire RH56 family at 19, and LEAP at 12. Eight further hands can be bought today or built from published designs and appear in the experiments of none. That is not a manufacturing lag, since the hardware exists and is purchasable, and section 3 takes it up: a field whose results rest on four platforms is more fragile than its publication volume suggests.

What this literature cannot currently do is compare its own results. Ninety-eight of the 112 methods state a success criterion, but they are not the same criterion, and only 70 say how many trials a reported rate rests on. The criteria run from never dropping the object to reaching a pose within a tolerance each paper picks for itself. Two success rates from two papers on nominally the same task are usually not measuring the same event, and none of the surveys preceding this one supplies a protocol that would make them comparable. Section 7 derives one, with the trial counts a stated confidence interval actually requires.

Every claim in this survey is traceable to a source. Two hundred and twenty-one bibliography entries were collected, 218 of them read into a structured record under a fixed template, with every value quoted from the source it came from and that source named. The counts printed here are recomputed from those records when the paper is built. Where a source is silent the record says so, which is why the coverage statistics in this paper are floors and not rates. The same discipline extended to released code: where a method published a repository, its reward was read from the repository as well as from the paper. Reading an implementation against its own description is established practice outside robotics, in repeatability and reproducibility studies [collberg\\_repeatability\\_2016](#) [raff\\_reproducibility\\_2019](#), in audits of how implementation detail moves reinforcement-learning results [engstrom\\_implementation\\_matters\\_2020](#), in reward-design review [knox\\_reward\\_misdesign\\_2023](#), in benchmark re-releases [metaworld\\_plus\\_2025](#), and most directly in recent work on paper-code alignment [scicoqa\\_2026](#) [biocon\\_2026](#). We are aware of none that does it for a robotics field's released reward functions, and section 5.6 reports what this one found.

The scope is single-hand and bimanual multi-fingered manipulation under learned control. Parallel-jaw work enters as a comparison and not as a subject. Two limits apply to every number here. This is a corpus of the learned era, which is a selection effect rather than a judgment on the analytic tradition; and a value this survey failed to extract is indistinguishable in the counts from a value its source never stated.

Figure 1 puts the field on one page. What follows works outward from the task: the task families, what makes them hard, and the terms used throughout (section 2), the hands and their sensing (section 3), the simulators and what they expose (section 4), how policies are trained (section 5), what changes with two hands (section 6), and how any of it is evaluated (section 7). Section 8 states what the corpus supports and what it does not. Appendix A gives the method and the corpus, Appendix B the hardware and engine tables, Appendix C every paper-against-code comparison in full, Appendix D the coding of the fourteen surveys that precede this one, and Appendix E the derivation behind every count in the protocol.

## 2. A taxonomy of the problem

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### 2.1 Task families

Dexterous manipulation is not one task, and what separates its tasks is what makes some of them hard: what is held, what moves, which way gravity points, and what counts as failure. Six families cover most of the corpus. They are read off the rows, and the last paragraph here says what they miss. Grasping is the largest. Fifty-seven of the 112 papers carry the grasp label, the labels are not exclusive, and one paper can sit in several families. Success is a lift that survives a hold, and the thresholds differ by more than an order of magnitude. `dexgrasplva_2025` requires the object "held 10 cm above the table for 20 s", while `omnigrasp_2024` requires it "held at least 0.5 s in simulation". What makes grasping hard at scale is the continuum of starting configurations. `unidexgrasp_pp_2023` states it in Sec. 4.3, that "we are dealing with an infinite number of tasks considering the initial object pose can change continuously".

Functional and tool use is second with 45 papers. It is the family where a stable grasp can still be the wrong answer. `dexterous_functional_grasping_2023` gives the case in Sec. 2.1, that "grabbing a hammer from the head or handle are both equally valid ways of using it", and only one of the two lets the tool be used. Success is defined against the tool's function, and the field has no shared way to state it.

In-hand reorientation has 31 papers and the most settled criteria, because the community inherited one number. `openai_dexterity_2018` declared the goal achieved below 0.4 rad of orientation error, `dextreme_2022` keeps 0.4 rad at test time against a 0.1 rad training tolerance, and `eureka_2023` counts consecutive successes at 0.1 rad. A shared tolerance is not a shared protocol, because the stopping rule differs. `visual_dexterity_2022` measures error "when the controller predicts it has reached the goal and stops", which makes the policy a judge of its own trial.

Tracking a human reference has 12 papers and inverts the problem. The target is a trajectory rather than an endpoint, so success is a per-frame error band. `maniptrans_2025` requires all four of 30 degrees of rotation, 3 cm of translation, 8 cm of joint error and 6 cm of fingertip error. The difficulty is that the reference came from a human hand and was never dynamically feasible for the robot. `dexmachina_2025` warns in App. B.4 that scoring by timesteps inside the thresholds makes "the results highly sensitive to the threshold values".

Bimanual coordination has 43 papers and handover has 10. Handover is the smallest family and the one whose failure has a single moment, because the giver must release only after the receiver has the object. `dynamic_handover_2023` reports a hit rate above its success rate and blames the gap in Sec. 5.4 on "occasional challenges encountered during the grasping phase of the catcher". Not every paper here learns both sides. In `dexterous_handover_2025` only the receiver is learned and the giver is a scripted arm.

Those six families do not cover the corpus. Twenty-four of the 112 papers carry a label from outside them and eight carry no label from the six at all. Twenty fall in a catch-all class, mostly generalist policies evaluated on a task suite rather than on a dexterous task family, among them `pi0_2024`, `pi05_2025`, `pistar06_2025`, `openvla_2024` and `gemini_robotics_15_2025`. Three adjacent families are named here instead of being folded into the six. Locomanipulation is `humanplus_2024`, `omnih2o_2024` and `groot_n16_2025`, where the base is not fixed and the gravity argument below changes character. Piano playing is `robopianist_2023`, `rp1m_2024` and `pianomime_2024`, discussed in Section 6, where success is a per-timestep F1 against a MIDI score rather than an object pose. Deformable manipulation is `dexdeform_2023`, whose object carries its own state and its own physics. `ferrari_canny_1992` carries no task label because it is a grasp-quality measure rather than a task. Table 1's six rows are the families with enough papers to compare, and not a partition of the corpus.

### 2.2 Sources of difficulty

Contact is non-smooth, and that is a property of the problem rather than of any solver. `bicchi_grasping_chapter_2001` states in Sec. 1.2 that contact constraints are unilateral, and that losing a contact "involves an abrupt change of the structure of the model under consideration". Its Sec. 1.4 gives the sharper version, that a rod sliding on rough ground has configurations

with no consistent solution and configurations with more than one. Simulators inherit the difficulty. The re-implementation study [contact\\_models\\_comparison\\_2023](#) concludes in Sec. V that "there is no fully satisfactory approach at the moment, as all existing solutions compromise either accuracy, robustness, or efficiency". [pang\\_global\\_planning\\_2022](#) treats non-smoothness as the thing to remove, curating every contact pair in Sec. III-D so that contact points and normals change smoothly.

The hand has more actuators than the object has degrees of freedom and is still short of authority over it. The object moves only through contacts, and the contacts are what cannot be commanded. [bicchi\\_grasping\\_chapter\\_2001](#) Sec. 1.4 states that in multifingered grasps "the number of independent contact forces is much larger than the number of actuators. Thus, from a controllability standpoint, not all the contact forces are controllable".

Vision is occluded by the hand doing the work. [bai\\_unified\\_manip\\_survey\\_2025](#) names it in Sec. 4.3, that "occlusion hampers object tracking". [dexpoint\\_2022](#) keeps vision and adds imagined hand points to the point cloud. [rotating\\_without\\_seeing\\_2023](#) removes vision entirely and rotates objects from 16 binary touch sensors over the palm, links and fingertips.

Gravity direction changes the task rather than scaling it. [visual\\_dexterity\\_2022](#) states that reorientation with the hand below the object "is much easier", because "with a downward-facing hand, the hand must manipulate the object while simultaneously counteracting gravity". [anyrotate\\_2024](#) makes the direction a randomised variable by "randomly initializing hand orientations between episodes", and its Sec. 5.3 measures the cost, with performance dropping progressively from palm up and palm down, through base up and base down, to thumb up and thumb down.

**The second hand, and three counts.** Three counts describe two hands and they measure different things. Fifty-three of the 112 papers record two hands on the robot, which is all the corpus's two-hand flag claims. Forty-three carry the task label for bimanual coordination, the narrower claim that coordinating the hands is the task. Section 6 narrows again, to the 28 papers whose notes place a learned closed-loop controller on two multi-fingered hands, and its opening paragraph names every exclusion that takes the 53 down to the 28. That 28 is the denominator for every architecture count in this survey. Every bimanual claim here names which of the three it uses.

**What the second hand adds.** Four things genuinely change when the second hand arrives. Contact stays non-smooth, occlusion stays, and gravity stays the same problem.

Role asymmetry is one. [asymdex\\_2024](#) assigns a dominant hand with full finger and wrist control and a facilitating hand with 6-DoF base pose only, so that "the facilitating hand repositions and reorients one object, while the dominant hand performs complex manipulations".

Two hands on one object also close a kinematic chain through the object. In [bicchi\\_grasping\\_chapter\\_2001](#) Sec. 1.4 the hand and object dynamics are separate and are "linked through the n rigid-body contact constraints". A second hand adds a second constraint set on the same object rather than a second independent problem. That is where physical plausibility fails first. [bimangrasp\\_2024](#) is the corpus row that measures it, rejecting a synthesised grasp when "total penetrations exceeded 1.5 mm" and reporting in Sec. IV-C that "penetration remains the primary cause of grasp failure".

Two-arm collision does not exist for one hand at all. [dydexhandover\\_2025](#) resets an episode when "any unintended arm contact with the environment or self-collisions" occurs. [bunny\\_visionpro\\_2024](#) pays for it in the controller, where adding a sphere-approximated self-collision cost raises motion-control time from 0.74 ms to 7.85 ms. [bidex\\_teleop\\_2024](#) solves it in hardware, mounting the hands so the human arm and the teacher arm "are perpendicular to each other and do not collide". [dexmimicgen\\_2024](#) states plainly that it does not handle inter-arm collision at all.

And the action space doubles. [bidexhands\\_2022](#) runs 20 tasks on two Shadow Hands and finds single-agent PPO beating multi-agent RL on most of them, offering in Sec. 5.2 that "PPO algorithm is able to use all observations for training the policy, while MARL can only use partial observations". [asymdex\\_2024](#) attacks the dimensionality from the other side, halving the observation and action dimension through its role split and a frame relative to the facilitating hand's object.

## 2.3 Terms used throughout

Six terms recur in every section that follows, and each is used here in one sense only.

**Method row.** One method paper read into this survey's structured record. It is the unit every count is taken over: 112 of the corpus's 218 rows are method rows, the rest being hands, simulators, datasets, benchmarks, surveys, tactile sensors and evaluation protocols.

**Privileged state.** Information a simulator can supply and a robot cannot: object pose and velocity, contact forces, mass, friction. A policy trained on it cannot be deployed as it stands, which is what makes the next term necessary.

**Teacher-student distillation.** Training one policy on privileged state, then fitting a second that sees only deployable observations, usually camera images and joint encoders, to reproduce the first. Section 5 finds 23 of the 112 methods do this, and it is the second half of the recipe section 1 describes.

**Reference and rollout.** A reference is a pose or a trajectory scored before anything executes: a retargeted human motion, a synthesized grasp, a planned path. A rollout is what a trained policy actually did. The distinction matters because a quantity can be measured carefully at the reference and never at the rollout, which section 7 finds is the case for interpenetration.

**Interpenetration.** The distance by which two bodies that should be touching instead overlap, a consequence of solving contact constraints approximately rather than exactly. Section 4 covers what each engine exposes; the depth is a property of the compliance someone chose, not of the engine.

**Success criterion and per-cell trial count.** The criterion is the predicate a paper uses to call an episode a success: a pose within a tolerance, an object never dropped, a task completed. The per-cell count is how many trials one reported rate is computed from. Section 7 finds 98 of them state a criterion and no two state the same one, and that 70 state a count.

### Table 1. Task families against the properties that define success

| task family                   | what is held                                                                   | what moves                                                                  | what gravity does                                                                                                                            | what counts as failure                                                                                         | shortest honest success criterion                                                                                                                                                                                                                                                                                           |
|-------------------------------|--------------------------------------------------------------------------------|-----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| In-hand reorientation         | one rigid object, held by the fingers for the whole episode                    | the object's orientation in the palm frame, while contacts break and remake | a disturbance whose direction is set by hand orientation, and a load to counteract with the palm down, <a href="#">visual_dexterity_2022</a> | the object leaves the hand, e.g. the fall reset at goal distance 0.24 m in <a href="#">dextreme_2022</a>       | orientation error below a stated tolerance under a stated stopping rule, 0.4 rad in <a href="#">openai_dexterity_2018</a> and <a href="#">dextreme_2022</a> , 0.1 rad in <a href="#">eureka_2023</a>                                                                                                                        |
| Grasping                      | nothing at the start, the object rests on a support                            | the hand closes, then the object is lifted clear                            | the test itself, since the grasp exists to resist it                                                                                         | no lift, or a drop inside the hold window                                                                      | a stated lift height held for a stated duration, 10 cm for 20 s in <a href="#">dexgraspvla_2025</a> , 10 cm to episode end in <a href="#">graspxl_2024</a> , 0.5 s in <a href="#">omnigrasp_2024</a>                                                                                                                        |
| Functional and tool use       | a tool, held in the pose its function requires rather than the most stable one | the tool's own joint, or the tool acting on a second object                 | secondary, the function dominates                                                                                                            | a stable grasp in the wrong place, <a href="#">dexterous_functional_grasping_2023</a>                          | no agreed criterion. Criteria range from never dropping the object, <a href="#">dexterous_functional_grasping_2023</a> , to a four-point rubric, <a href="#">dexvla_2025</a> . The one mechanical threshold is holding the tool joint past a commanded angle for the episode, <a href="#">articulated_tools_inhand_2025</a> |
| Tracking a human reference    | whatever the human held in the captured clip                                   | hand and object follow a recorded trajectory under physics                  | full, and the reference never obeyed it, so the reference is not dynamically feasible                                                        | per-frame error leaves the band, or the object drops                                                           | all stated per-frame errors below threshold for the whole clip, 30 deg and 3 cm and 8 cm joint and 6 cm fingertip in <a href="#">maniptrans_2025</a> . Thresholds are not comparable across papers, and <a href="#">dexmachina_2025</a> calls the results "highly sensitive to the threshold values"                        |
| Bimanual coordination         | one object by two hands, or one object per hand                                | both hands, the object, and the relative pose of the hands                  | can be carried by one hand while the other acts, <a href="#">asymdex_2024</a>                                                                | the object drops, the arms collide, or the hands work against each other                                       | no agreed criterion. Task-specific thresholds dominate, e.g. 0.035 m block-to-cup distance in <a href="#">asymdex_2024</a> . The one criterion with a physical-plausibility gate is 2.0 s of hold with total penetration under 1.5 mm, <a href="#">bimangrasp_2024</a>                                                      |
| Handover and in-hand transfer | by the giver at the start, by the receiver at the end                          | the object crosses between hands, released only after it is regained        | unopposed during the exchange, and total in a thrown transfer, <a href="#">dynamic_handover_2023</a>                                         | a drop at the moment of release, and the catcher's grasp phase, <a href="#">dynamic_handover_2023</a> Sec. 5.4 | the receiver holds the object and moves it clear, more than 10 cm from the first hand in <a href="#">hato_visuotactile_2024</a> , or holds it to episode end in <a href="#">dydexhandover_2025</a>                                                                                                                          |

Two cells say the field has no agreed criterion, and both are in families where a second body is involved. A criterion naming a distance, a duration and a trial count can be re-run by someone else. A rubric cannot, and Section 7 takes up what follows from that.

## 2.4 What has changed since 2022

The corpus roughly tripled between 2022 and 2024, and its composition changed while it grew. Three shifts are visible in the rows, and none of them is small.

**The demonstration branch overtook the simulator branch in 2024.** Sorting each method by whether it trains in simulation, through reinforcement learning or distillation, or from human demonstration, through behavior cloning, a vision-language-action model, a diffusion or flow policy, or a collection rig, the simulator branch led 13 papers to 4 in 2023 and trailed 15 to 18 in 2024. It has trailed since: demonstration-trained work is 22 of the 34 papers from 2025 and 8 of the 13 so far from 2026. The recipe section 1 describes is this field's mature pipeline rather than its growth area, and a reader who knows only that recipe is reading the field as it was in 2023.

**In-hand reorientation is being abandoned as the canonical task.** It was 8 of the 10 method rows in 2022 and 3 of the 34 in 2025. The absolute count falls while the corpus triples, which is not a shift in emphasis but a change in what the field treats as worth publishing. Bimanual coordination moved the other way over the same span, from 1 row to 16, and functional and tool use from 2 to 15. The cube-rotation benchmark that section 5 shows the reward literature organized itself around is the task the newer literature has left.

**The migration off Isaac Gym began in 2025 and not before.** No row earlier than 2025 uses Isaac Lab or Isaac Sim at all. Three do in 2025 and four in the partial 2026, against nine and two respectively on Isaac Gym, so on small numbers the successors already carry more of the most recent year than the original does. A survey assembled twelve months earlier would have recorded a field that had not moved, which is worth stating plainly: the concentration this survey reports on one simulator is a fact about a corpus with a cutoff, not a permanent property of the field.

## 3. Hands and who makes them

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### 3.1 Design axes

A policy is only as dexterous as the hand it runs on, and the hands available divide sharply: a few that a laboratory can buy, a few more it can build from published designs, and a growing number that exist only in an announcement. What can be said about them is less than it looks. No degree-of-freedom, force, weight or price figure in Table 2 or Table 3 was measured by anyone outside the maker. Six rows are the exception, from peer-reviewed papers with stated protocols: ILDA, Pisa/IIT, ORCA, RUKA, LEAP and BiDexHand. Table 2 dates every other claim.

The degree-of-freedom count is the first number a vendor states and the least comparable one. Shadow's specification of 4 December 2024 reads "20 actuated DOF and a further 4 under-actuated movements for a total of 24 joints" [shadow\\_dexterous\\_hand\\_2005](#). Table 2 prints joints first and actuated DoF second, and the gap is the informative number. Where a vendor's own DoF figure differs from the joint count, the joint count is what the table prints, as with Inspire's "Degrees of freedom 6, Numbers of joints 12" [inspire\\_rh56dfx\\_2023](#). An actuator count is neither of those columns and has its own. DexHand states no joint count at all, only 16 finger micro-servos and two wrist servos, so 18 servos is the only count quotable for it [dexhand\\_open\\_source\\_2023](#). Daxo states 120 actuators, no DoF figure, and a tendon structure with no rigid joints, which has far fewer kinematic degrees of freedom than actuators [daxo\\_muscle\\_v0\\_2025](#).

The actuation ratio is the real decision. The Pisa/IIT SoftHand drives 19 joints from one motor and its authors state the cost plainly: "No in-hand dexterous manipulation is required for this prototype" [pisa\\_iit\\_softhand\\_2014](#). Tendon drive moves the motors off the fingers and the mass follows them. Shadow's hand plus forearm weighs 4.3 kg, against 1.1 kg for the ILDA hand alone with every motor in its palm [shadow\\_dexterous\\_hand\\_2005](#) [ilda\\_hand\\_2021](#). ILDA states an 18 kg payload against Shadow's 4 kg in a power grasp, and Shadow states no fingertip force at all. What tendons cost is state estimation. The Faive Hand estimated joint angles from tendon length through a Kalman filter and its cube reorientation failed on hardware, which its authors blamed on "poor joint angle measurement from the EKFs, especially when there is contact" [faive\\_hand\\_2023](#). Ruka-v2 adds detachable AS5600 encoders to measure that problem rather than close the loop on it, and those encoders put its open-loop linear joint-to-motor map at 8.26 degrees of error over seven joints [ruka\\_v2\\_2026](#).

Direct drive trades torque density for transparency. LEAP's Dynamixel joints resist 19.5 N in a pull-out test against the Allegro Hand's 8.5 N, at 595 g for the LEAP hand [leap\\_hand\\_2023](#), and the Allegro's own weight has no reachable source. Linkage drive is the argument against the forearm. ILDA puts 15 motors, drivers and ball screws inside the palm and reports 34 N at the fingertip in the bent pose and 28 N stretched [ilda\\_hand\\_2021](#). Linkages buy coupling for free, as in BiDexHand's four-bar that drives each DIP off its PIP, whose paper claims 16 actuated DoF and 21 joints while its README describes 15 servos driving 15 joints [bidexhand\\_2025](#).

Fingertip force is the number a hand buyer reads first, and across these rows it is six different measurements. Table 2 puts the quantity beside the figure: pull-out resistance for LEAP's 19.5 N, pinch for RUKA's 2.74 N and BrainCo's 15 N, fingertip normal force under a 1 cm indenter for Unitree's 10 N, a calibrated five-trial fingertip force for BiDexHand's 2.14 N, a peak vendor claim for 1X's 45 N, and two unlabelled spec columns for Sharpa's "20 N | 12 N". BrainCo's whole-fist grip of 50 N is a separate figure and is no longer in the same column as a pinch. ORCA's 19.6 N was none of these. It is the newton equivalent of a 2 kg index-finger payload at a fixed 600 mA motor current, so its force cell is empty and its payload cell carries that protocol [orca\\_hand\\_2025](#). No two hands in Table 2 report payload under the same test either.

Compliance is repairability seen twice over, and both are mechanical fuses: the Pisa/IIT SoftHand's rolling-contact joints return to assembly after over-extension and ORCA's "poppable" pin joints dislocate rather than break [pisa\\_iit\\_softhand\\_2014](#) [orca\\_hand\\_2025](#). ORCA also reports tendon drive as maintenance: "Prolonged use requires manual re-tensioning to maintain performance". What fails is rarely the motors. Its durability run found silicone skin degrading on two fingertips after about 2,000 to 4,000 grasp cycles and sensor wires snapping on three after about 4,500 to 7,000. The sensing wore out an order of magnitude sooner than the hand.

Two columns matter before any of the above and were missing. The first is the rate a loop can be closed at, which spans 250 Hz on a Tesollo to 1 kHz on a Shadow, a Unitree or a Wuji, with the Allegro at 333 Hz on its own CAN clock and Inspire stating no rate at all. LEAP's 500 Hz is a serial query ceiling and its own sim-to-real policy runs at 20 Hz, which answer different questions `leap_hand_2023`. A reader deploying torque control is bitten by that first. The second column says whether a URDF or MJCF exists and who ships it, which is the bridge to section 4 and to this survey's recommendation to pick a hand a simulator already carries. Shadow, Sharpa, Wuji, the Ability Hand, RUKA, Faive and the Allegro driver ship one. LEAP is the warning: its paper says the URDF is released and the released API repository contains none `leap_hand_2023`.

## Table 2. Hands that can be obtained

| hand                           | maker                            | joints | act. DoF | actuators | actuation                                                                      | weight g | force N | what the force is                                                                                                           | payload                                                  | control rate                                                                           | URDF or MJCF                                                     | tactile                                                                                                                                                                                                                                                                         | price USD | open HW | status | source      | claim date                                       | corpus methods using it                                |
|--------------------------------|----------------------------------|--------|----------|-----------|--------------------------------------------------------------------------------|----------|---------|-----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|----------------------------------------------------------------------------------------|------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------|--------|-------------|--------------------------------------------------|--------------------------------------------------------|
| shadow_dexterous_hand_2005 [1] | Shadow Robot Company             | 24     | 20       | 20        | tendon-driven, motors in the forearm (20 Smart Motor nodes, Maxon motors, PWM) | 4300     |         |                                                                                                                             | 4 kg in a power grasp, vendor claim, protocol not stated | EtherCAT at 1 kHz to the host, with a 5 kHz tendon-force loop inside each motor module | MuJoCo models for Hand E and variants, plus Gazebo, in sr_common | Shadow Tactile Fingertips (STF): 17x3 DoF Hall-effect taxels, 1000 Hz, uncalibrated; up to 5 fingertips; software also supports BioTac/MST/UBIO/PST fingertip options                                                                                                           |           | no      | sold   | datasheet   | specification dated 2024-12-04                   | 21: asymdex_2024, bidexgrasp_2026, bimangrap_2024, ... |
| sharpa_wave_2026 [2]           | Sharpa Robotics (Sharpa Pte Ltd) | 22     | 22       | 22        |                                                                                | 1300     | 20 / 12 | vendor spec table with two unlabelled columns, probably two configurations; grip force is listed separately as 150 N / 90 N | 40 kg / 24 kg (the same two unlabelled columns)          | 500 Hz over 1000BASE-T Ethernet                                                        | MuJoCo and Isaac Sim models                                      | 'Dynamic Tactile Array' (DTA), camera-type array at the fingertip: resolution 240x240 / 60x60, force resolution 0.02 N / 0.05 N, spatial resolution 1 mm / 2 mm, frame rate 180 fps / 30 fps, force range 0-30 N, latency 20 ms, 6-D F/T; number of sensors per hand not stated |           | no      | sold   | vendor page | undated page, copyright 2026, fetched 2026-09-17 | 6: dexteleop0_2026, egoscale_2026, metis_2025, ...     |
| proception_prohand_2026 [3]    | Proception Inc (YC W25)          | 22     |          |           | tendon-driven, motors pull cables to move the fingers                          |          |         |                                                                                                                             |                                                          |                                                                                        |                                                                  | integrated skin-like sensors detecting contact, supporting grip control; same 'sensor skin' used on the                                                                                                                                                                         |           | no      | sold   | press       | launch post and press, 2026-06-29                | 0                                                      |

| hand               | maker                                                                                                                                                                           | joints | act. DoF | actuators | actuation                                                                                                                                                                   | weight g | force N | what the force is                                                               | payload                                 | control rate                                                                    | URDF or MJCF                                                          | tactile                                                                                       | price USD | open HW | status      | source      | claim date                                       | corpus methods using it            |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|----------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------|---------------------------------------------------------------------------------|-----------------------------------------|---------------------------------------------------------------------------------|-----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|-----------|---------|-------------|-------------|--------------------------------------------------|------------------------------------|
|                    |                                                                                                                                                                                 |        |          |           |                                                                                                                                                                             |          |         |                                                                                 |                                         |                                                                                 |                                                                       | companion ProGlove; type/count not stated                                                     |           |         |             |             |                                                  |                                    |
| bidexhand_2025 [4] | Zhengyang Kris Weng, Center for Robotics and Biosystems, Northwestern University                                                                                                | 21     | 16       | 15        | cable-driven, N-configuration tendon routing (FeeTech servos, endless-loop antagonistic pull-pull); thumb CMC flexion driven via a 4-bar linkage instead of a direct tendon |          | 2.14    | fingertip normal force, calibrated sensor, five trials per finger, average peak | 4.54 kg (10 lb) lifted                  | serial to an onboard ESP32, ROS 2 position control; rate not stated             | URDF, consumed by the ROS 2 and MoveIt stack; no physics simulator    | none mentioned                                                                                |           | yes     | open-source | paper       | paper, 2025 (ICRA Dexterity workshop abstract)   | 0                                  |
| wuji_hand_2025 [5] | Wuji Technology (founded 2019)                                                                                                                                                  | 20     | 20       | 20        | direct-drive rotary, backdrivable                                                                                                                                           |          |         |                                                                                 |                                         | 1000 Hz across 20 axes over 100BASE-TX Ethernet, MIT force-position hybrid mode | MuJoCo and Isaac Sim models                                           | 'Multi-Axis Force/Torque Fingertip Sensing' confirmed present; no type, count or range given  |           | no      | sold        | vendor page | undated page, copyright 2026, fetched 2026-09-17 | 2: toporetar get_2026, unidex_2026 |
| ruka_v2_2026 [6]   | Xinqi (Lucas) Liu, Ruoxi Hu, Alejandro Ojeda Olarte, Zhuoran Chen, Kenny Ma, Charles Cheng Ji, Lerrel Pinto, Raunaq Bhirangi, Irmak Guzey, New York University and NYU Shanghai | 20     | 16       | 16        | tendon-driven, forearm-mounted actuators; decoupled parallel 2-DoF wrist via a passive spherical ball joint; single-tendon-plus-spring abduction/adduction                  |          |         |                                                                                 |                                         | Dynamixel over USB serial; no rate stated                                       | MuJoCo XML in the shared v1 repo; not confirmed to carry the v2 wrist | optional e-flesh fingertips (magnetic touch-sensing form factor); not part of the base design | 1500      | yes     | open-source | paper       | paper, 2026                                      | 0                                  |
| tesollo_dg5f_2024  | Tesollo Inc. (Incheon HQ, Gwangmye                                                                                                                                              | 20     | 20       | 20        | one integrated actuator per joint ('high-                                                                                                                                   | 1763     |         |                                                                                 | pinching 2.5 kg rated and 5 kg maximum; | 250 Hz over Modbus RTU and TCP and                                              |                                                                       | none standard; optional fingertip                                                             |           | no      | sold        | vendor page | undated page, copyright 2026,                    | 0                                  |

| hand               | maker                                                                                                                                                                                                                                             | joints | act. DoF | actuators | actuation                                                                                                                                                              | weight g | force N | what the force is                                                                               | payload                                                                                              | control rate                                                        | URDF or MJCF                                                                     | tactile                                                                                                                                     | price USD | open HW | status      | source    | claim date                                            | corpus methods using it |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|----------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------|-------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------|-------------|-----------|-------------------------------------------------------|-------------------------|
|                    | ong R&D/factory )                                                                                                                                                                                                                                 |        |          |           | torque actuation', absolute encoder); page does not say direct-drive                                                                                                   |          |         |                                                                                                 | enveloping 10 kg rated and 20 kg maximum                                                             | Ethernet TCP/IP                                                     |                                                                                  | sensors (6-axis F/T, 3-axis force, or tactile) available; count/type not given                                                              |           |         |             |           | download links dated 2026-08                          |                         |
| unitree_dex5_2025  | Unitree Robotics (Yushu Technology Co., Ltd.)                                                                                                                                                                                                     | 20     | 16       | 16        | in-joint geared motor ('hollow-cup motor' + high-precision encoder + low-damping small-clearance reducer), backdrivable; page does not use the words tendon or linkage | 1100     | 10      | fingertip normal force under a 1 cm diameter cylinder pressed vertically down (vendor footnote) | 3.5 kg palm down and 4.5 kg palm left, on a 5 cm round hard object                                   | 1000 Hz over USB 2.0, with per-joint stiffness and damping commands |                                                                                  | Dex5-1: none. Dex5-1P: 94 pressure sensors per hand (2x5 palm + 2x3x5 finger pad + 2x3x5 fingertip + 2x3x4 finger root), range 10 g-2500 g. |           | no      | sold        | datasheet | undated page, copyright 2016-2025, fetched 2026-09-17 | 0                       |
| orca_hand_2025 [7] | Clemens C. Christoph, Maximilian Eberlein, Filipp Katsimalis, Arturo Roberti, Aristotelis Sympetheros, Michel R. Vogt, Davide Liconti, Chenyu Yang, Barnabas Gavin Cangan, Ronan J. Hinchet, Robert K. Katzschnann, Soft Robotics Lab, ETH Zurich | 17     | 17       | 17        | tendon-driven (antagonistic fishing-line tendon pairs per joint); wrist uses a GT2 timing belt drive                                                                   | 1200     |         |                                                                                                 | 10.5 kg on all four fingers and 2 kg on the index finger alone, both at a fixed 600 mA motor current | serial to Dynamixel or Feetech motors; rate not stated              | URDF-derived kinematic constants only; no URDF or MJCF file in the released tree | yes: FSR-based binary tactile sensing on all 5 fingertips (RP-C7.6-ST), absolute detection threshold as low as 0.05 N                       |           | yes     | open-source | paper     | paper, 2025                                           | 0                       |
| leap_hand_2023     | Kenneth Shaw, Ananye                                                                                                                                                                                                                              | 16     | 16       | 16        | direct-drive (Dynamixel servos, e.g.                                                                                                                                   | 595      | 19.5    | pull-out resistance: the outward                                                                |                                                                                                      | up to 500 Hz querying over USB                                      | URDF claimed in the paper;                                                       | none (future work only: 'we plan to                                                                                                         | 2000      | yes     | open-source | paper     | paper, 2023                                           | 12: bidex_tel_eop_2024, |

| hand                     | maker                                                                                                                      | joints | act. DoF | actuators | actuation                                                                                                            | weight g | force N | what the force is                                                                           | payload                                                                                                          | control rate                                                                         | URDF or MJCF                                                    | tactile                                                                                                                          | price USD | open HW | status      | source              | claim date                                                                | corpus methods using it                                                      |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------|--------|----------|-----------|----------------------------------------------------------------------------------------------------------------------|----------|---------|---------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|-----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|-----------|---------|-------------|---------------------|---------------------------------------------------------------------------|------------------------------------------------------------------------------|
|                          | Agarwal, Deepak Pathak, Carnegie Mellon University                                                                         |        |          |           | XC330-M288), joint velocity ~8 rad/s                                                                                 |          |         | force a flexed finger resists before slipping or deviating more than 15 degrees (Table III) |                                                                                                                  | serial; the paper's own sim-to-real policy runs at 20 Hz                             | the released LEAP_Hand_API repo contains no URDF, xacro or MJCF | develop and integrate LEAP Hand with low-cost touch sensors')                                                                    |           |         |             |                     |                                                                           | bidexhd_2024, cross_embodiment_world_mode ls_2025, ...                       |
| allegro_hand_v4_2016 [8] | Wonik Robotics Co. Ltd. (Seoul, South Korea); earlier versions 1.0/2.0 made by SimLab Co. Ltd.                             | 16     | 16       | 16        |                                                                                                                      |          |         |                                                                                             |                                                                                                                  | 333 Hz, CAN, the hand's own real-time clock (driver README)                          | URDF and xacro, left and right, in the ROS driver repo          | none                                                                                                                             |           | no      | sold        | driver repo         | driver repo, versions 1.0-4.0 undated, fetched 2026-09-17                 | 35: anyrotate_2024, anyteLeop_2023, asymdex_2024, ...                        |
| ruka_2025                | Anya Zorin, Irmak Guzey, Billy Yan, Aadithya Iyer, Lisa Kondrich, Nikhil X. Bhattachali, Lerrel Pinto, New York University | 15     | 11       | 11        | tendon-driven (11 Dynamixel actuators in the forearm: XM430-W210T for the thumb, XL330-M288-T for the other fingers) |          | 2.74    | pinch force, best of three trials, averaged over the left and right hands (Table III)       | 6.0 kg: weight added to a curled cloth-bag grip until joint-angle error exceeds 15 degrees, best of three trials | Dynamixel over a USB-to-serial bridge; rate not stated, data collection ran at 15 Hz | MJCF                                                            | none (stated limitation: 'lacks tactile sensing')                                                                                | 1300      | yes     | open-source | paper               | paper, 2025                                                               | 0                                                                            |
| roboteraxhand1_2024 [9]  | ROBOTER A                                                                                                                  | 12     | 12       |           | gear-driven force-controlled joint modules per finger segment, back-drivable                                         | 1100     |         |                                                                                             | 25 kg, tracker's 'Strength' figure, over 25 kg gripping palm-up                                                  | not stated                                                                           | URDF exists but is licensed; maniptrans_2025 withholds it       | tactile/force sensors on every fingertip confirmed present (senses contact, force, temperature); count and array size not stated | 14000     | no      | sold        | third-party tracker | undated tracker page, fetched 2026-09-17; the vendor page was not fetched | 8: cross_embodiment_world_mode ls_2025, dextrimit_2026, dexmachina_2025, ... |
| shadow_dex_ee_2024       | Shadow Robot Company, in collaboration with Google DeepMind                                                                | 12     |          |           |                                                                                                                      | 4100     |         |                                                                                             |                                                                                                                  | not stated; the page claims 'high bandwidth torque and position control loops'       |                                                                 | stereo camera-based fingertip tactile sensors (hundreds of taxels each); multi-taxel                                             |           | no      | sold        | vendor page         | undated page, copyright 2026, fetched 2026-09-17                          | 1: demostart_2024                                                            |

| hand                      | maker                                       | joints | act. DoF | actuators | actuation | weight g | force N | what the force is                                                                                 | payload      | control rate                                                      | URDF or MJCF                                          | tactile                                                                                                                                              | price USD | open HW | status | source      | claim date                                                    | corpus methods using it                                                 |
|---------------------------|---------------------------------------------|--------|----------|-----------|-----------|----------|---------|---------------------------------------------------------------------------------------------------|--------------|-------------------------------------------------------------------|-------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------|--------|-------------|---------------------------------------------------------------|-------------------------------------------------------------------------|
|                           |                                             |        |          |           |           |          |         |                                                                                                   |              |                                                                   |                                                       | 3-DoF arrays on middle and proximal phalanges; counts not stated                                                                                     |           |         |        |             |                                                               |                                                                         |
| inspire_rh56dfx_2023 [10] | Beijing Inspire Robots Technology Co., Ltd. | 12     | 6        |           |           | 540      | 10      | fingertip strength, vendor spec, four fingers; thumb 15 N; force resolution 0.50 N                |              | RS485; rate not stated                                            |                                                       | none (this variant is the 'without tactile sensors' table; a tactile FTP variant exists per bib but is not on this page)                             |           | no      | sold   | vendor page | undated page, fetched 2026-09-17                              | 19: ace_teleop_2024, articulated_tools_inhand_2025, being_h05_2026, ... |
| brainco_revo2_2025        | BrainCo Inc.                                | 11     | 6        |           |           | 383      | 15      | pinch force, vendor minimum ( $\geq 15$ N); the same page states a whole-fist grip of $\geq 50$ N | $\geq 20$ kg | RS485 and CAN FD; EtherCAT on Pro and Touch; rate not stated      |                                                       | Touch variant only: multi-dimensional fingertip tactile module with 'Tactile Adaptive Control'; count/type not stated. Basic and Pro variants: none. |           | no      | sold   | datasheet   | undated page, fetched 2026-09-17                              | 1: bidexgrasp_2026                                                      |
| agibot_omnihand_2025      |                                             |        |          |           |           |          |         |                                                                                                   |              |                                                                   |                                                       |                                                                                                                                                      |           |         |        | unavailable | no source: both store.agibot.com URLs 404, fetched 2026-09-17 | 1: clutterdexgrasp_2025                                                 |
| linkerbot_l20_2025        | LinkerBot (SDK author 'CHIUS INC')          |        |          |           |           |          |         |                                                                                                   |              | USB-to-CAN at 1 Mbit/s, optional Modbus or RS485; rate not stated | URDF, with PyBullet and Isaac Gym examples in the SDK | L10/L20: per finger 4 channels (normal force, tangential force, tangential direction, proximity), 5 sites x 4 channels, 0-255 range, sensor type     |           | no      | sold   | vendor page | undated page, fetched 2026-09-17                              | 1: being_h05_2026                                                       |

| hand                           | maker                                                                                              | joints | act. DoF | actuators | actuation                                                                                                                                                       | weight g | force N | what the force is | payload | control rate                                           | URDF or MJCF                                                                                | tactile                                                                                                    | price USD | open HW | status      | source                         | claim date                                                    | corpus methods using it                                     |
|--------------------------------|----------------------------------------------------------------------------------------------------|--------|----------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------|-------------------|---------|--------------------------------------------------------|---------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|-----------|---------|-------------|--------------------------------|---------------------------------------------------------------|-------------------------------------------------------------|
|                                |                                                                                                    |        |          |           |                                                                                                                                                                 |          |         |                   |         |                                                        |                                                                                             | not stated. O6 (reseller): 16 capacitive tactile regions.                                                  |           |         |             |                                |                                                               |                                                             |
| psyonic_ability_hand_2021 [11] | PSYONIC (San Diego, CA, USA)                                                                       |        | 6        | 6         | linkage (repo file/function names indicate a four-bar finger linkage; not stated on the vendor page itself)                                                     |          |         |                   |         | BLE, I2C, UART or RS485; rate not stated               | URDF (left, right, large, small, no-FSR) plus MuJoCo and Isaac Sim paths in the vendor repo | 30 touch sensor values streamed (FSR-based per API and URDF 'no_fsr' variant); per-finger split not stated |           | no      | sold        | vendor page                    | undated page, fetched 2026-09-17                              | 7: ace_teleop_2024, asymdex_2024, bunny_visionpro_2024, ... |
| dexhand_open_source_2023 [12]  | Rob Knight, The Robot Studio (design); electronics/firmware/ROS 2 by Trent Shumay, IoT Design Shop |        |          | 18        | tendon (fishing line, Suffix 832 80lb / 0.8mm kiteline); Emax ES3301/ES3302/ES3351/ES3352 micro-servos for fingers, Feetech SCS2332 or PWM servos for the wrist |          |         |                   |         | Arduino SCServo or SBUS firmware, BLE; rate not stated | URDF, in the IoT Design Shop ROS 2 packages                                                 | none mentioned                                                                                             | 300       | yes     | open-source | project page and GitHub README | page posts 2023-08-08 to 2023-10-01; no hardware release date | 0                                                           |

20 rows. 95 of 240 specification cells (39%) over the 12 specification columns are values no source stated; the key, maker, provenance and usage columns are excluded because they are never blank. Every figure here is the maker's or the authors' own claim. Nobody outside the maker has measured any DoF, force, weight or price cell in this table, except the rows sourced to a peer-reviewed paper with a stated protocol. The force column is not a ranking: read 'what the force is' first, because pull-out resistance, pinch force, a fingertip normal force under an indenter and an unlabelled vendor spec are different measurements. A blank cell means no source stated the value. No cell is truncated: a value wider than its column is wrapped at a word boundary, so a cell that runs to several rendered lines is one value and not several.

Figures with no reachable source, and other caveats on individual rows:

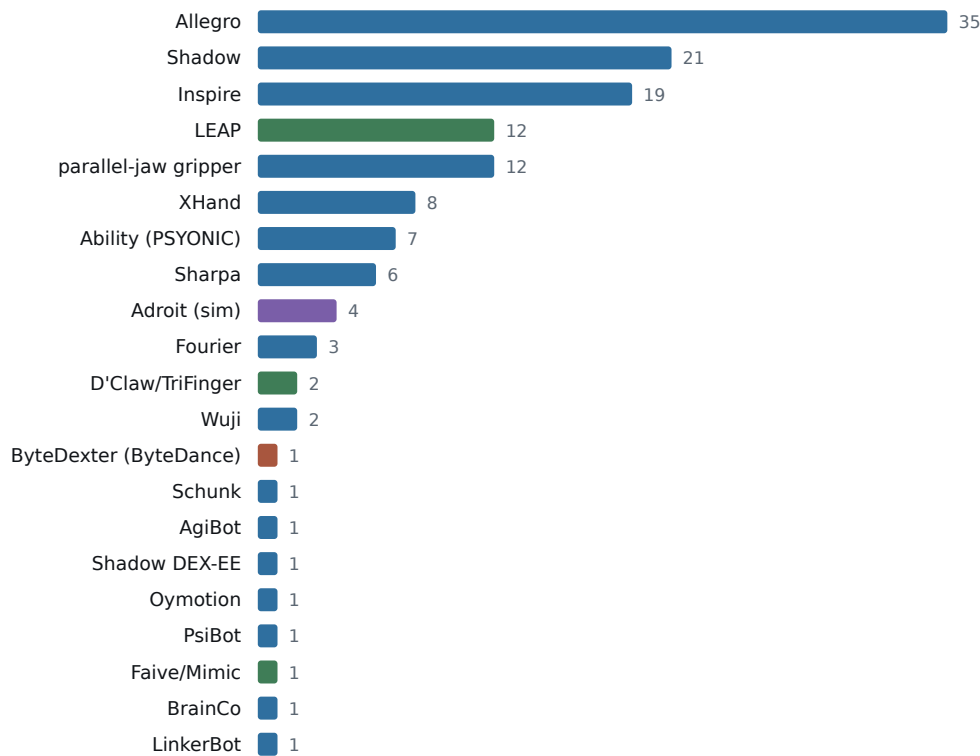
1. `shadow_dexterous_hand_2005` : the four distal joints are coupled and not independently controllable, so the '24 DoF' quoted in learning papers is the joint count
2. `sharpa_wave_2026` : the page footnotes 'Specifications may vary between products', and no column of the spec table is labelled
3. `proception_prohand_2026` : the evidence is a shipping announcement, not a datasheet: no weight, force, actuated-DoF count or control rate is stated anywhere
4. `bidexhand_2025` : the paper states 16 independently actuated DoF and 21 joints; the README states 15 servos driving 15 joints
5. `wuji_hand_2025` : the page footnotes that these figures 'reflect the current Beta1 product and the SPEC will continue to iterate'
6. `ruka_v2_2026` : the AS5600 encoders are attachable and detachable and are read for calibration and measurement only, not in the control loop
7. `orca_hand_2025` : the 19.6 N previously tabulated as a fingertip force is the newton equivalent of the 2 kg index-finger payload at a control-imposed current limit, and no source states it as a force
8. `allegro_hand_v4_2016` : allegrohand.com/v4 returned HTTP 404 and the Wonik wiki timed out; every cell here comes from the ROS driver repository, and weight, joint torque, payload and price have no reachable source
9. `robotera_xhand1_2024` : the 15 N in circulation comes from a survey table, not from the tracker or the vendor; the tracker states no fingertip force
10. `inspire_rh56dfx_2023` : the vendor's own page reads 'Degrees of freedom 6, Numbers of joints 12'; the joints column here prints the joint count, so the vendor's DoF figure is the actuated-DoF column
11. `psyonic_ability_hand_2021` : the four-bar finger linkage rests on repository file names and the five fingers on URDF meshes; the vendor page states neither
12. `dexhand_open_source_2023` : no joint count is given anywhere, so no DoF figure can be quoted; the servo count is 16 finger and thumb micro-servos plus 2 wrist servos, with a third optional. The \$300 is 'additional total cost of components', excluding printing and the wrist servos

### 3.2 Hands in use, and what they cost

**Figure 2. Which hands the surveyed methods actually run on**

One count per method paper whose own experiments use the hand, from 112 method rows.

Colour: blue sold commercially, green open hardware, purple simulation-only model, orange built in-house and not sold.



announced, not sold (Table 3) | 0 — none of the 9 company-announced hands appears in any method row

research prototypes (Table 3) | 3, from 2 of the 4 lab hands in Table 3

The two oldest designs in Table 2 carry 49 of the 103 papers that name a hand at all, and seven rows use both. Neither design's date is confirmed by its own sources, so 2005 and 2016 are the bibliography's. Figure 2 counts, per hand, the method papers whose own experiments use it. Of 112 papers, 103 name a hand. The Allegro accounts for 35, Shadow for 21, the Inspire RH56 family for 19, a parallel-jaw gripper for 12 and LEAP for 12. Seventy-five of the 103 name an Allegro, a Shadow or Adroit model, LEAP or an Inspire.

The Allegro's position is the uncomfortable part. Its product page at [allegrohand.com/v4](http://allegrohand.com/v4) returned HTTP 404 while the Wonik wiki timed out [allegro\\_hand\\_v4\\_2016](http://allegro_hand_v4_2016). Everything Table 2 confirms about the most-used hand in the corpus comes from its ROS driver: 16 joints, four fingers, a torque interface, a 333 Hz CAN clock, 12 V, no tactile sensing. Its weight, joint torque, payload and price have no reachable source. The field's most common platform cannot be specified from a page a reader can open.

Concentration would matter less if the hand did not move the result, and it does. On the same simulated cube rotation, LEAP reaches 0.2288 rad/s against the Allegro's 0.0828 rad/s [leap\\_hand\\_2023](http://leap_hand_2023), and RUKA reports a 2.74 N pinch against the Allegro's 1.60 N under the same three-trial pinch test [ruka\\_2025](http://ruka_2025). A method compared only on Allegro hardware is compared at one point in a space where a single axis moves the headline number two or three times over. Inspire, XHand and Sharpa take 29 of the 103 papers between them, four rows use two of the three, and none is earlier than 2024. Eight take none at all: ORCA, RUKA, Ruka-v2, BiDexHand, DexHand, the Proception ProHand, the Tesollo DG-5F and the Unitree Dex5.

**Open hardware and the collapse in cost.** Six rows in Table 2 are open hardware, and LEAP Hand V2 in Table 3 is a seventh. Five of the seven state a dollar cost: \$2,000 for LEAP, \$3,000 for LEAP Hand V2, \$1,500 for Ruka-v2, \$1,300 for RUKA and \$300 for DexHand. ORCA states a material cost below 2,000 CHF that the price column leaves unconverted, and BiDexHand states none [orca\\_hand\\_2025](#) [bidexhand\\_2025](#). The Faive Hand states neither a cost nor a license that could be read, so it is not counted as open hardware here [faive\\_hand\\_2023](#). Two of the stated bases need saying. DexHand's \$300 is "additional total cost of components", excluding the printing and the wrist servos [dexhand\\_open\\_source\\_2023](#), and ORCA's own figure sits against Ruka-v2's table listing ORCA at about \$3.5K [ruka\\_v2\\_2026](#).

The expensive end of the collapse is secondhand throughout. The only six-figure numbers on disk are RUKA's comparison table at \$100,000 for a Shadow Hand and Faive's "steep price tag of 110k GBP" [ruka\\_2025](#) [faive\\_hand\\_2023](#). Shadow's own page says to discuss pricing and Table 2's price cell for it is empty. The collapse is real at the cheap end and secondhand at the expensive one, and it has barely moved the literature. Twelve of the 103 hand-naming papers use an open-hardware hand, and all twelve are LEAP.

What the cheap hands give up is sensing. LEAP has none and names touch sensors as future work [leap\\_hand\\_2023](#), RUKA states its design "lacks tactile sensing" [ruka\\_2025](#), and BiDexHand's sources never mention touch [bidexhand\\_2025](#). ORCA is the exception, with binary FSR fingertips whose threshold was measured as low as 0.05 N on a fresh fingertip, against the sensor's rated 0.29 N, and 6.38 N on a degraded one [orca\\_hand\\_2025](#).

### 3.3 Tactile sensing

DIGIT set the cost floor. Its paper states a "total estimated manufacturing cost is approximately 15 USD per sensor ... when manufactured in a batch of 1000" [digit\\_2020](#), which is where the price of touch sensing stopped being the obstacle. Durability was as much of the contribution: its gel degraded by 0.3 percent over fifteen abrasion passes, where the two gels compared against it degraded by more than 800.

Digit 360 is the same lineage at a different operating point, claiming about 8.3 million taxels, spatial features to 7 um, normal and shear force resolution of 1.01 mN and 1.27 mN, and on-device processing that cuts event-to-action latency from 6 ms to 1.2 ms [digit360\\_2024](#). It is also not for sale, and no corpus paper's parsed text names it.

XELA's uSkin is the non-camera alternative and the one that bolts onto hands the field already uses. Its curved fingertip kit for the Allegro V4 and LEAP carries 30 three-axis sensing points, a full Allegro integration reaches 368, resolution is 0.1 gram-force, and the kit runs at 275 Hz [xela\\_uskin\\_2020](#). The 500 Hz figure often quoted belongs to the product family, not the curved fingertip. One corpus paper names it, [sparsh\\_2024](#), whose self-supervised encoders are pre-trained on 462.7k tactile images and beat matched end-to-end models by 95.1 percent when both see 33 to 50 percent of the labels. Its own bead-maze policies never complete the maze on the real robot.

The usage number is the one to keep, and it needs its inclusion rule stated: a sensor physically on the hand, read by the deployed policy. Eight papers meet it: [anyrotate\\_2024](#), [articulated\\_tools\\_inhand\\_2025](#), [dexteleop0\\_2026](#), [dexumi\\_2025](#), [hato\\_visuotactile\\_2024](#), [robot\\_synesthesia\\_2023](#), [rotateit\\_2023](#) and [rotating\\_without\\_seeing\\_2023](#). The rule excludes [penspin\\_2024](#), whose 20 binary contacts are simulated on an Allegro that has no tactile hardware and whose released config sets `enable_tactile: False`, and it excludes [dexndm\\_2025](#) and [dexplore\\_2025](#) for the same reason. Sixty-three of the 112 method papers use the word tactile somewhere in their parsed text and 35 carry it in their structured note. Against the eight that meet the rule above, either number is the gap worth quoting.

Almost none of the eight uses a high-resolution sensor. [rotating\\_without\\_seeing\\_2023](#) removes vision entirely and rotates objects from 16 binary touch sensors over the palm, links and fingertips, deployed zero-shot to a real Allegro, and Robot Synesthesia uses 16 force-sensing resistors read as binary [robot\\_synesthesia\\_2023](#). Two papers exclude touch deliberately, HORA reporting rotation "even without the usage of vision and tactile sensing" [hora\\_2022](#), and ORCA's authors dropping it from their reinforcement learning "due to the additional complexity involved in accurately modeling them" [orca\\_hand\\_2025](#). Taxel counts have risen by three orders of magnitude while the policies consuming them have stayed at binary contact.

### 3.4 The catalog against the literature

**Three classes of evidence.** Behind Table 3's rows sit three kinds of evidence, and conflating them is how a DoF figure with no source ends up in a survey. Vendor prose carrying numbers is the strongest, as on 1X's page of 9 July 2026 `onex_neo_hand_2026`. Video is second and carries none. Press or bibliography assertion is third, as with Tesla's V3 hand, known here only through a paraphrase of patents because the USPTO PDF parsed empty `tesla_optimus_hand_2025`. Table 3 blanks the cells resting on the third class and footnotes who did the arithmetic.

A survey can go one step past recording that a claim is unverified, which is to say which claims are implausible on their face. Daxo's 120 actuators in 750 g is about 6 g per actuator including structure, tendons, routing and skin `daxo_muscle_v0_2025`. Clone's 27 degrees of freedom under 2 pounds excludes a 500 W pump the source does not confirm is excluded `clone_robotics_hand_2024`. Figure 03's "Degrees of freedom, hands | 20" sits on the same tracker page as "Number of fingers | 10", so it is almost certainly the pair `figure_03_hand_2025`. Tesla's repeated 22 is one article's arithmetic of four DoF on each of five fingers plus two at the wrist, and Gen 2's own figure was 11 `tesla_optimus_hand_2025`.

Skepticism belongs to the evidence class, not to which table a row lands in. Sharpa's 22 of 22, Wuji's 20 of 20 and Tesollo's 20 of 20 are vendor claims about unmeasured hardware and they sit in Table 2, where Sharpa's page footnotes "Specifications may vary between products" and Wuji's says the spec "will continue to iterate" on a Beta1 product `sharpa_wave_2026 wuji_hand_2025`. Even the best-specified vendor page in the corpus leaves its fingertip-force columns unlabelled.

**Table 3. Hands announced but not purchasable**

| hand                         | maker                                                  | joints | act. DoF | actuators | actuation                                                                                                                                                                                                                                                                                   | weight g | force N | what the force is                       | payload | control rate | URDF or MJCF | tactile                                                                                                                                           | price USD | open HW | status        | source      | claim date                                      | corpus methods using it |
|------------------------------|--------------------------------------------------------|--------|----------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------|-----------------------------------------|---------|--------------|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------|---------------|-------------|-------------------------------------------------|-------------------------|
| clone_robotics_hand_2024 [1] | Clone Robotics                                         | 27     |          |           | hydraulic artificial muscle ('Myofiber'; small water-filled tubes that contract when pressurized), carbon-fiber bones and ligament-style tethers, driven by a 500 W water pump and 36 electro-hydraulic valves; each Myofiber generates up to 1 kg grip force, survived 650,000 test cycles | <910     |         |                                         |         |              |              | pressure pads in the palm detect grip firmness (count not given); 70 inertial sensors track angle/speed (proprioception, not tactile)             |           | no      | internal-only | press       | press, 2025-11-24, on a video of 2025-11-15     | 0                       |
| onex_neo_hand_2026           | 1X Technologies                                        | 25     | 25       |           | quasi-direct-drive tendons via the '1X Tendon Drive' at low gear ratios (~5:1-15:1); motors in the forearm, tendons pulled through the wrist                                                                                                                                                |          | 45      | distal flexion force, peak vendor claim |         | not stated   |              | high-resolution tactile sensing across fingertips and finger surfaces (normal force, contact location, shear); sensor count/technology not stated |           | no      | announced     | vendor page | vendor page, 2026-07-09                         | 0                       |
| sanctuary_phoenix_hand_2024  | Sanctuary AI (Sanctuary Cognitive Systems Corporation) | 21     |          |           | hydraulic, 'unique miniaturized hydraulic valves'                                                                                                                                                                                                                                           |          |         |                                         |         |              |              | not in the press release text; sidebar-only headlines mention new                                                                                 |           | no      | internal-only | press       | undated press release; trade reprint 2024-12-17 | 0                       |

| hand                      | maker                                                                                                     | joints | act. DoF | actuators | actuation                                                                                                                                                | weight g | force N | what the force is                                                                                                     | payload                                                                                           | control rate                                                                             | URDF or MJCF                                                  | tactile                                                                                                                                        | price USD | open HW | status    | source      | claim date                                                            | corpus methods using it             |
|---------------------------|-----------------------------------------------------------------------------------------------------------|--------|----------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------|-----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|---------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------|-----------|-------------|-----------------------------------------------------------------------|-------------------------------------|
|                           |                                                                                                           |        |          |           |                                                                                                                                                          |          |         |                                                                                                                       |                                                                                                   |                                                                                          |                                                               | touch/tactile sensors as a separate later announcement, not confirmed as part of this hand                                                     |           |         |           |             |                                                                       |                                     |
| leap_hand_v2_adv_2025 [2] | Kenneth Shaw, Deepak Pathak, Carnegie Mellon University                                                   | 21     | 17       | 17        | tendon (PIP/DIP coupled by a single tendon per finger); motor type not stated                                                                            |          |         |                                                                                                                       |                                                                                                   | not stated                                                                               | URDF, loaded into PyBullet by the released teleoperation node | none mentioned                                                                                                                                 | 3000      | yes     | announced | vendor page | undated project page; API repo at commit a0936196, fetched 2026-09-17 | 2: bidex_tel_eop_2024, dexwild_2025 |
| ilda_hand_2021            | Ajou University, Korea Institute of Machinery & Materials, Korea University                               | 20     | 15       | 15        | linkage-driven, direct linear drive (3 Maxon DCX8M motors per finger with GPX8 16:1 gearboxes driving ball screws via parallel+serial four-bar linkages) | 1100     | 34      | fingertip normal force in the bent pose; 28 N stretched, and 25 N per finger measured while crushing an aluminium can | 18 kg                                                                                             | CAN to a desktop; loop rate not stated                                                   |                                                               | 6-axis F/T sensor per fingertip (5 total), force resolution 62 mN, range +/-35 N; not a distributed taxel skin                                 |           | no      | prototype | paper       | paper, accepted 2021-11-05                                            | 0                                   |
| pisa_iit_softhand_2014    | Centro E. Piaggio, University of Pisa and IIT (Catalano, Grioli, Farnioli, Serio, Piazza, Bicchi)         | 19     | 1        | 1         | tendon-driven adaptive synergy, single motor via differential gears (6 W Maxon RE-max21, 84:1 reduction)                                                 |          |         |                                                                                                                       | holding force about 20 N along z and holding torque 2 N m, 28 N and 3.5 N m with a stronger motor | not stated                                                                               | URDF and Gazebo models in the ROS repo (BSD 3-Clause)         | none (motor encoder only; a padded work glove supplied contact compliance during experiments)                                                  |           | no      | prototype | paper       | paper, IJRR 2014                                                      | 0                                   |
| faive_hand_2023 [3]       | Yasunori Toshimitsu, Benedek Forrai, Barnabas Gavin Cangan, Ulrich Steger, Manuel Knecht, Stefan Weirich, | 16     | 11       | 16        | tendon-driven, rolling-contact joints (16 Dynamixel XC330-T288-T servos, 6 antagonistic pairs)                                                           | 1100     |         |                                                                                                                       | 10 kg, whole-hand downward power grasp on a dumbbell                                              | Dynamixel over serial; joint angles estimated from tendon length by EKF; rate not stated | MJCF, used directly by the released Isaac Gym training repo   | none on the physical hand; a simulated 'fingertip force' (15-dim, critic-only/privileged) exists only in the RL training observation, not real |           |         | prototype | paper       | paper, 2023                                                           | 1: graspxl_2024                     |

| hand                            | maker                                                | joints | act. DoF | actuators | actuation                                                                                                                                                           | weight g | force N | what the force is | payload | control rate | URDF or MJCF | tactile                                                                                                                                                | price USD | open HW | status    | source              | claim date                                                                                   | corpus methods using it |
|---------------------------------|------------------------------------------------------|--------|----------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------|-------------------|---------|--------------|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------|-----------|---------------------|----------------------------------------------------------------------------------------------|-------------------------|
|                                 | Robert K. Katzschmann, Soft Robotics Lab, ETH Zurich |        |          |           |                                                                                                                                                                     |          |         |                   |         |              |              | tactile hardware                                                                                                                                       |           |         |           |                     |                                                                                              |                         |
| paxini_dexh13_2024              | PaXini Tech                                          |        |          |           |                                                                                                                                                                     |          |         |                   |         |              |              | 1,140 ITPU multidimensional tactile processing units (press-release claim; unclear if PX-6AX-GEN3 sensor family)                                       |           |         | announced | press               | press, CES 2026-01-07                                                                        | 0                       |
| tesla_optimus_hand_2025 [4]     | Tesla                                                |        |          |           | V3: tendon-driven from forearm actuators, three tendons per finger through a crosstalk-managing wrist. Gen 2: in-hand actuators/sensors (unlocated in the article). |          |         |                   |         |              |              | Gen 2: tactile sensing on all fingers, demonstrated matching contact location and force in an egg-pickup video; type/count not stated. V3: not stated. |           | no      | prototype | press               | Gen 2 video 2023-12-13; V3 patents filed 2024-10, relayed by press 2026-04                   | 0                       |
| boston_dynamics_atlas_hand_2026 | Boston Dynamics                                      |        |          |           | electric (robot is fully electric); specific hand mechanism not stated                                                                                              |          |         |                   |         |              |              | tactile sensing in the fingers and palms, confirmed present; type and count not stated                                                                 |           | no      | prototype | press               | trade press, 2026-01-05 (CES); the Boston Dynamics blog of the same date has no hand content | 0                       |
| figure_03_hand_2025 [5]         | Figure AI                                            |        |          |           |                                                                                                                                                                     |          |         |                   |         |              |              | not stated on page for the hand itself; palm cameras in each hand are vision, not tactile                                                              |           | no      | prototype | third-party tracker | undated tracker page, fetched 2026-09-17; figure.ai launch page returned HTTP 404            | 0                       |

| hand                      | maker         | joints | act. DoF | actuators | actuation                                                                                                                                                                                         | weight g | force N | what the force is | payload | control rate | URDF or MJCF | tactile                                                                                                                                               | price USD | open HW | status    | source              | claim date                                                                  | corpus methods using it |
|---------------------------|---------------|--------|----------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------|-------------------|---------|--------------|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------|-----------|---------------------|-----------------------------------------------------------------------------|-------------------------|
| daxo_muscle_v0_2025 [6]   | Daxo Robotics |        |          | 120       | ultra-redundant tendon-driven; compliant structure with no rigid joints, flexible materials and tendon routing                                                                                    | 750      |         |                   |         |              |              |                                                                                                                                                       |           |         | prototype | third-party tracker | third-party catalogue post, 2026-09-16; daxo-robotics.com returned HTTP 404 | 0                       |
| xiaomi_cyberone_hand_2026 | Xiaomi        |        |          |           | motors located in the hand (compact motors generate heat requiring liquid cooling); no tendon/linkage statement, though contrast with tendon-driven hands' cycle life implies a non-tendon design |          |         |                   |         |              |              | full-palm tactile sensing, area ~8,200 sq mm, detects pressure and contact across the whole palm not just fingertips; taxel count and type not stated |           | no      | prototype | press               | press, 2026-03-30                                                           | 0                       |

13 rows. 96 of 156 specification cells (61%) over the 12 specification columns are values no source stated; the key, maker, provenance and usage columns are excluded because they are never blank. Every figure here is the maker's or the authors' own claim. Nobody outside the maker has measured any DoF, force, weight or price cell in this table, except the rows sourced to a peer-reviewed paper with a stated protocol. The force column is not a ranking: read 'what the force is' first, because pull-out resistance, pinch force, a fingertip normal force under an indenter and an unlabelled vendor spec are different measurements. A blank cell means no source stated the value. No cell is truncated: a value wider than its column is wrapped at a word boundary, so a cell that runs to several rendered lines is one value and not several.

Figures with no reachable source, and other caveats on individual rows:

1. `clone_robotics_hand_2024`: 'under 2 pounds' is an upper bound, and the source does not say whether the 500 W pump and 36 valves are inside it
2. `leap_hand_v2_adv_2025`: the page claims 21 DoF with 17 powered motors and the README calls the same hand 17-DOF
3. `faive_hand_2023`: the row's open-hardware status is unconfirmed: the repo has LICENSE and LICENSE-NVIDIA files whose contents were not captured, and no open-hardware claim appears in the sources
4. `tesla_optimus_hand_2025`: the 22 DoF in circulation is a Teslarati article's arithmetic of four DoF on each of five fingers plus two at the wrist, not a Tesla statement; Gen 2's own figure is 11 DoF. This row's actuation cell describes V3 and its tactile cell describes Gen 2
5. `figure_03_hand_2025`: the tracker's 'Degrees of freedom, hands | 20' sits beside 'Number of fingers | 10', so the 20 is most likely the pair and about 10 per hand; the 16-per-hand figure in circulation has no reachable source
6. `daxo_muscle_v0_2025`: 120 actuators in 750 g is about 6 g per actuator including structure, tendons, routing and skin; no DoF number is stated, and a redundant tendon architecture with no rigid joints has fewer kinematic DoF than actuators

**What is sold against what is published on.** The bottom rows of Figure 2 carry the finding. None of the nine company-announced hands in Table 3 appears in a single paper whose own experiments use it. Tesla, Figure, 1X, Sanctuary, Boston Dynamics, Xiaomi, Clone, Daxo and PaXini account for zero of the 112 method papers' experiments. The nearest thing to a counterexample is `helix_2025`, a Figure blog post claiming a 35-DoF whole-upper-body action space at 200 Hz that includes individual finger control. It never names the hand, gives no per-hand DoF count, and reports no success rate or trial count for any task. It is the maker describing its own unreleased hand, which is the evidence class the finding is about. The four research prototypes in Table 3 are in a different position, since the Faive Hand and LEAP Hand v2 Advanced account for three papers between them `graspxl_2024` `bidex_teleop_2024`.

Table 3's emptiness is measurable and part of the same finding. Over the twelve specification columns, 61 percent of its cells are values no source stated, against 39 percent for the hands that can be bought. The hands with the highest advertised DoF counts have the least specification behind them.

The one hand that crosses the gap crosses it because its maker published rather than announced. ByteDexter reaches a paper through a ByteDance technical report stating 21 DoF per hand, 16 piezoresistive fingertip channels in the action vector and real-robot success rates `gr_dexter_2025`, not through a product page. That mechanism is available to every vendor in Table 3 and none has used it.

The gap runs the other way too, and the last column of Table 2 shows it. Eight hands that can be bought or built from published designs take zero papers each: the Unitree Dex5 with 94 pressure sensors on its P variant `unitree_dex5_2025`, the fully actuated Tesollo DG-5F `tesollo_dg5f_2024`, the 22-DoF Proception ProHand `proception_prohand_2026`, ORCA, RUKA, Ruka-v2, BiDexHand and DexHand. Cheapness is established only for RUKA, Ruka-v2 and DexHand, because Table 2's price cell is empty for Unitree, Tesollo, the ProHand, ORCA and BiDexHand, and the only Dex5 price on disk is Ruka-v2's secondhand "\$25K" `ruka_v2_2026`. A reader choosing a hand on this corpus's evidence has four options with published precedent behind them: the Allegro at 35 papers, a Shadow at 21 and 4 more for its Adroit model, the Inspire RH56 family at 19, and LEAP at 12, of which only LEAP has both a published design and a stated price. Below those four the record thins fast: eight rows for the XHand, seven for the Ability Hand, six for Sharpa, two each for Wuji and LEAP Hand

v2, one each for five more, and none at all for the other 19 hands of Tables 2 and 3. Those 19 sit in the tables on their specifications alone, and no paper in this corpus runs on any of them.

**What would close the gap.** Nineteen of the 33 hands in Tables 2 and 3 appear in no method row, and 11 of those are neither sold nor open, so they appear in none for that reason and 33 is not the denominator for a software-lag claim; the eight hands named just above are. What decides used from unused is one regular expression per hand run against the papers' own hand field, in `tools/hand_usage.py`, so the partition is recomputed and not argued about. The lag runs through the tooling too. Seven of the fifteen simulator rows ship a first-party hand model at all, six of those hands are real, Brax's being a synthetic claw, and every one of the six is an Allegro or a Shadow; only four of the 33 tabulated hands appear in any engine. `bench2dex_2026` meanwhile compares 12 hands and `dexverse_2026` six without stating a DoF count for any. What would close it is a conformance suite for hand models: one URDF or MJCF per hand, fixed joint-limit, mass and collision checks, and a published pass or fail per engine.

## 4. Simulators and the physics underneath

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### 4.1 Requirements of a dexterous simulation

The clearest demonstration that hands are the hard case came from a benchmark that was not about hands. Erez et al. built a 35-DOF arm modeled on the Shadow Hand, closed it around a capsule with fixed spring-dampers, and asked four of the five engines for the largest timestep at which the object was still in the hand. MuJoCo held the grasp at 16 ms, PhysX at 2 ms, ODE at 0.25 ms and Bullet at 0.03 ms, a spread of a factor of 500 ([physics\\_engine\\_comparison\\_2015](#), Sec. IV-D). Havok is the fifth and was excluded for want of a working PD controller. Three caveats travel with that spread. The engines run a deliberately restricted common model, hinge joints with sphere and capsule geometry, no boxes and no meshes (Sec. II). The authors call it "an open question whether this model system can be tuned to work better in the gaming engines" (Sec. IV-D). The timesteps are log-spaced and good only to a factor of two, and the authors wrote MuJoCo and disclose it.

A grasp is hard for nameable reasons. It is many contacts at once, all persistent, all near stiction, on an object much lighter than the mechanism holding it. Persistence and multiplicity make the problem hyperstatic, and Le Lidec et al. show that per-contact solvers of the projected Gauss-Seidel family, and RaiSim's, then inject spurious jamming forces at stiction that vanish only once the object slides. The mass ratio makes it ill-conditioned, and in their stacked-cube test at a  $10^3$  to  $10^{-3}$  kg ratio those same methods fail to converge. Global methods with proximal regularisation stay robust in both cases ([contact\\_models\\_comparison\\_2023](#), Sec. IV-A). What degrades a truncated solve is therefore conditioning and redundancy, not stiffness, and a grasp supplies both by construction. Locomotion is forgiving by comparison. On flat ground the contact model and solver choice "hardly affects" their quadruped's tracked base velocity, and only on rough, slippery terrain do RaiSim and CCP deviate. A walking robot makes and breaks a few contacts against ground far heavier than itself. A hand does neither.

Speed enters by the same door. MuJoCo Playground reports that contact time scales with the number of possible contacts rather than the active ones, because JAX requires static shapes, which is why its tasks override the bound on contact points and the bound on geometry pairs by hand ([mujoco\\_playground\\_2025](#), Sec. VI). Every override printed in that paper is a locomotion port. A hand's bound would have to be set the same way, for the worst case, but no hand configuration is printed.

Figure 3 sets out the stages of one simulation step. Three of them make overlap, and they do not answer to the same knob. In an engine that enforces non-penetration at the velocity level, integration turns any residual approach velocity into overlap of order  $v$  times  $\Delta t$ ; an engine that enforces the gap at the next configuration carries no such term, which is why Dojo's hard-contact NCP keeps its feet above the floor at every timestep it was tested at. Constraint assembly fixes the compliance a loaded contact then rests at. A truncated solver leaves a residual that grows with conditioning. Which of the three dominates in a grasp is not measured anywhere in this corpus, and the three are not ordered here. A fourth thing is not on the figure, because it is not a stage of the step and not a source of overlap: a mismatch between the geometry the solver uses and the geometry the renderer draws, which runs in both directions.

## 4.2 Contact models and solvers

**Figure 3. One simulation step, and where engines differ**

The stages every rigid-body engine runs. Under each, the choices that separate the engines used for hands. Dashed red marks the three stages that make overlap. They do not answer to the same knob.

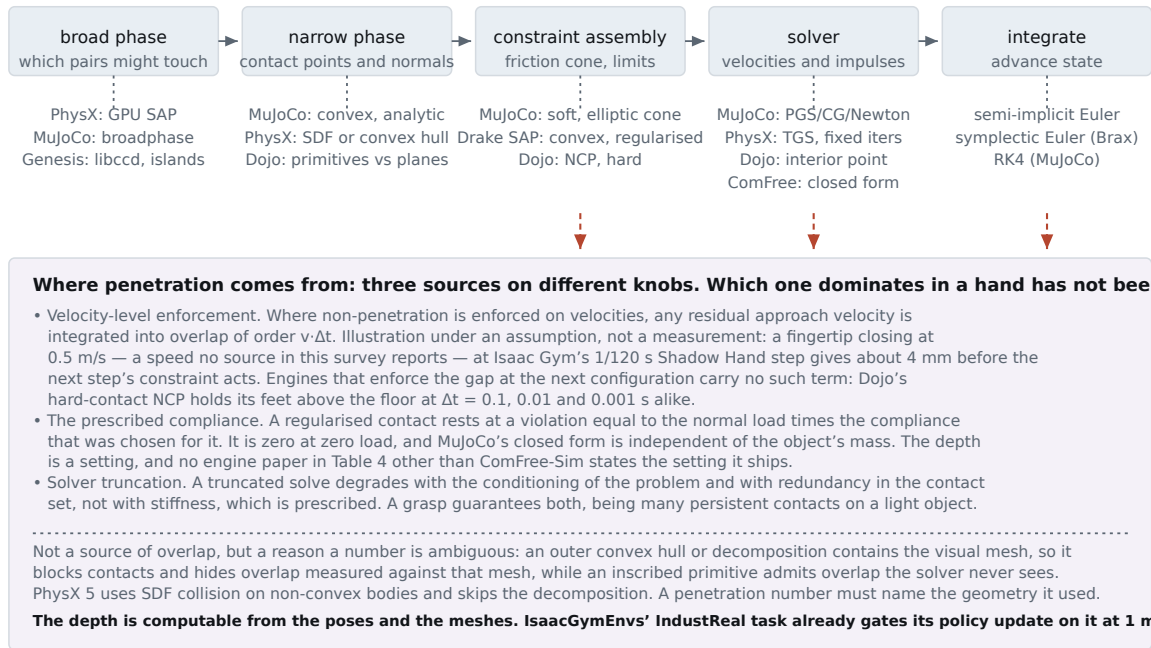


Table 4 is the engine-by-engine comparison, one row per simulator, built only from what a note confirmed. Le Lidec et al. supply the taxonomy that organises it, checking each formulation against the Signorini condition, Coulomb's law, and the maximum dissipation principle. Linear complementarity, the family of Bullet, ODE, PhysX and DART, satisfies Signorini alone, because linearising the friction cone to a pyramid biases friction toward its corners. The cone complementarity problem satisfies the other two but relaxes Signorini, so contact acts at a distance of size  $\Delta t \cdot \mu \cdot l_c \cdot T$ . The full nonlinear problem satisfies all three and is non-convex ([contact\\_models\\_comparison\\_2023](#), Table II). A contact model is not the algorithm that solves it. PhysX linearises the cone, which places its model in the LCP family, and it does not solve an LCP. Its Temporal Gauss-Seidel scheme folds substepping into the Gauss-Seidel sweep and exposes position and velocity iteration counts separately ([isaacgym\\_2021](#), Sec. 3).

MuJoCo is inside that taxonomy. Le Lidec et al. classify it as CCP-MuJoCo, a convex relaxation solved by a Newton method on the primal QCQP, in the same family as CCP-Drake, the SAP-style scheme ([contact\\_models\\_comparison\\_2023](#), Table III). The relaxation makes two errors of opposite sign in one engine. A loaded contact carries a violation, and a sliding contact carries force at a positive gap. Drake's SAP inherits the same pair, and its gliding effect at distance  $\varphi \approx \delta t \cdot \mu \cdot l_v \cdot t_l$  "unfortunately does not go away as  $\delta t \rightarrow 0$ " ([castro\\_sap\\_contact\\_2021](#), Sec. V-A). Le Lidec et al. call that compliance a "numerical trick designed to circumvent the issues due to hyper-staticity or ill-conditioning at the cost of impairing the simulation".

The depth a loaded contact carries is a chosen number, not a property of the engine. In a regularised formulation the steady-state violation at a contact is the normal load times the compliance: zero at zero load, growing with what the contact carries. MuJoCo drives that violation with a critically damped stabiliser whose steady-state depth, for an object resting under gravity, has a closed form independent of the object's mass ([mujoco\\_convex\\_contact\\_2014](#), Sec. V). Whoever sets the stiffness sets the depth.

Published penetration figures do not compare with each other, and the reasons are instructive. Dojo solves a hard-contact complementarity problem and stays above the floor at every timestep it was tested at, where MuJoCo penetrates tens of millimetres on the same Atlas drop ([dojo\\_2022](#), Sec. V-A). But the MuJoCo column is not a trend: a ten-times-smaller step

produces more overlap, which no timestep-independent stabiliser does, so either that configuration ties compliance to the timestep or the quantity is an impact transient rather than a steady-state depth. Drake's SAP reports figures orders of magnitude smaller again, and they are analytical bounds for a point mass at rest on a plane under a near-rigid stiffness rule, not measured depths ([castro\\_sap\\_contact\\_2021](#), Sec. V-B). ComFree-Sim does report penetration directly, as an explicit tuning parameter, on 5 cm convex primitives dropped under their own weight ([comfree\\_sim\\_2026](#), Sec. IV-A). A point-mass bound, a humanoid drop transient and a dropped primitive differ in load, inertia and regime, so the distances between them measure nothing. What the set does establish is the part that matters for a hand: in a compliant formulation the depth follows from a stiffness somebody chose, and the GPU era moved that knob rather than removing it. [physics\\_engine\\_comparison\\_2015](#) puts the consequence plainly, reporting that engines pushed past their stable step "effectively simulate a different physics model which can no longer hold the object" (Sec. IV-D). No paper in this corpus reports the depth for a fingertip loaded by a grasp, which is the one configuration a dexterous result depends on.

That brings Table 4's most important column, and the claim it does not support. The column records whether a parsed source reported a penetration depth, not what an engine can compute. Two of the 15 engines report one, Dojo and ComFree-Sim, and both are engines whose paper is about contact accuracy. Four are recorded as not reporting it: Brax, Isaac Gym, Isaac Lab and Orbit. Nine rows are blank. The word "penetration" appears nowhere in the Isaac Gym paper, and Isaac Lab's contact sensor reports force, duration and an average contact point with no contact-quality metric. The Isaac family carries 42 of the 112 method papers in this corpus.

**The tooling exists and the number is not recorded.** NVIDIA's own IsaacGymEnvs computes interpenetration depth in simulation: its IndustReal tasks sample points on one mesh, query them against the other, and reduce to a per-environment maximum, then gate the policy update on it, scaling reward down as the maximum approaches a one-millimetre threshold ([code/md/isaacgym\\_2021.md](#)). That is a shipped task measuring penetration per environment during training and acting on it, in the engine Table 4 records as not exposing the quantity. The depth is computable from the poses and the meshes in a few lines of Warp, and the field's own benchmark repository already does it.

One knob in that stack bears on penetration and nothing in the corpus explains its setting. PhysX's [max\\_depenetration\\_velocity](#) caps how fast the solver pushes overlapping bodies apart, so it governs how long an overlap persists rather than how deep it gets, and it appears across unrelated stock IsaacGymEnvs tasks at five different values ([dextrack\\_2025](#)). What the corpus shows is the pattern of values; why any of them was chosen is not in the record.

The rest of Table 4 is largely empty, and the emptiness is a result. Sixty-nine of its 165 cells are values no parsed source stated, 41 percent, which is the share its own footer counts; over the 150 cells outside the engine-key column the share is 46 percent. No engine paper states a default physics timestep. Three report one for a named experiment, and the timestep column reports those experiment settings. Isaac Gym's cell is its Shadow Hand step, from the only per-task timestep table any engine paper here publishes, which runs 1/120 s for Shadow Hand and Allegro, 1/200 s for ANYmal and TriFinger and 1/60 s for Franka ([isaacgym\\_2021](#), Table 4). MuJoCo's 0.01 s is the 27-DoF humanoid test's step and ComFree-Sim's 0.002 s is its benchmark step, against a stated stability limit near 0.02 s. Four engines state a solver iteration count and seven ship any dexterous hand at all. The license column answers a question a reader choosing an engine actually has, and thirteen of fifteen rows do not answer it.

## Table 4. Simulators and physics engines

| engine           | contact model                                                                                                                                                                                                                                        | solver                                                                                                                                                                                                                                                                                                                   | iters                                                                                                                                                    | diff. | GPU | dt s  | penetration exposed | throughput                                                                                                                                                                                                                                                                                                                                                                                                         | hands shipped                                                                                                                               | licence |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----|-------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|---------|
| brax_2021        | rigid bodies in maximal coordinates; naive/quadratic-scaling pairwise collision detection (no broad-phase acceleration structure); no convex-decomposition or mesh-vs-primitive distinction described                                                | not LCP-based; velocity-level collision updates with Baumgarte stabilization (inspired by the Tiny Differentiable Simulator); joints modeled as spring constraints rather than Featherstone-style articulated-body methods                                                                                               |                                                                                                                                                          | yes   | yes |       | no                  | millions of simulation steps/sec on a single accelerator (e.g. MuJoCo Ant equivalent on a single chip); scales to hundreds of millions of steps/sec distributed across a 4x2 TPUv3 and other TPU topologies; Ant on a TPUv3 8x8 estimated at ~hundreds of millions of steps/sec vs ~thousands of steps/sec for single-threaded CPU MuJoCo Gym (all figures qualitative/order-of-magnitude, no exact numeric table) | synthetic 4-fingered claw hand (Grasp environment, non-commercial, from-scratch morphology; not confirmed present in current repo snapshot) |         |
| comfree_sim_2026 | complementarity-free, analytical (closed-form) contact resolution in the dual cone of the Coulomb friction cone via a prediction-correction/impedance-style update, extended to a unified 6D model (normal, tangential, torsional, rolling friction) | no complementarity solve; a 4-kernel GPU pipeline (Algorithm 1: smooth-velocity prediction, per-contact/per-face dual-cone solve, generalized-impulse accumulation, velocity correction) computes the impulse in closed form per contact facet                                                                           | none (closed-form, no per-step iterative solve)                                                                                                          |       | yes | 0.002 | yes                 | AMD 32-core CPU + NVIDIA RTX 4090 GPU: ~3x faster (near-linear vs. MJWarp's superlinear) step time vs. contact count at 512 parallel envs; ~2x parallel-environment throughput vs. MJWarp at 256-4096 envs (Allegro-hand cube-grasping benchmark); on real LEAP-hand MPC (AMD 32-core CPU + RTX 4090), MPPI compute 13.9-28.2 ms vs. MJWarp's 34.0-68.8 ms (~2.4x average speedup)                                 |                                                                                                                                             |         |
| dojo_2022        | hard-contact nonlinear complementarity problem (NCP) with an exact nonlinear (second-order-cone) friction cone, avoiding pyramidal/linearized approximation, to avoid interpenetration and creep artifacts                                           | custom primal-dual interior-point solver (Algorithm 2), based on Mehrotra's predictor-corrector algorithm, extended for non-Euclidean quaternion variables, with cone-handling borrowed from CVXOPT; gradients via implicit differentiation (implicit function theorem) of the KKT/complementarity system, with central- | converges within 15 iterations for all three robots tested in the convergence study (Sec. V-A); default tolerances $r_{tol} = \kappa_{appa\_tol} = 1e-5$ | yes   | no  |       | yes                 | Intel Core i9-10885H, 32GB RAM, CPU only (no GPU support; described as future work). Table V, 1000-step forward+gradient wall-clock time at $dt=0.01s$ : Humanoid - Dojo 1.750+/-0.135s vs MuJoCo 1.512+/-0.045s vs Drake 5.463+/-0.078s vs Brax 6.975+/-0.485s; Unitree A1 - Dojo 5.235+/-0.071s vs MuJoCo                                                                                                        |                                                                                                                                             |         |

| engine        | contact model                                                                                                                                                                                                                                                                                                                                                                                                               | solver                                                                                                                                                                                                                                                                                                                                                                            | iters | diff. | GPU | dt s                 | penetration exposed | throughput                                                                                                                                                                                                                                                                                                                 | hands shipped                                                                                                          | licence |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|-----|----------------------|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|---------|
|               |                                                                                                                                                                                                                                                                                                                                                                                                                             | path parameter kappa trading gradient smoothness for physical accuracy                                                                                                                                                                                                                                                                                                            |       |       |     |                      |                     | 1.114+/-0.007s vs Drake 3.870+/-0.024s vs Brax 11.064+/-0.521s; Franka Panda - Dojo 1.159+/-0.077s vs MuJoCo 0.335+/-0.001s vs Drake 2.352+/-0.055s vs Brax 10.954+/-0.395s; Skydio X2 - Dojo 0.807+/-0.003s vs MuJoCo 0.047+/-0.002s vs Drake 0.571+/-0.011s vs Brax 7.953+/-0.484s. MuJoCo is fastest on every system.   |                                                                                                                        |         |
| genesis_2024  | Constraint-based rigid solver (equality/inequality constraints, plus an explicit noslip() post-pass, architecturally similar to MuJoCo's solver+noslip design) combined with multi-material coupling: FEM, MPM, particle-based PBD and SPH, a libuipc Incremental Potential Contact solver, an explicit inter-solver coupler, and a SAP (semi-analytic-primal-style) compliant-contact path used for grasp-coupling scenes. | ConstraintSolver with Newton-style line-search iterative resolve (linereach.py) and constraint-island decomposition for parallelism (island.py); multiple pluggable physics solvers (Rigid, FEM, MPM, PBD, SPH, libuipc, SAP) compiled via the Quadrants (Taichi-derived) backend.                                                                                                |       | yes   | yes |                      |                     |                                                                                                                                                                                                                                                                                                                            | Shadow Hand                                                                                                            |         |
| isaacgym_2021 | Rigid-body contacts via PhysX; contact geometry can be primitive shapes or meshes loaded from URDF/MJCF (Sec. 2.2); net contact forces exposed per-rigid-body via the Tensor API (Table 1, Get-only). No explicit statement of convex-vs-mesh contact generation or per-pair contact-point count. Shadow Hand tendons simulated via PhysX Fixed Tendon mechanics:                                                           | Temporal Gauss-Seidel (TGS) solver (Sec. 3, ref. 18), not classic PGS/LCP: folds sub-stepping into a per-body accumulated velocity-delta buffer projected onto the constraint Jacobians. Table 3 exposes two user-tunable iteration knobs, Position iterations (biased, velocity+positional-error-correcting) and Velocity iterations (unbiased, velocity-error-only-correcting); |       | no    | yes | 0.008333333333333333 | no                  | Shadow Hand: 150,000 parallel environment steps/sec at 16,384 environments, single NVIDIA A100 GPU (Sec. 5.3); Shadow Hand OpenAI reproduction (feed-forward): >20 consecutive successes in <1 hour on 1x A100 vs. 30 hours on OpenAI's cluster of 384x16-core CPUs (6144 cores total) + 8x NVIDIA V100 GPUs (Sec. 6.4.1). | Shadow Dexterous Hand, Allegro Hand, TriFinger (3-finger, 9-DoF manipulator; the paper itself says this is not a hand) |         |

| engine          | contact model                                                                                                                                                                                                                                                                                                                                                                                                                                                        | solver                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | iters                                                                                                                                        | diff. | GPU | dt s | penetration exposed | throughput                                                                                                                                                                                                                                                                                                                                                                                            | hands shipped                                                                                                                                                                                                                                                                                          | licence    |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|-------|-----|------|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
|                 | spring+damping force proportional to deviation from rest length, propagated through a tendon-joint tree (Sec. 3, Appendix A.1).                                                                                                                                                                                                                                                                                                                                      | no default numeric values for these are given in the main text.                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                              |       |     |      |                     |                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                        |            |
| isaacsim_2025   | Rigid contacts by default via PhysX 5 (SDF collisions, Featherstone articulation solver, Sec. 2.2); Factory assembly envs specifically use 'SDF-based contact generation, a contact reduction technique, and a Gauss-Seidel solver' (Sec. 6.4). ContactSensor exposes net normal force per body, optional filtered pairs, contact duration, average contact point, and a short history (Sec. 3.3.1); no contact-quality/penetration metric is defined (survey note). | NVIDIA PhysX 5's internal rigid-body/Featherstone-articulation solver, used as-is; Factory assembly envs specifically named as using a Gauss-Seidel solver with SDF-based contact generation and contact reduction (Sec. 6.4). No timestep, decimation, or solver-iteration values are stated anywhere in the paper for the general (dexterous) suite; the only note is qualitative, that the Digit humanoid 'requires a higher solver iteration count for stable simulation' (Fig. 14). |                                                                                                                                              | no    | yes |      | no                  | DextrAH teacher task (state-based grasp-and-lift): over 900,000 FPS training throughput with 8 GPUs (RTX Pro 6000, the paper's only multi-GPU platform) and 16,384 environments (Sec. 4.1.1); Franka cabinet drawer task: over 1.6 million FPS at the same 8-GPU/16,384-environment setting. Single-GPU dexterous-suite numbers are not stated.                                                       | KUKA Allegro hand (first-party dexterous suite, Sec. 7.2.3), ShadowHand (third-party GraspQP grasp evaluation only, Sec. 6.4), AbilityHand (third-party GraspQP grasp evaluation only, Sec. 6.4), Fourier GR1-T2 dex hands, Unitree Inspire hand, Unitree trihand (XR teleop retargeters in code only) |            |
| maniskill3_2024 | PhysX (via SAPIEN) rigid-body contacts on GPU; no statement of convex-vs-mesh contact generation or contact-point count in the note. AllegroHandRightTouch reads FSR contact impulse (get_fsr_impulse) for touch sensing (App. VII-H, code).                                                                                                                                                                                                                         | PhysX (via SAPIEN), used as-is; no LCP/PGS/TGS/Newton solver-algorithm name given. Benchmark (cartpole) settings state 'Solver Position Iterations: 4 / Solver Velocity Iterations: 0' (App. XI-A), but per-task _default_sim_config values are not exported in the parsed source.                                                                                                                                                                                                       | 4 position / 0 velocity iterations (benchmarked cartpole configuration only, App. XI-A); per-task defaults not exported in the parsed source |       | yes |      |                     | Up to 30,000+ FPS (RGBD+segmentation) on a single RTX 4090 GPU, environment count for that headline figure not stated; at 128 parallel environments with cameras, ManiSkill3 uses 3.5GB GPU memory vs. Isaac Lab's 14.1GB (Sec. III-B); PPO uses up to 4096 parallel environments for state-based training and 256-1024 for RGB training (App. IX-A), all on RTX 4090 (App. XI-A benchmark hardware). | Allegro Hand (incl. touch-sensor variant, AllegroHandRightTouch), Ability Hand, Inspire Hand, Delta 3-finger hand, TriFinger                                                                                                                                                                           | Apache-2.0 |
| mujoco_2012     | soft, convex velocity-stepping contact; three interchangeable solvers replace the standard LCP-with-friction-pyramid                                                                                                                                                                                                                                                                                                                                                 | implicit-complementarity solver (customized non-smooth Newton method, most accurate); convex                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                              | no    | no  |      |                     | up to ~400,000 dynamics evaluations/sec on a 12-physical-core machine (2x 6-core Intel X5860 3.33GHz,                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                        |            |

| engine                     | contact model                                                                                                                                                                                                                                                                 | solver                                                                                                                                                                                                                                                                                                                                                  | iters                                     | diff. | GPU | dt s | penetration exposed | throughput                                                                                                                                                                                                                                                                                                                 | hands shipped | licence |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|-------|-----|------|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|---------|
|                            | approach: implicit-complementarity, convex (kinetic-energy minimization with a soft non-penetration cost), and diagonal (mass-aware spring-damper)                                                                                                                            | solver (interior-point method used for reported timings; projected Newton/CG/Gauss-Seidel also mentioned as unverified-faster alternatives); diagonal solver (per-contact critically-damped spring-damper using the diagonal of the A matrix, fastest/least accurate)                                                                                   |                                           |       |     |      |                     | 24 threads via hyper-threading), 3D humanoid with 18 DOF and 6 active contacts (Abstract, Table 3)                                                                                                                                                                                                                         |               |         |
| mujoco_convex_contact_2014 | soft, convex, complementarity-free; a unified impulse vector covers joint dry friction, joint/tendon/distance limits, and frictional contacts together, with an elliptical friction cone whose dimensionality per contact is 2 (sliding), 3, or 5 (sliding+torsional+rolling) | GPJS (generalized projected Gauss-Seidel, described as the authors' own unpublished-at-the-time method handling cone and pyramid constraints) for the contact/limit impulse; a separate 'soft Gauss principle' derivation for equality/holonomic constraints; an analytically-invertible closed-form solver for a restricted circular-cone special case | 5 and 50 (both tested, Fig. 3, Sec. VI-B) | no    | no  | 0.01 |                     | single-core Intel i7-3930K (Windows 7): forward dynamics of a 27-dof humanoid with 10 contacts evaluated in 0.1 ms (100x real-time at a 10 ms timestep); inverse dynamics of the same humanoid under 100 impulses in 30 microseconds (10 microseconds with no impulses); no parallel/multi-environment throughput reported |               |         |
| mujoco_warp_2025           | MuJoCo's native solref/solimp-parameterized soft-constraint contact model ported to Warp; per-contact Jacobian construction (dense and sparse variants, with flex variants) and equality/limit/friction constraint rows; not a re-derived contact model                       | MuJoCo's default (implicit CCP-style) solver and Newton constraint solver (via a 'newton' flag threaded through equality/limit/friction kernels) are supported; legacy PGS solver and noslip post-processing pass are 'not yet supported' per the README                                                                                                |                                           | no    | yes |      |                     |                                                                                                                                                                                                                                                                                                                            |               |         |
| newton_2025                | Solver-dependent: MuJoCo-style soft-constraint (solref/solimp-style) contact via SolverMuJoCo/MJWarp; Proximal-ADMM / Dual-Variational-Inequality contact resolution via SolverKamino; position-based-dynamics-style constraint projection                                    | Multi-solver architecture: SolverMuJoCo (wraps MJWarp), SolverKamino (Proximal-ADMM + DVI), XPBD (rigid), VBD (cloth/deformables), Style3D (cloth), plus a separate IKSolver (LM and L-BFGS backends).                                                                                                                                                  |                                           |       | yes |      |                     |                                                                                                                                                                                                                                                                                                                            | Allegro Hand  |         |

| engine        | contact model                                                                                                                                                                                                                                                                            | solver                                                                                                              | iters | diff. | GPU | dt s | penetration exposed | throughput                                                                                                                                                                                                                                                                                                                                        | hands shipped | licence                                                                                                                                                               |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|-------|-------|-----|------|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               | for the XPBD and VBD solvers.                                                                                                                                                                                                                                                            |                                                                                                                     |       |       |     |      |                     |                                                                                                                                                                                                                                                                                                                                                   |               |                                                                                                                                                                       |
| orbit_2023    | PhysX SDK 5 signed-distance-field (SDF) collision checking for rigid bodies (handles non-convex geometry such as screw threads); FEM-based solver using a constraint-based stable neo-hookean material formulation for deformable bodies; PBD (position-based dynamics) solver for cloth | PhysX SDK 5's internal rigid-body and FEM solvers, used as-is; no LCP/PGS/TGS, Newton, or CG iteration counts given |       | no    | yes |      | no                  | 125,000 FPS physics-only ceiling (no env count stated); ~10x rigid-body throughput vs. CPU-vectorized frameworks (robosuite, ManiSkill2, IsaacGymEnvs) swept 256-4096 envs; ~3x deformable/cloth throughput vs. DEDO; 270 FPS aggregate for 10 cameras at 640x480; hardware: 16-core AMD Ryzen 5950X, 64GB RAM, NVIDIA RTX 3090 (Sec. V-E-b, VII) | Allegro hand  |                                                                                                                                                                       |
| pybullet_2016 |                                                                                                                                                                                                                                                                                          |                                                                                                                     |       |       | yes |      |                     |                                                                                                                                                                                                                                                                                                                                                   |               |                                                                                                                                                                       |
| raisim_2018   |                                                                                                                                                                                                                                                                                          |                                                                                                                     |       |       |     |      |                     |                                                                                                                                                                                                                                                                                                                                                   |               | requires a valid license and activation key from the RaiSim Tech website; the repo README also states RaiSim itself is no longer supported in favor of a RaiSim2 repo |
| sapien_2020   | Rigid-body contact with convex-decomposed collision meshes (PhysX 4.1); three joint systems offered (kinematic, dynamic, and PhysX articulation) trading control accuracy against speed.                                                                                                 | PhysX 4.1's internal rigid-body solver, used as-is; no LCP/PGS/TGS name or iteration count given in the paper.      |       |       | no  |      |                     | ~5000 Hz engine, ~700 Hz OpenGL render; single-instance CPU-physics figure with no parallel environments, on a laptop with a 2.2 GHz Intel i7-8750 CPU and an Nvidia GeForce RTX 2070 GPU.                                                                                                                                                        |               |                                                                                                                                                                       |

15 rows; 69 of 165 cells (41%) are values no source stated. No cell is truncated. A value wider than its column is wrapped at a word boundary, so a cell that runs to several rendered lines is one value and not several.

### 4.3 GPU-parallel engines and their throughput

**The GPU-parallel turn.** Isaac Gym set the pattern. Physics, observations, rewards and actions stay on the GPU, and PhysX resolves contacts with the Temporal Gauss-Seidel sweep described above. Its per-task timesteps are published, which is rare: the Shadow Hand runs a 1/120 s physics step under a 1/60 s control step, or 1/20 s in the OpenAI variant. The result that reorganised the field is that reproducing OpenAI's Shadow Hand cube reorientation took under an hour on one A100, against 30 hours on 6144 CPU cores and 8 V100s ([isaacgym\\_2021](#), Sec. 6.4.1). Thirty-five of the papers examined run on it.

Orbit and Isaac Lab moved the stack to PhysX 5, and the dexterous offering is thinner than the predecessor's: the first-party suite is lifting, grasping and reorienting with the KUKA Allegro hand, while Shadow and Ability hands appear only through third-party grasp evaluation ([isaacgym\\_2025](#), Sec. 7.2.3). MJX and MuJoCo Playground took the other route, putting MuJoCo's own solver on the GPU through JAX at the cost described above. MuJoCo Warp ports the same solref and solimp soft-constraint rows to NVIDIA Warp, and its README states that the legacy PGS solver and the noslip pass are unsupported and that differentiability "is not yet available" ([mujoco\\_warp\\_2025](#)). GPU MuJoCo is not a differentiable MuJoCo. Genesis is the broadest, pairing a rigid constraint solver with FEM, MPM, PBD and SPH solvers and a differentiable backward path, and it ships a Shadow Hand. Its README quotes no throughput number at all, so the large FPS figures that circulate for Genesis trace to nothing in [genesis\\_2024](#).

Newton changes how these names should be read. It is explicitly multi-solver: SolverMuJoCo wraps MuJoCo Warp, SolverKamino is a proximal-ADMM and dual-variational-inequality contact solver, and XPBD and VBD do position-based constraint projection ([newton\\_2025](#)). Its contact model is solver-dependent, not a property of the engine, which is how Table 4 records it. Isaac Lab's code already exposes a `--physics newton_mjwarp` backend switch in its hands demo, and its roadmap announces Newton integration. An experiment reported as Isaac Lab may be running PhysX 5 with TGS, or MuJoCo's soft constraint rows under Warp, and those two make different contact errors. The name also fails to fix the physics inside one engine. MuJoCo's convex solver is a family, interior point or projected Newton, conjugate gradient or Gauss-Seidel ([mujoco\\_2012](#), Sec. II-D), and MuJoCo Warp supports neither PGS nor the noslip pass. PhysX 4 and PhysX 5 differ in whether non-convex rigid bodies get SDF collision, which is a contact-geometry difference between Isaac Gym and Isaac Lab under one vendor name ([orbit\\_2023](#), [isaacgym\\_2025](#)). Naming a simulator no longer names its physics. Papers should report the backend and the solver beside the framework.

**Reported throughput, and why the numbers do not compare.** Four headline figures measure four different quantities. Isaac Gym reports parallel environment steps per second with physics, observations and rewards on device: 150,000 for the Shadow Hand at 16,384 environments on one A100. Isaac Lab reports frames per second in training, which includes the learning update: over 900,000 for the DextrAH teacher task at 16,384 environments on eight RTX Pro 6000 GPUs, with no single-GPU dexterous number anywhere in the paper. ManiSkill3 reports "up to 30,000+ FPS" with RGBD and segmentation on one RTX 4090, measured over 1000 random actions with reward and termination removed from the timing, and does not state the environment count behind the figure ([maniskill3\\_2024](#)). Orbit reports a 125,000 FPS physics-only ceiling on an RTX 3090, also with no environment count ([orbit\\_2023](#), Sec. VII).

The one number reported cleanly enough to reuse is MuJoCo Playground's, in PPO steps per second on a single A100 over five seeds. LeapCubeReorient runs at  $76,354 \pm 143$  and PandaRobotiqPushCube at  $487,341 \pm 4,346$ . Same hardware, same measurement, same codebase, a factor of 6.4 between two environments. It is not a factor from the task in isolation. Playground tunes solver iterations, line-search iterations, timestep and contact bounds per environment, with values as far apart as one and four solver iterations ([mujoco\\_playground\\_2025](#), Table III), and it does not print the two configurations side by side. That omission is what makes the two figures incomparable. The same confound sits inside Isaac Gym's paper on one A100: 700,000 environment steps per second for Ant, 200,000 for Humanoid, 150,000 for the Shadow Hand. A training-loop FPS is also mostly not a measurement of physics. Playground breaks the fractional cost down on an RTX 4090: for CartpoleBalance, physics is 0.02, rendering 0.06, inference 0.01 and the policy update 0.91, and the policy update still dominates on the Franka task.

Cross-framework comparisons add further free parameters. ManiSkill2's PickCube table takes the best result over 16 to 512 environments for each system (maniskill2\_2023, Table 1a), giving ManiSkill2 with a render server  $2487 \pm 24$  FPS at its optimum of 64 environments against Isaac Gym's  $865 \pm 35$  at its optimum of 512. The env-count tuning is the second problem here. The first is that the two systems are not the same kind of thing: ManiSkill2 runs rigid-body physics on CPU worker processes behind a shared GPU render server, Isaac Gym runs physics on the GPU, and its own paper says so plainly. The measured quantity includes  $128 \times 128$  rendering at 500 Hz simulation and 20 Hz control for both, so this is a visual sample-collection loop rather than two physics engines. Its authors add the fidelity caveat themselves, and Playground is equally explicit that its cross-simulator plot borrows its Isaac Lab and ManiSkill3 numbers from the ManiSkill3 paper. Several sources give no number at all: MuJoCo Warp's README points to an external nightly dashboard, and Newton's and Genesis's READMEs contain no FPS or speedup anywhere.

The failure reaches past the engine papers into the methods. Of the 47 corpus method papers that name a GPU-batched simulator, 19 state no environment count anywhere, and Isaac Lab's own paper states no physics timestep, no decimation and no solver iteration count for any task. A throughput figure is interpretable only with the environment count, the GPU, the timesteps, the solver iteration budget, and a statement of whether rendering and the learning update sit inside the measurement. Almost nobody reports all five.

#### 4.4 Tactile simulation and the sim-to-real gap

The three tactile simulators in this corpus calibrate against three different things, and none of them is a manipulation outcome. TACTO is a rendering layer over a host engine, by default PyBullet's rigid contact model. It reads post-solve link poses and the engine's reported normal force and maps that force to gel-mesh deformation at the rendering level, so it contributes no contact physics of its own. Its only sim-to-real number is a tactile pose-estimation task, at  $1.66 \pm 0.16$  mm with color-jitter augmentation against  $0.76 \pm 0.07$  mm for a model trained on 128 real datapoints (tactio\_2020, Table II).

Taxim is example-based rather than simulated, with an optical model calibrated from 50 real indentations, and it beats TACTO on every optical-similarity metric against real images. Its marker-motion model is calibrated against an ANSYS FEM, and the errors are  $1.00 \times 10^{-2}$  mm real against FEM,  $1.02 \times 10^{-2}$  mm real against Taxim, and  $3.96 \times 10^{-3}$  mm FEM against Taxim. Taxim tracks its own reference more tightly than that reference tracks reality, so its residual is bounded below by the calibration target. The authors also state that they simulate quasi-static contact only, with slip left as future work (taxim\_2021, Sec. V). Slip is what a hand needs.

Tactile Genesis inverts the thread. It implements two penetration-depth backends on the collision geometry, an analytic SDF query and a BVH raycast, and uses the depth as the sensor signal rather than as an error, so the quantity every other engine treats as a defect sets this one's operating point.

**The sim-to-real gap for hands.** Most of what is written about this gap is attribution without measurement. PenSpin asserts that the pure physics gap "cannot be bridged by extensive domain randomization alone" while reporting no experiment that isolates it (penspin\_2024). MuJoCo Playground attributes its LEAP hand failures to physical flex in low-cost hardware and names more accurate collision geometries as the fix, without measuring either. DeXtreme lists four candidate causes for its shortfall, including a malfunctioning Allegro thumb used in most trials, and disambiguates none by experiment (dextreme\_2022).

What has been measured is narrower, and section 7 collects the transfer numbers. The one result that links a quantified model error to transfer is a humanoid recipe paper, which ranks its system identification runs by dynamics-model MSE and pairs each with real success over 10 trials: the lowest-MSE model grasps 8 out of 10, the median-MSE model 3 out of 10, the highest-MSE model 0 out of 10 (humanoid\_sim2real\_recipe\_2025, Table 1). Where a cause has been pinned down elsewhere it is usually perception or actuation, not contact. OpenAI's pose estimator has 3.12 mm error on rendered images and  $9.27 \pm 4.02$  mm on 992 real ones, and PDDM reports a camera tracker with 5 mm average error and 20 ms latency as the unmodelled source in its real numbers (pddm\_2019, App. C).

The one paper that looks at the contact side is usually read backwards. DeXtreme's real-to-sim replay interpenetrated because the replayed poses carried the pose estimator's error, not because the contact model failed, and the paper says so: it could not

calibrate physics against the real system because the state it replayed was already wrong. The commonly cited reading, that the contact model was at fault, is not what the sentence says.

## 5. Training a dexterous policy

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### 5.1 Sources of supervision

A dexterous policy is defined by what supervises it, and four sources are in use across the corpus's 112 papers: a reward function supervises reinforcement learning, a human demonstration supervises imitation, a human reference trajectory supervises a physics-based tracker, which is imitation with a simulator in the loop, and nothing supervises a model-based planner, which is handed a cost and a model instead.

The labels do not partition the corpus. Reinforcement learning is the largest at 53 of the 112 papers, behavior cloning next at 27, distillation at 23, and ten further tags cover the rest. They sum to far more than 112, because most methods published since 2024 sit on two branches at once: a reinforcement-learning teacher and a behavior-cloned student are one paper, not two. Figure 4 draws the tree and the cross-links. Table 7 is the row-by-row version of the same thing, and is the table to scan when looking for work comparable to your own.

What the deployed policy looks like once training is done matters more than the name a paper gives itself, and on that axis the field has converged hard. Section 5.5 shows why.

**What the tabulation does not record.** Table 7's emptiest columns are the ones a reader most needs: only 31 rows state an environment count, only 70 state how many real trials are behind the headline number, and only 39 state how many unseen objects were tested. The trial and unseen-object figures are the audited ones, after section 7.1 recovered 15 trial counts and 7 unseen-object counts that the notes carried and the extraction had dropped. A mostly empty row is not a weak method, but it is one that cannot be compared with any other row here.

Table 7. Methods

| method                  | yr   | task                                    | paradigm         | algorithm                                                                                                                                                                                                                                                                            | hand                                                                                                                          | DoF | bi | sim                                                         | real | trials | unseen obj | penetration   | code |
|-------------------------|------|-----------------------------------------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|-----|----|-------------------------------------------------------------|------|--------|------------|---------------|------|
| ferrari_canny_1992      | 1992 |                                         |                  |                                                                                                                                                                                                                                                                                      |                                                                                                                               |     |    |                                                             |      |        |            |               | no   |
| dapg_2017               | 2017 | grasp, reorient, functional/tool, other | RL+demo          | NPG (Natural Policy Gradient) + BC pretraining (DAPG)                                                                                                                                                                                                                                | ADROIT                                                                                                                        | 24  | no | MuJoCo                                                      | no   |        |            | not addressed | yes  |
| openai_dexterity_2018   | 2018 | reorient                                | RL               | PPO                                                                                                                                                                                                                                                                                  | ShadowRobot Dexterous Hand (EDC electric-motor version)                                                                       | 24  | no | MuJoCo                                                      | yes  | 10     |            | not addressed | no   |
| openai_rubiks_cube_2019 | 2019 | reorient                                | RL, distillation | PPO with Automatic Domain Randomization (ADR); Policy Cloning (DAGGER/distillation-style) for architecture changes                                                                                                                                                                   | Shadow Dexterous E Series Hand (E3M5R)                                                                                        | 20  | no | MuJoCo (+ ORRB/Unity3D for rendered vision training images) | yes  | 10     |            | not addressed | no   |
| pddm_2019               | 2019 | reorient, other                         | MPC              | PDDM: ensemble of feed-forward dynamics models + MPPI-style filtering/reward-weighted-refinement trajectory optimizer (no policy network, no policy-gradient training)                                                                                                               | Shadow Hand (24-DoF, real and sim); D'Claw (9-DoF, sim only, valve turning)                                                   |     | no | MuJoCo                                                      | yes  |        |            | not addressed | yes  |
| dexpilot_2020           | 2020 | reorient, grasp, functional/tool, other | teleop-system    | DART model-based hand tracker bootstrapped by two learned neural priors (GloveNet ResNet-50 keypoint regressor; PointNet++-based keypoint network + JointNet FC mapping), online fingertip-vector kinematic retargeting solved via SLSQP (NLopt) warm-started each frame, Riemannian | Wonik Robotics Allegro hand (16-DoF, 4x4-joint fingers), retrofitted with 4 SynTouch BioTac tactile sensors at the fingertips | 16  | no |                                                             | yes  | 150    |            | not addressed | no   |

| method                  | yr   | task                             | paradigm                   | algorithm                                                                                                                                                                                            | hand                            | DoF | bi  | sim               | real | trials | unseen obj | penetration   | code |
|-------------------------|------|----------------------------------|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|-----|-----|-------------------|------|--------|------------|---------------|------|
|                         |      |                                  |                            | Motion Policies for arm motion generation                                                                                                                                                            |                                 |     |     |                   |      |        |            |               |      |
| castro_sap_contact_2021 | 2021 | bimanual-coord                   |                            | SAP (Semi-Analytic Primal) solver: a custom Newton-type solver on an unconstrained convex primal reformulation of compliant contact, with analytic gradients, line search, and sparsity exploitation | Allegro hand (16 DoF each), two | 16  | yes | Drake             | no   |        |            | measured      | yes  |
| dexmv_2021              | 2021 | grasp, functional/tool           | RL+demo, data-collection   | DAPG / SOIL / GAIL+ on top of TRPO, using demonstrations translated from human video via TSV retargeting + inverse-dynamics action estimation                                                        | Adroit Hand                     | 30  | no  | MuJoCo            | no   |        | 500        | not addressed | yes  |
| dexpoint_2022           | 2022 | grasp, other                     | RL                         | PPO                                                                                                                                                                                                  | Allegro Hand                    | 16  | no  | SAPIEN            | yes  | 260    | 40         | not addressed | yes  |
| dextreme_2022           | 2022 | reorient                         | RL                         | PPO (rl-games implementation, LSTM actor/critic, asymmetric/state-privileged critic)                                                                                                                 | Allegro Hand                    | 16  | no  | Isaac Gym (PhysX) | yes  | 10     |            | not addressed | yes  |
| dexvip_2022             | 2022 | grasp                            | RL                         | PPO with a video-mined consensus grasp-pose auxiliary reward (R_pose) combined with an affordance reward (R_aff) and a lift-success reward (R_succ)                                                  | Adroit Hand                     | 30  | no  | MuJoCo            | no   |        |            | not addressed | no   |
| dime_2022               | 2022 | reorient, grasp                  | teleop-system, BC, RL+demo | INN/VINN (nearest-neighbor imitation) vs. BC; DAPG/PPO/BCRL (simulation RL-finetuning)                                                                                                               | Allegro Hand                    |     | no  | MuJoCo            | yes  | 10     |            | not addressed | yes  |
| holo_dex_2022           | 2022 | reorient, grasp, functional/tool | teleop-system, BC          | VINN (BYOL nearest-neighbor) vs.                                                                                                                                                                     | Allegro Hand                    | 16  | no  |                   | yes  | 10     | 10         | not addressed | yes  |

| method                    | yr   | task                                   | paradigm            | algorithm                                                                                                                                                                                                                                                                                                                 | hand                                                                                                                                                                | DoF | bi  | sim                                                                                                                                                                                       | real | trials | unseen obj | penetration   | code |
|---------------------------|------|----------------------------------------|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|--------|------------|---------------|------|
|                           |      |                                        |                     | Behavior Cloning / BC-Rep                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                     |     |     |                                                                                                                                                                                           |      |        |            |               |      |
| hora_2022                 | 2022 | reorient                               | RL, distillation    | PPO                                                                                                                                                                                                                                                                                                                       | Allegro Hand (Wonik Robotics)                                                                                                                                       | 16  | no  | IsaacGym                                                                                                                                                                                  | yes  | 240    | 30         | not addressed | yes  |
| mjpc_2022                 | 2022 | reorient                               | MPC                 | Predictive Sampling: zero-order, derivative-free sampling-based shooting MPC over a spline-parameterised control sequence; the framework (MJPC) also implements iLQG and Gradient Descent as alternative planners, but the Hand result is attributed to Predictive Sampling                                               | Shadow Hand (MuJoCo Menagerie model)                                                                                                                                |     | no  | MuJoCo                                                                                                                                                                                    | no   |        |            | not addressed | yes  |
| pang_global_planning_2022 | 2022 | reorient, grasp, bimanual-coord, other | trajopt             | iMPC (iLQR-style trajectory optimizer over a locally-smoothed convex quasi-dynamic contact model, CQDC) for local case studies; an RRT variant (RRT-through-contact) using a smoothed Mahalanobis distance metric, single-step dynamically-consistent extension, and contact sampling for global planning on harder tasks | Allegro Hand (full 3D model, in-hand rotation / pen placement / plate pickup / door opening); a 2-DoF-per-finger planar hand (2 fingers) for 2D reorientation tasks |     | yes | Custom Convex Quasi-Dynamic Differentiable Contact (CQDC) model implemented in Drake (MultibodyPlant/SceneGraph), validated against Drake's own high-fidelity second-order contact solver | yes  |        |            | constrained   | yes  |
| videodex_2022             | 2022 | grasp, reorient, functional/tool       | BC, data-collection | two-stream Neural Dynamic Policy (NDP) trajectory regression (L1 loss); pretrained on human-video-retargeted trajectories, fine-tuned on real                                                                                                                                                                             | LEAP Hand                                                                                                                                                           | 16  | no  |                                                                                                                                                                                           | yes  |        |            | not addressed | no   |

| method                | yr   | task                                   | paradigm               | algorithm                                                                                                                                                                                                                                | hand                                                                    | DoF | bi  | sim                                                                                                 | real | trials | unseen obj | penetration   | code |
|-----------------------|------|----------------------------------------|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|-----|-----|-----------------------------------------------------------------------------------------------------|------|--------|------------|---------------|------|
|                       |      |                                        |                        | teleoperated demonstrations; R3M visual encoder                                                                                                                                                                                          |                                                                         |     |     |                                                                                                     |      |        |            |               |      |
| visual_dexterity_2022 | 2022 | reorient                               | RL, distillation       | PPO (teacher, privileged state) + DAgger imitation distillation to a depth-point-cloud student (two-stage: synthetic-PC pretrain then rendered-occluded-PC fine-tune)                                                                    | D'Claw (open-source, low-cost; 3-finger and modified 4-finger versions) |     | no  | Isaac Gym (PhysX)                                                                                   | yes  | 20     | 12         | not addressed | yes  |
| aloha_act_2023        | 2023 | bimanual-coord                         | BC                     | ACT (Action Chunking with Transformers, CVAE)                                                                                                                                                                                            | parallel-jaw gripper (not dexterous), two                               |     | yes | MuJoCo                                                                                              | yes  | 25     |            | not addressed | yes  |
| anyteleop_2023        | 2023 | grasp, functional/tool, handover       | teleop-system, RL+demo | DAPG (downstream IL); keypoint-vector retargeting optimization (dex_retargeting)                                                                                                                                                         | Allegro Hand                                                            |     | no  | SAPIEN                                                                                              | yes  | 100    |            | not addressed | yes  |
| artigrasp_2023        | 2023 | grasp, functional/tool, bimanual-coord | RL                     | PPO (D-Grasp-based implementation)                                                                                                                                                                                                       | MANO hand model, two                                                    | 51  | yes | RaiSim                                                                                              | no   |        | 1          | not addressed | yes  |
| dexdeform_2023        | 2023 | bimanual-coord, other                  | BC, trajopt            | latent skill model (VAE over demonstrations) for skill-planned trajectory initialization, refined by gradient-based trajectory optimization through a differentiable MPM simulator, with refined rollouts fed back as new demonstrations | Shadow Dexterous Hand (simulated)                                       | 28  | yes | PlasticineLab (MLS-MPM, CUDA), built on Hu et al. 2018's Moving Least Squares Material Point Method | no   |        |            | not addressed | yes  |
| dexpbt_2023           | 2023 | grasp, reorient, bimanual-coord        | RL                     | PPO (rl_games) with decentralized Population-Based Training (PBT) over RL hyperparameters and reward-shaping coefficients                                                                                                                | Allegro Hand                                                            | 16  | yes | Isaac Gym (PhysX)                                                                                   | no   |        |            | not addressed | yes  |

| method                             | yr   | task                                          | paradigm                          | algorithm                                                                                                                | hand                                                                                | DoF | bi  | sim                                                                                                                                                                                                        | real | trials | unseen obj | penetration   | code |
|------------------------------------|------|-----------------------------------------------|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|--------|------------|---------------|------|
| dexterous_functional_grasping_2023 | 2023 | functional/tool                               | RL                                | PPO                                                                                                                      | LEAP hand                                                                           | 16  | no  | IsaacGym                                                                                                                                                                                                   | yes  | 70     | 5          |               | no   |
| diffusion_policy_2023              | 2023 | grasp, functional/tool, bimanual-coord, other | BC, diffusion                     | conditional DDPM over action sequences (CNN or transformer noise-prediction network), DDIM at inference                  |                                                                                     |     | yes | task-specific: Robomimic (MuJoCo-backed), Push-T (custom 2D physics per IBC), BlockPush (per BET), Franka Kitchen (per Relay Policy Learning); no single engine/timestep stated for the combined benchmark | yes  | 20     |            | not addressed | yes  |
| dynamic_handover_2023              | 2023 | handover, bimanual-coord                      | RL                                | MAPPO                                                                                                                    | Allegro Hand                                                                        | 16  | yes | IsaacGym                                                                                                                                                                                                   | yes  | 15     | 14         | not addressed | no   |
| eureka_2023                        | 2023 | reorient, bimanual-coord, other               | RL                                | PPO with LLM-authored (GPT-4) evolutionary reward search                                                                 | Shadow Hand (also Allegro Hand; bimanual pairs of Shadow Hands for Dexterity tasks) | 20  | yes | IsaacGym                                                                                                                                                                                                   | yes  |        |            | not addressed | yes  |
| pgdm_2023                          | 2023 | grasp, functional/tool                        | RL                                | PPO (Stable-Baselines3) with an object-trajectory-tracking reward plus pre-grasp-based exploration initialization (PGDM) | ShadowHand                                                                          | 24  | no  | MuJoCo                                                                                                                                                                                                     | yes  |        |            | not addressed | no   |
| physhoi_2023                       | 2023 | track-human-ref                               | RL                                | PPO                                                                                                                      | SMPL-X humanoid (simulated, not a robot hand)                                       | 90  | no  | Isaac Gym                                                                                                                                                                                                  | no   |        |            | not addressed | yes  |
| robot_synesthesia_2023             | 2023 | reorient                                      | RL, distillation                  | PPO                                                                                                                      | Allegro Hand                                                                        | 16  | no  | Isaac Gym                                                                                                                                                                                                  | yes  | 5      |            | not addressed | no   |
| rotateit_2023                      | 2023 | reorient                                      | RL, distillation                  | PPO                                                                                                                      | AllegroHand (Wonik Robotics)                                                        | 16  | no  | IsaacGym                                                                                                                                                                                                   | yes  |        | 15         | not addressed |      |
| rotating_without_seeing_2023       | 2023 | reorient                                      | RL                                | PPO                                                                                                                      | Allegro Hand                                                                        | 16  | no  | IsaacGym                                                                                                                                                                                                   | yes  | 30     | 5          | not addressed |      |
| unidexgrasp_2023                   | 2023 | grasp                                         | grasp-synthesis, RL, distillation | PPO (teacher) + DAgger (student distillation)                                                                            | ShadowHand                                                                          | 26  | no  | Isaac Gym                                                                                                                                                                                                  | no   |        | 241        | penalised     | yes  |
| unidexgrasp_p_2023                 | 2023 | grasp                                         | RL, distillation                  | PPO + DAgger (with critic distillation)                                                                                  | Shadow Hand                                                                         | 24  | no  | Isaac Gym                                                                                                                                                                                                  | no   |        |            | not addressed | yes  |

| method               | yr   | task                                      | paradigm                   | algorithm                                                                                                                                                                     | hand                                                                                                                                                                                    | DoF | bi  | sim               | real | trials | unseen obj | penetration   | code |
|----------------------|------|-------------------------------------------|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-------------------|------|--------|------------|---------------|------|
| ace_teleop_2024      | 2024 | grasp, functional/tool                    | teleop-system, BC          | 3D Diffusion Policy (DP3) for xArm platforms; ACT for H1 humanoid (downstream IL); IK-based scale/recenter end-effector mapping + AnyTeleop fingertip-vector hand retargeting | Ability Hand / Inspire Hand / parallel-jaw gripper (embodiment-dependent)                                                                                                               |     | yes |                   | yes  | 10     | 18         | not addressed | no   |
| anyrotate_2024       | 2024 | reorient                                  | RL, distillation           | PPO                                                                                                                                                                           | Allegro Hand                                                                                                                                                                            | 16  | no  | IsaacGym          | yes  |        | 10         | not addressed | no   |
| asymdex_2024         | 2024 | bimanual-coord, grasp                     | RL                         | PPO                                                                                                                                                                           | Shadow Hand (30-DoF: 24-DoF hand + 6-DoF floating wrist) in simulation; 16-DoF Allegro hand + 6-DoF Ability Hand (mismatched, hardware-availability artifact) in real-world experiments | 30  | yes | NVIDIA Isaac Gym  | yes  | 20     |            | not addressed | yes  |
| bidex_teleop_2024    | 2024 | bimanual-coord, handover, functional/tool | teleop-system, BC          | ACT (Action Chunking Transformer)                                                                                                                                             | LEAP Hand (16 DoF); LEAP Hand V2 (21 DoF) used for 'extreme dexterity' experiments                                                                                                      | 16  | yes | none              | yes  | 20     |            | not addressed | no   |
| bidexhd_2024         | 2024 | functional/tool, bimanual-coord           | RL, distillation           | IPPO (teacher) + DAgger (student distillation)                                                                                                                                | LEAP Hand, left and right                                                                                                                                                               | 16  | yes | Isaac Gym (PhysX) | no   |        |            | not addressed | yes  |
| bimangrasp_2024      | 2024 | grasp, bimanual-coord                     | grasp-synthesis, diffusion | MALA (Metropolis-adjusted Langevin) energy optimisation + DDPM (BimanGrasp-DDPM)                                                                                              | Shadow Hand, pair                                                                                                                                                                       | 22  | yes | Isaac Gym         | no   |        | 225        | penalised     | no   |
| bunny_visionpro_2024 | 2024 | grasp, functional/tool, bimanual-coord    | teleop-system, BC          | ACT, Diffusion Policy, DP3 (downstream IL); SQP fingertip-vector hand retargeting + unified IK/singularity/coll                                                               | Ability Hand                                                                                                                                                                            | 24  | yes |                   | yes  | 10     |            | not addressed | yes  |

| method           | yr   | task                                                    | paradigm                           | algorithm                                                                                                                                                                                                                                        | hand                                                                                                                                                                        | DoF | bi  | sim                                                                                                         | real | trials | unseen obj | penetration   | code |
|------------------|------|---------------------------------------------------------|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-------------------------------------------------------------------------------------------------------------|------|--------|------------|---------------|------|
|                  |      |                                                         |                                    | ision arm-motion optimization                                                                                                                                                                                                                    |                                                                                                                                                                             |     |     |                                                                                                             |      |        |            |               |      |
| cyberdemo_2024   | 2024 | grasp, reorient, functional/tool                        | BC, teleop-system, data-collection | ACT (Action Chunking with Transformers) BC policy, trained on sim teleoperation demonstrations augmented via Sensitivity-Aware Kinematics Augmentation, automatic curriculum learning, and action aggregation, fine-tuned on real demonstrations | Allegro Hand                                                                                                                                                                | 16  | no  | SAPIEN                                                                                                      | yes  | 20     | 2          | not addressed | no   |
| demostart_2024   | 2024 | grasp, functional/tool, reorient, other                 | RL, distillation                   | MPO (teacher, auto-curriculum) + BC distillation (student, Perceiver-Actor-Critic)                                                                                                                                                               | DEX-EE Hand                                                                                                                                                                 | 12  | no  | MuJoCo                                                                                                      | yes  | 100    |            | not addressed | no   |
| dexcap_2024      | 2024 | grasp, bimanual-coord, functional/tool                  | BC, diffusion, data-collection     | Diffusion Policy with Perceiver point-cloud encoder (DP-perc)                                                                                                                                                                                    | LEAP Hand                                                                                                                                                                   | 16  | yes | none                                                                                                        | yes  | 60     | 9          |               | yes  |
| dexmimicgen_2024 | 2024 | bimanual-coord, grasp, functional/tool                  | data-collection, BC                | MimicGen-style per-arm subtask transform/replay generation; downstream BC-RNN / BC-RNN-GMM / Diffusion Policy                                                                                                                                    | Inspire dexterous hand (6-DoF, real-world GR1); unnamed 'dexterous hands' in the two simulated dexterous-hand embodiments                                                   | 6   | yes | RoboSuite (MuJoCo)                                                                                          | yes  | 20     |            |               | yes  |
| dp3_2024         | 2024 | reorient, grasp, functional/tool, bimanual-coord, other | BC, diffusion                      | Diffusion Policy backbone (DDPM training / DDIM inference) conditioned on a compact 3D point-cloud feature (small MLP + max-pool encoder) instead of image features                                                                              | Shadow Hand (Adroit, Bi-DexHands, DexDeform, DexMV domains); Allegro Hand (DexArt, HORA domains; also real Roll-Up/Dumpling/Drill); parallel gripper (MetaWorld; real Pour) |     | yes | MuJoCo (Adroit, MetaWorld), IsaacGym (Bi-DexHands, HORA), Sapien (DexArt, DexMV), PlasticineLab (DexDeform) | yes  | 40     | 5          | not addressed | yes  |
| dreureka_2024    | 2024 | reorient                                                | RL                                 | PPO with LLM-authored (GPT-                                                                                                                                                                                                                      | LEAP hand                                                                                                                                                                   | 16  | no  | Isaac Gym                                                                                                   | yes  |        |            | not addressed | yes  |

| method                 | yr   | task                                   | paradigm             | algorithm                                                                                                                                                                      | hand                                                             | DoF | bi  | sim                               | real | trials | unseen obj | penetration   | code |
|------------------------|------|----------------------------------------|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|-----|-----|-----------------------------------|------|--------|------------|---------------|------|
|                        |      |                                        |                      | 4) reward search + reward-aware physics prior domain randomization                                                                                                             |                                                                  |     |     |                                   |      |        |            |               |      |
| egomimic_2024          | 2024 | grasp, functional/tool, bimanual-coord | BC                   | shared-backbone ACT variant co-trained on paired egocentric human video and teleoperated bimanual robot demonstrations, with dual pose-space and joint-space action heads      | parallel-jaw gripper (not a dexterous hand; see note scope flag) | 1   | yes |                                   | yes  | 135    |            | not addressed | yes  |
| graspxl_2024           | 2024 | grasp                                  | RL                   | PPO                                                                                                                                                                            | MANO (also Shadow Hand, Allegro Hand, Faive Hand)                | 45  | no  | RaiSim                            | no   |        | 503409     | not addressed | yes  |
| hato_visuotactile_2024 | 2024 | handover, bimanual-coord               | diffusion            | Diffusion Policy (DDPM, CNN-based)                                                                                                                                             | Psyonic Ability Hand, two                                        | 6   | yes |                                   | yes  | 10     |            | not addressed | yes  |
| hudor_2024             | 2024 | grasp, functional/tool                 | RL                   | open-loop IK replay of a single retargeted human fingertip trajectory plus online residual RL (DrQv2) trained with an object-centric point-tracking trajectory-matching reward | Allegro Hand                                                     | 16  | no  |                                   | yes  | 10     |            | not addressed | no   |
| humanplus_2024         | 2024 | track-human-ref, locomanipulation      | RL, BC               | PPO (HST shadowing policy) + decoder-only Transformer BC (HIT)                                                                                                                 | Inspire-Robots RH56DFX, two                                      | 6   | yes | Isaac Gym-based legged_gym/rsi_rl | yes  | 10     |            | not addressed | yes  |
| objdex_2024            | 2024 | track-human-ref, bimanual-coord        | BC, RL, distillation | BC (Transformer high-level planner) + PPO (low-level controller)                                                                                                               | Shadow Hand                                                      |     | yes |                                   | yes  | 80     | 1          | not addressed | no   |
| okami_2024             | 2024 | grasp, functional/tool, bimanual-coord | trajopt, BC          | single-video imitation: SLAHMR-extended SMPL-H body+hand reconstruction, GPT-4V/Grounded-SAM/Cutie                                                                             | Inspire Hand (x2)                                                | 6   | yes | RoboSuite (2 of 6 tasks only)     | yes  | 12     |            | not addressed | yes  |

| method                   | yr   | task                                 | paradigm             | algorithm                                                                                                                                                                                                                                                                                      | hand                                                                                                                                    | DoF | bi  | sim                                                                                                              | real | trials | unseen obj | penetration   | code |
|--------------------------|------|--------------------------------------|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|-----|-----|------------------------------------------------------------------------------------------------------------------|------|--------|------------|---------------|------|
|                          |      |                                      |                      | object-centric<br>reference-plan<br>construction,<br>factorized arm IK<br>(Pink) + dex-<br>retargeting hand-<br>joint mapping,<br>SE(3) trajectory<br>warping to new<br>object poses at<br>test time;<br>optional<br>downstream ACT<br>BC policy trained<br>on OKAMI-<br>generated<br>rollouts |                                                                                                                                         |     |     |                                                                                                                  |      |        |            |               |      |
| omnigrasp_2024           | 2024 | grasp, track-<br>human-ref           | RL, distillation     | PPO<br>(hierarchical:<br>PHC-X finger<br>imitator →<br>PULSE-X<br>distilled latent<br>prior → PPO<br>over latent action<br>space)                                                                                                                                                              | SMPL-X whole-<br>body humanoid<br>(simulated, not a<br>robot hand)                                                                      | 90  | yes | Isaac Gym                                                                                                        | no   |        | 5          | not addressed | yes  |
| omnih2o_2024             | 2024 | track-human-ref,<br>locomanipulation | RL, distillation     | PPO (teacher) +<br>DAGger (student)<br>+ diffusion policy<br>(autonomy layer)                                                                                                                                                                                                                  | Inspire hands<br>(open-loop, DoF<br>not stated)                                                                                         |     | yes |                                                                                                                  | yes  | 20     |            |               | yes  |
| open_televisi<br>on_2024 | 2024 | bimanual-coord,<br>functional/tool   | teleop-system,<br>BC | ACT (Action<br>Chunking<br>Transformer)<br>with DinoV2 ViT<br>backbone                                                                                                                                                                                                                         | Inspire Robots<br>hand (6 actuated<br>DoF, 12 total);<br>Fourier GR-1<br>embodiment<br>instead uses a 1-<br>DoF parallel-jaw<br>gripper | 6   | yes | none                                                                                                             | yes  | 20     |            | not addressed | yes  |
| openvla_2024             | 2024 | other                                | VLA                  | autoregressive<br>next-token<br>prediction over<br>discretized (256-<br>bin per-<br>dimension,<br>quantile-based)<br>action tokens,<br>single-step (no<br>action chunking)                                                                                                                     |                                                                                                                                         |     | no  | LIBERO<br>benchmark<br>(simulated<br>Franka) for fine-<br>tuning<br>experiments;<br>engine/version<br>not stated | yes  | 230    |            | not addressed | yes  |
| penspin_2024             | 2024 | reorient                             | RL, distillation     | PPO + BC                                                                                                                                                                                                                                                                                       | Allegro Hand                                                                                                                            | 16  | no  | Isaac Gym                                                                                                        | yes  | 50     | 7          | not addressed | yes  |
| pi0_2024                 | 2024 | other                                | VLA, flow            | conditional flow<br>matching (linear-<br>Gaussian/optima<br>l-transport<br>probability path),<br>forward-Euler<br>integration at<br>inference (10<br>steps), shifted-                                                                                                                          |                                                                                                                                         |     | yes |                                                                                                                  | yes  | 10     |            | not addressed | yes  |

| method                        | yr   | task                                   | paradigm                                      | algorithm                                                                                                                                                                                                              | hand                                                                                                                                                               | DoF | bi  | sim                  | real | trials | unseen obj | penetration   | code |
|-------------------------------|------|----------------------------------------|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|----------------------|------|--------|------------|---------------|------|
|                               |      |                                        |                                               | Beta noisy-timestep sampling                                                                                                                                                                                           |                                                                                                                                                                    |     |     |                      |      |        |            |               |      |
| pianomime_2024                | 2024 | music                                  | RL+demo, distillation                         | PPO (per-song specialists); DDPM diffusion (generalist)                                                                                                                                                                | Shadow Hand E3M5                                                                                                                                                   | 23  | yes | ROBOPIANIST (MuJoCo) | no   |        |            | not addressed | yes  |
| rdt1b_2024                    | 2024 | bimanual-coord                         | diffusion                                     | RDT (DiT + cross-attention, QKNorm/RMSNorm, MLP decoder, Alternating Condition Injection)                                                                                                                              | parallel-jaw gripper (ALOHA-style, Cobot Mobile ALOHA), two                                                                                                        |     | yes |                      | yes  | 139    | 2          | not addressed | yes  |
| resdex_2024                   | 2024 | grasp                                  | RL, distillation                              | PPO + DAgger                                                                                                                                                                                                           | ShadowHand                                                                                                                                                         | 18  | no  | IsaacGym             | no   |        | 241        | not addressed | yes  |
| twisting_lids_2024            | 2024 | functional/tool, bimanual-coord        | RL                                            | PPO (asymmetric actor-critic)                                                                                                                                                                                          | Allegro Hand, two                                                                                                                                                  | 16  | yes | Isaac Gym            | yes  | 20     | 15         | not addressed | no   |
| umi_2024                      | 2024 | functional/tool, bimanual-coord, other | BC, diffusion, data-collection, teleop-system | Diffusion Policy (unmodified loss/decoder); contribution is the hand-held data-collection hardware and the observation/action interface (relative-trajectory representation, inference-time latency matching)          |                                                                                                                                                                    |     | yes |                      | yes  | 260    |            |               | yes  |
| articulated_tools_inhand_2025 | 2025 | functional/tool                        | RL, BC, distillation                          | PPO (oracle) + BC (student + CATFA online adaptation)                                                                                                                                                                  | Inspire Hand                                                                                                                                                       | 6   | no  | IsaacLab             | yes  | 50     |            | not addressed | no   |
| being_h0_2025                 | 2025 | grasp, functional/tool                 | VLA, BC                                       | autoregressive next-token prediction over discretized MANO motion tokens (Grouped Residual Quantization) for pretraining; L1-loss MLP regression head over learnable query tokens for post-training imitation learning | 6-DoF Inspire hand mounted on a 7-DoF Franka Research 3 arm (real robot); MANO parametric hand model used as the pretraining representation (no physical hardware) | 6   | no  |                      | yes  | 20     |            | not addressed | yes  |
| clutterdexgrasp_2025          | 2025 | grasp                                  | RL, diffusion, distillation                   | PPO (teacher) + DP3 diffusion                                                                                                                                                                                          | AgiBot dexterous hand                                                                                                                                              | 6   | no  | Isaac Gym            | yes  | 167    | 2029       | constrained   | no   |

| method                             | yr   | task                            | paradigm         | algorithm                                                                                                                                                                                                                                                                                                                                        | hand                                                                                                                                                                                                                        | DoF | bi  | sim                                                                                                                      | real | trials | unseen obj | penetration   | code |
|------------------------------------|------|---------------------------------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|--------------------------------------------------------------------------------------------------------------------------|------|--------|------------|---------------|------|
|                                    |      |                                 |                  | policy (student)                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                             |     |     |                                                                                                                          |      |        |            |               |      |
| cross_embodiment_world_models_2025 | 2025 | other                           | world-model, MPC | DPI-Net (graph neural network particle-dynamics world model) + sampling-based receding-horizon MPC                                                                                                                                                                                                                                               | PSYONIC Ability Hand (6-DoF) and Robot Era XHand (12-DoF) for real-world deployment; trained across 6 simulated hands (Ability, Allegro, XHand, Leap, Shadow, forearm-less Shadow variant) for the embodiment-scaling study |     | no  | SAPIEN (rigid-body Object Pushing data) and Rewarped (differentiable multiphysics, deformable Plasticine Reshaping data) | yes  | 20     |            | not addressed | no   |
| dexgraspvla_2025                   | 2025 | grasp, other                    | VLA, diffusion   | hierarchical VLA: frozen VLM high-level planner (bounding-box affordance) + DIT diffusion-policy controller conditioned on frozen DINOv2 features and a SAM+Cutie object mask, trained end-to-end from scratch by imitation learning (MSE noise-prediction loss); Immiscible Diffusion noise assignment; DDIM sampling (16 of 50 training steps) | PsiBot G0-R, 6-DoF, single (right) hand                                                                                                                                                                                     | 6   | no  |                                                                                                                          | yes  | 1287   | 360        | not addressed | yes  |
| dexmachina_2025                    | 2025 | track-human-ref, bimanual-coord | RL               | PPO (rl-games)                                                                                                                                                                                                                                                                                                                                   | six URDFs: Inspire, Allegro, XHand, Schunk (main), Ability, DexRobot Dex Hand                                                                                                                                               |     | yes | Genesis                                                                                                                  | no   |        |            | constrained   | yes  |
| dexman_2025                        | 2025 | track-human-ref, bimanual-coord | RL, trajopt      | PPO (RL_GAMES) residual policy on top of IK-retargeted human motion                                                                                                                                                                                                                                                                              | Shadow Dexterous Hand                                                                                                                                                                                                       |     | yes | NVIDIA Isaac Gym                                                                                                         | no   |        |            |               | no   |
| dexndm_2025                        | 2025 | reorient                        | RL, distillation | PPO (oracle) + BC (generalist) + supervised residual policy                                                                                                                                                                                                                                                                                      | LEAP hand                                                                                                                                                                                                                   |     | no  | Isaac Gym                                                                                                                | yes  |        |            | not addressed | no   |
| dexplore_2025                      | 2025 | track-human-ref                 | RL, distillation | PPO (teacher) + DAgger-style                                                                                                                                                                                                                                                                                                                     | Inspire hand (also Allegro)                                                                                                                                                                                                 | 6   | no  | Isaac Gym                                                                                                                | yes  |        | 2          | not addressed | yes  |

| method                  | yr   | task                                   | paradigm            | algorithm                                                                                                                                                                                                                                             | hand                                                                                                                                                                                             | DoF | bi | sim                                                            | real | trials | unseen obj | penetration   | code |
|-------------------------|------|----------------------------------------|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----|----------------------------------------------------------------|------|--------|------------|---------------|------|
|                         |      |                                        |                     | VAE distillation (student)                                                                                                                                                                                                                            | hand)                                                                                                                                                                                            |     |    |                                                                |      |        |            |               |      |
| dexremoe_2025           | 2025 | reorient                               | RL                  | PPO                                                                                                                                                                                                                                                   | GX11 three-fingered dexterous hand (custom)                                                                                                                                                      | 11  | no | IsaacGym                                                       | no   |        | 50         | not addressed | no   |
| dexteritygen_2025       | 2025 | reorient, functional/tool, grasp       | RL, diffusion       | RL-generated Anygrasp-to-Anygrasp transitions used to train a UNet DDPM diffusion foundation controller (DexGen) with gradient-guided diffusion sampling that projects an external (teleoperated) command onto a safe, high-likelihood learned action | Allegro Hand                                                                                                                                                                                     | 16  | no |                                                                | yes  | 240    |            |               | no   |
| dexterous_handover_2025 | 2025 | handover, grasp                        | RL                  | PPO                                                                                                                                                                                                                                                   | Allegro Hand                                                                                                                                                                                     |     | no | IsaacLab                                                       | no   |        | 3          | not addressed |      |
| dextrack_2025           | 2025 | track-human-ref                        | RL+demo             | PPO + IL action-supervision loss                                                                                                                                                                                                                      | Allegro hand (sim); LEAP hand (real)                                                                                                                                                             | 16  | no | Isaac Gym                                                      | yes  |        |            | measured      | yes  |
| dexumi_2025             | 2025 | grasp, functional/tool                 | BC, data-collection | diffusion policy (DDPM-style action-chunking) trained on exoskeleton-collected demonstrations converted to robot-hand-looking video via a SAM2 segment/inpaint/composite pipeline                                                                     | Inspire Hand (12 DoF, 6 active) and XHand (12 active DoF)                                                                                                                                        |     | no |                                                                | yes  | 20     |            |               | yes  |
| dexvla_2025             | 2025 | grasp, functional/tool, bimanual-coord | VLA, diffusion      | ScaleDP (Scale Diffusion Policy, transformer-based diffusion action head, up to 1B parameters, multi-head per embodiment) plugged into a Qwen2-VL-2B VLA backbone, trained with a 3-stage 'Embodied                                                   | Robotiq parallel-jaw gripper (Franka rig a, Bimanual UR5e rig c); Inspire multi-fingered dexterous hand, 6-DoF hand joint space (Franka rig b); AgileX bimanual arm end-effector unnamed (rig d) |     |    | LIBERO (secondary, App. A.2 only; main results are real-robot) | yes  | 10     | 30         | not addressed | yes  |

| method                  | yr   | task                                   | paradigm            | algorithm                                                                                                                                                                                               | hand                                                                                                                                                                                                                                                                                                         | DoF | bi  | sim                                                                                                              | real | trials | unseen obj | penetration   | code |
|-------------------------|------|----------------------------------------|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|------------------------------------------------------------------------------------------------------------------|------|--------|------------|---------------|------|
|                         |      |                                        |                     | Curriculum Learning' recipe; loss $L = L_{diff} + \alpha L_{ntp}$ ( $\alpha=1$ )                                                                                                                        |                                                                                                                                                                                                                                                                                                              |     |     |                                                                                                                  |      |        |            |               |      |
| dexwild_2025            | 2025 | grasp, functional/tool, bimanual-coord | BC, data-collection | diffusion U-Net policy (also compared against ACT) co-trained on human wearable-rig demonstrations and teleoperated robot demonstrations via fixed-ratio batch sampling                                 | LEAP Hand / LEAP Hand V2 Advanced                                                                                                                                                                                                                                                                            | 17  | yes |                                                                                                                  | yes  |        | 11         |               | yes  |
| dydexhandover_2025      | 2025 | handover, bimanual-coord               | RL                  | MAPPO (human-regularized, CTDE, with hybrid advantage estimation)                                                                                                                                       | 11-DoF (6 actuated) dexterous hand, per side (vendor not stated)                                                                                                                                                                                                                                             | 11  | yes | NVIDIA Isaac Sim / Isaac Lab                                                                                     | no   |        | 9          | not addressed | no   |
| egozero_2025            | 2025 | grasp, functional/tool                 | BC                  | closed-loop Transformer policy (BC, Gaussian NLL loss) over a morphology-agnostic 3D-point state-action space (triangulated object keypoints + thumb/index 3D coordinates + a thresholded grasp scalar) | Franka Panda parallel-jaw gripper (not a dexterous hand; see note scope flag)                                                                                                                                                                                                                                | 1   | no  |                                                                                                                  | yes  | 15     |            | not addressed | yes  |
| gemini_robotics_15_2025 | 2025 | other                                  | VLA                 |                                                                                                                                                                                                         | ALOHA parallel gripper (by platform convention, not re-specified); Bi-arm Franka parallel gripper; Apollo humanoid multi-finger dexterous hand (grasp types shown: single-finger push, cylindrical side grasp, medium wrap, five-finger power grasp, bimanual grasp/rotate; no DoF or vendor model given for |     | yes | MuJoCo (used to generate evaluation scenes at scale, not for training; engine timestep/contact model not stated) | yes  |        |            | not addressed | no   |

| method                     | yr   | task                                                     | paradigm          | algorithm                                                                                                                                                                                                                                 | hand                                                                                                                                                                                        | DoF | bi  | sim                                                                                                | real | trials | unseen obj | penetration   | code |
|----------------------------|------|----------------------------------------------------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|----------------------------------------------------------------------------------------------------|------|--------|------------|---------------|------|
|                            |      |                                                          |                   |                                                                                                                                                                                                                                           | any embodiment)                                                                                                                                                                             |     |     |                                                                                                    |      |        |            |               |      |
| gemini_robotics_2025       | 2025 | other, handover                                          | VLA, distillation |                                                                                                                                                                                                                                           | ALOHA 2 parallel gripper, two fingers (primary embodiment); bi-arm Franka parallel gripper; Apollo humanoid five-fingered dexterous hand (no model/DoF given, evaluated only qualitatively) |     | yes |                                                                                                    | yes  | 20     |            | not addressed | no   |
| geometric_retargeting_2025 | 2025 | grasp                                                    | teleop-system     | per-finger MLP retargeting network (GeoRT) trained offline against five geometric losses: motion-direction preservation, C-space coverage (Chamfer), flatness, pinch correspondence, and a learned self-collision classifier              | Allegro Hand                                                                                                                                                                                |     | no  |                                                                                                    | yes  |        |            | not addressed | yes  |
| gr_dexter_2025             | 2025 | bimanual-coord, grasp                                    | VLA               | GR-Dexter (Mixture-of-Transformer VLA, flow matching + next-token prediction, following GR-3)                                                                                                                                             | ByteDexter V2, two                                                                                                                                                                          | 21  | yes |                                                                                                    | yes  | 125    | 23         | not addressed | no   |
| groot_n16_2025             | 2025 | grasp, functional/tool, bimanual-coord, locomanipulation | VLA               | flow-matching DiT action head (32 layers, per page and cross-confirmed by code changelog) fed by a Cosmos-2B VLM backbone (per page); predicts state-relative action chunks for most embodiments; no algorithm name beyond "DiT" is given |                                                                                                                                                                                             |     | yes | Galaxea R1 Pro simulated on the BEHAVIOR suite; physics engine, timestep, and env count not stated | yes  |        |            | not addressed | yes  |
| groot_n1_2025              | 2025 | grasp, functional/tool, bimanual-coord, handover         | VLA, flow         | dual-system architecture: Eagle-2 VLM (System 2,                                                                                                                                                                                          | Fourier dexterous hands, on the Fourier GR-1 humanoid;                                                                                                                                      |     | yes | RoboCasa (Kitchen and GR-1 Tabletop benchmarks);                                                   | yes  | 10     | 5          | not addressed | yes  |

| method                            | yr   | task                                         | paradigm | algorithm                                                                                                                                                                                                                                                                                                                                                     | hand                                                                                                                                                                                                                                               | DoF | bi  | sim                                                                                                                                                     | real | trials | unseen obj | penetration   | code |
|-----------------------------------|------|----------------------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------|------|--------|------------|---------------|------|
|                                   |      |                                              |          | reasoning)<br>feeding a DiT<br>flow-matching<br>action head<br>(System 1,<br>acting) via<br>alternating<br>cross-/self-<br>attention; K=4-<br>step forward-<br>Euler denoising<br>at inference;<br>latent-action VQ-<br>VAE for labeling<br>action-less video                                                                                                 | DoF/finger count<br>not stated                                                                                                                                                                                                                     |     |     | DexMimicGen<br>(cross-<br>embodiment<br>suite and sim<br>pretraining-data<br>generation);<br>underlying<br>physics engine<br>and timestep not<br>stated |      |        |            |               |      |
| h_rdt_2025                        | 2025 | grasp,<br>functional/tool,<br>bimanual-coord | flow     | 2B-parameter<br>flow-matching<br>diffusion<br>transformer (H-<br>RDT), pretrained<br>on EgoDex<br>human hand-<br>pose trajectories<br>then fine-tuned<br>per robot<br>embodiment by<br>reinitializing the<br>state/action<br>adaptors and<br>action decoder                                                                                                   | parallel-jaw / 2-<br>jaw grippers at<br>deployment<br>(Aloha-Agilex,<br>ARX5, Franka-<br>Panda,<br>UR5+UMI); no<br>dexterous hand<br>deployed, though<br>pretraining action<br>space includes<br>bimanual<br>fingertip<br>positions from<br>EgoDex |     | yes | RoboTwin 2.0                                                                                                                                            | yes  | 25     |            | not addressed | no   |
| helix_2025                        | 2025 | grasp,<br>functional/tool,<br>handover       | VLA      | dual-network<br>"System 1,<br>System 2" VLA:<br>S2 (7B VLM, 7-<br>9Hz) produces a<br>continuous latent<br>conditioning<br>vector consumed<br>by S1 (80M-<br>param cross-<br>attention<br>encoder-decoder<br>transformer,<br>200Hz); trained<br>end-to-end with<br>a "standard<br>regression loss"<br>(no<br>diffusion/flow-<br>matching, no<br>formula given) |                                                                                                                                                                                                                                                    |     | yes |                                                                                                                                                         | yes  |        |            | not addressed | no   |
| human2sim2robot_2025              | 2025 | track-human-ref                              | RL       | PPO                                                                                                                                                                                                                                                                                                                                                           | Allegro Hand                                                                                                                                                                                                                                       | 16  | no  | IsaacGym                                                                                                                                                | yes  | 70     |            | not addressed | yes  |
| humanoid_policy_human_policy_2025 | 2025 | grasp,<br>functional/tool                    | BC       | Human Action<br>Transformer<br>(HAT), an ACT-<br>style action-<br>chunking<br>transformer co-                                                                                                                                                                                                                                                                 | Inspire Hand (x2,<br>5-fingered)                                                                                                                                                                                                                   | 6   | yes |                                                                                                                                                         | yes  | 230    |            | not addressed | yes  |

| method                            | yr   | task                                   | paradigm          | algorithm                                                                                                                                                                                                                                                        | hand                                                                                                                                  | DoF | bi  | sim              | real | trials | unseen obj | penetration   | code |
|-----------------------------------|------|----------------------------------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|-----|-----|------------------|------|--------|------------|---------------|------|
|                                   |      |                                        |                   | trained jointly on unified human (PH2D) and teleoperated-robot demonstrations in a shared 54-D wrist-pose + fingertip state-action space                                                                                                                         |                                                                                                                                       |     |     |                  |      |        |            |               |      |
| humanoid_sim2<br>real_recipe_2025 | 2025 | grasp, bimanual-coord, handover        | RL, distillation  | PPO (asymmetric actor-critic) for per-task specialist policies, distilled into a generalist via Diffusion Policy trained on filtered successful rollouts                                                                                                         | Fourier GR1 hand (6 actuated + 5 underactuated DoF); cross-embodiment check also uses Inspire hand (6 actuated + 6 underactuated DoF) | 6   | yes | NVIDIA Isaac Gym | yes  | 10     |            |               | no   |
| maniptrans_2025                   | 2025 | track-human-ref, bimanual-coord        | RL, BC, diffusion | PPO (two-stage: frozen generalist imitator + per-task residual policy)                                                                                                                                                                                           | Inspire Hand (12-DoF sim); also Shadow(22), MANO(22), Allegro(16)                                                                     | 12  | yes | Isaac Gym        | yes  |        |            | not addressed | yes  |
| metis_2025                        | 2025 | grasp, functional/tool, bimanual-coord | VLA               | Autoregressive next-token cross-entropy over discretized 'motion-aware dynamics' tokens (VQ-VAE visual dynamics + RQ-VAE motion dynamics), plus a supervised continuous-action regression head (Action Decoder); combined loss $L = L_{ar} + \lambda L_{action}$ | Inspire dexterous hand (paired, main real-robot eval); cross-embodiment eval uses 22-DoF SharpaWave Dexterous Hand                    | 6   | yes |                  | yes  | 20     |            | not addressed | no   |
| pi05_2025                         | 2025 | other                                  | VLA, flow         | hierarchical high-level discrete-subtask prediction (FAST tokenizer, autoregressive) + low-level flow-matching action expert; two-stage schedule pretrain ( $\alpha=0$ , 280k steps) then post-train                                                             |                                                                                                                                       |     | yes |                  | yes  | 40     |            | not addressed | yes  |

| method                   | yr   | task                  | paradigm         | algorithm                                                                                                                                                                                                                                                                                                    | hand                                                                                                                                      | DoF | bi  | sim                                           | real | trials | unseen obj | penetration   | code |
|--------------------------|------|-----------------------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-----------------------------------------------|------|--------|------------|---------------|------|
|                          |      |                       |                  | (alpha=10.0, 80k steps)                                                                                                                                                                                                                                                                                      |                                                                                                                                           |     |     |                                               |      |        |            |               |      |
| pistar06_2025            | 2025 | other                 | VLA, RL, flow    | RECAP: advantage-conditioned policy extraction on a flow-matching VLA, using a distributional value function (cross-entropy over 201 discretized return bins) to compute an N-step advantage and a binarized improvement indicator, with classifier-free-guidance-style advantage-conditioning dropout (30%) |                                                                                                                                           |     | yes |                                               | yes  | 750    |            | not addressed | yes  |
| wm_dex_human_videos_2025 | 2025 | grasp                 | world-model, MPC | DexWM: a deterministic latent-space (DINOv2) world model with a CDiT-based predictor conditioned on MANO fingertip-keypoint-difference actions, trained with a state-prediction loss plus an auxiliary hand-consistency heatmap loss; controlled via CEM/MPC planning in latent space                        | Allegro Hand                                                                                                                              |     | no  | RoboCasa                                      | yes  | 12     |            |               | no   |
| being_h05_2026           | 2026 | bimanual-coord, other | VLA, flow        | Rectified Flow (continuous action velocity-field prediction) + Masked Motion Token Prediction (discrete cross-entropy channel) inside a Mixture-of-Transformers / Mixture-of-Flow (MoF) action                                                                                                               | Inspire Hand (Franka FR3); LinkerBot O6 (Unitree G1); unnamed 6-DoF dexterous hands on PND Adam-U and BeingBeyond D1; parallel gripper on | 6   |     | LIBERO, RoboCasa (engine/timestep not stated) | yes  | 20     |            | not addressed | yes  |

| method          | yr   | task                                   | paradigm                                      | algorithm                                                                                                                                                                                                                                                                                                                                              | hand                                                                                                                        | DoF | bi  | sim                                                                                                                       | real | trials | unseen obj | penetration   | code |
|-----------------|------|----------------------------------------|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|-----|-----|---------------------------------------------------------------------------------------------------------------------------|------|--------|------------|---------------|------|
|                 |      |                                        |                                               | expert with shared 'Foundation Expert' layers and routed Top-K specialized experts                                                                                                                                                                                                                                                                     | LeRobot SO-101 (not dexterous)                                                                                              |     |     |                                                                                                                           |      |        |            |               |      |
| bidexgrasp_2026 | 2026 | grasp, bimanual-coord                  | grasp-synthesis, diffusion                    | bi-level QP-ADMM optimisation (region-pair init + decoupled per-hand force closure) + DDPM generator                                                                                                                                                                                                                                                   | Shadow Hand pair (sim); Inspire and BrainCo hands (real)                                                                    |     | yes | MuJoCo                                                                                                                    | yes  | 260    | 30         | measured      | no   |
| deximit_2026    | 2026 | bimanual-coord, grasp, functional/tool | data-collection, BC, grasp-synthesis, trajopt | 4D hand-object reconstruction (ST2+FPose) + LLM-based (Qwen3-VL) subtask decomposition with Action-Centric Scheduling + BODex-style force-closure grasp synthesis and keyframe motion planning, feeding a 3D Diffusion Policy (DP3)                                                                                                                    | XHands (real-world deployment); exact simulation hand not explicitly named                                                  |     | yes |                                                                                                                           | yes  |        |            | constrained   | no   |
| dexora_2026     | 2026 | grasp, functional/tool, bimanual-coord | VLA, diffusion, teleop-system                 | Decoder-only diffusion transformer (28 layers, hidden size 1024, 16 attention heads) with T5 (language) and SigLIP (vision) encoders feeding conditional tokens; DDPM training, DPMSolver++ sampling for inference; training clips reweighted by a discriminator trained with a positive-unlabeled objective (DWBC-style score-to-weight mapping) that | XHAND dexterous hand (paired), 12 fully actuated joints per hand, thumb and index additionally support lateral ab/adduction | 12  | yes | MuJoCo (digital twin of the real platform, used to generate the synthetic corpus and mirrored in real-time teleoperation) | yes  | 20     |            | not addressed |      |

| method                    | yr   | task                                   | paradigm                   | algorithm                                                                                                                                                                                                                              | hand                                                                               | DoF | bi  | sim                 | real | trials | unseen obj | penetration   | code |
|---------------------------|------|----------------------------------------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-----|-----|---------------------|------|--------|------------|---------------|------|
|                           |      |                                        |                            | scores demonstration quality                                                                                                                                                                                                           |                                                                                    |     |     |                     |      |        |            |               |      |
| dexteleop0_2026           | 2026 | grasp, functional/tool, bimanual-coord | teleop-system, MPC         | box-constrained QP shared-autonomy residual controller (force tracking + multi-contact force-torque balance + nominal stabilization terms) on top of DexPilot-style vector retargeting                                                 | Sharpa Wave (x2)                                                                   | 22  | yes | NVIDIA IsaacSim 4.5 | yes  | 35     |            |               | no   |
| egoscale_2026             | 2026 | grasp, functional/tool, bimanual-coord | flow, VLA                  | flow-based VLA (GR00T N1-style): VLM backbone + DiT action expert trained with a flow-matching objective, in a three-stage recipe (large-scale human-video pretraining, aligned human-robot mid-training, task-specific post-training) | Sharpa Wave (22-DoF); cross-embodiment target Unittree G1 (7-DoF tri-finger hand)  | 22  | yes |                     | yes  | 10     |            |               | no   |
| force_grasp_sim2real_2026 | 2026 | grasp, reorient                        | RL                         | PPO (asymmetric actor-critic)                                                                                                                                                                                                          | xHand                                                                              | 12  | no  | IsaacLab            | yes  | 70     | 2          |               | no   |
| poise_2026                | 2026 | reorient                               | RL                         | asymmetric PPO                                                                                                                                                                                                                         | Sharpa Wave hand                                                                   | 22  | no  | Isaac Lab           | yes  | 10     | 1          | not addressed | no   |
| simtoolreal_2026          | 2026 | functional/tool                        | RL                         | SAPG (PPO variant)                                                                                                                                                                                                                     | Sharpa five-fingered hand                                                          | 22  | no  | IsaacGym            | yes  | 120    | 12         | not addressed | no   |
| teledexter_2026           | 2026 | reorient, functional/tool              | RL, teleop-system          | single-stage RL (SAPG) with consecutive subgoal co-tracking reward                                                                                                                                                                     | SharpaWave (22-DoF, headline); also LeapHand (16-DoF)                              | 22  | no  | Isaac Gym           | yes  | 15     |            | penalised     | no   |
| toporetaret_2026          | 2026 | track-human-ref                        | trajopt, RL                | constrained Laplacian-optimization retargeting + PPO tracking                                                                                                                                                                          | Wuji Hand                                                                          |     | no  |                     | yes  |        |            | constrained   | no   |
| unidex_2026               | 2026 | grasp, functional/tool                 | flow, VLA, data-collection | UniDex-VLA: a pi0-style flow-matching VLA with a Uni3D pointcloud                                                                                                                                                                      | Inspire, Leap, Shadow, Allegro, Ability, Oymotion, XHand, Wuji (8 hands in UniDex- |     | no  |                     | yes  | 20     | 1          |               | no   |

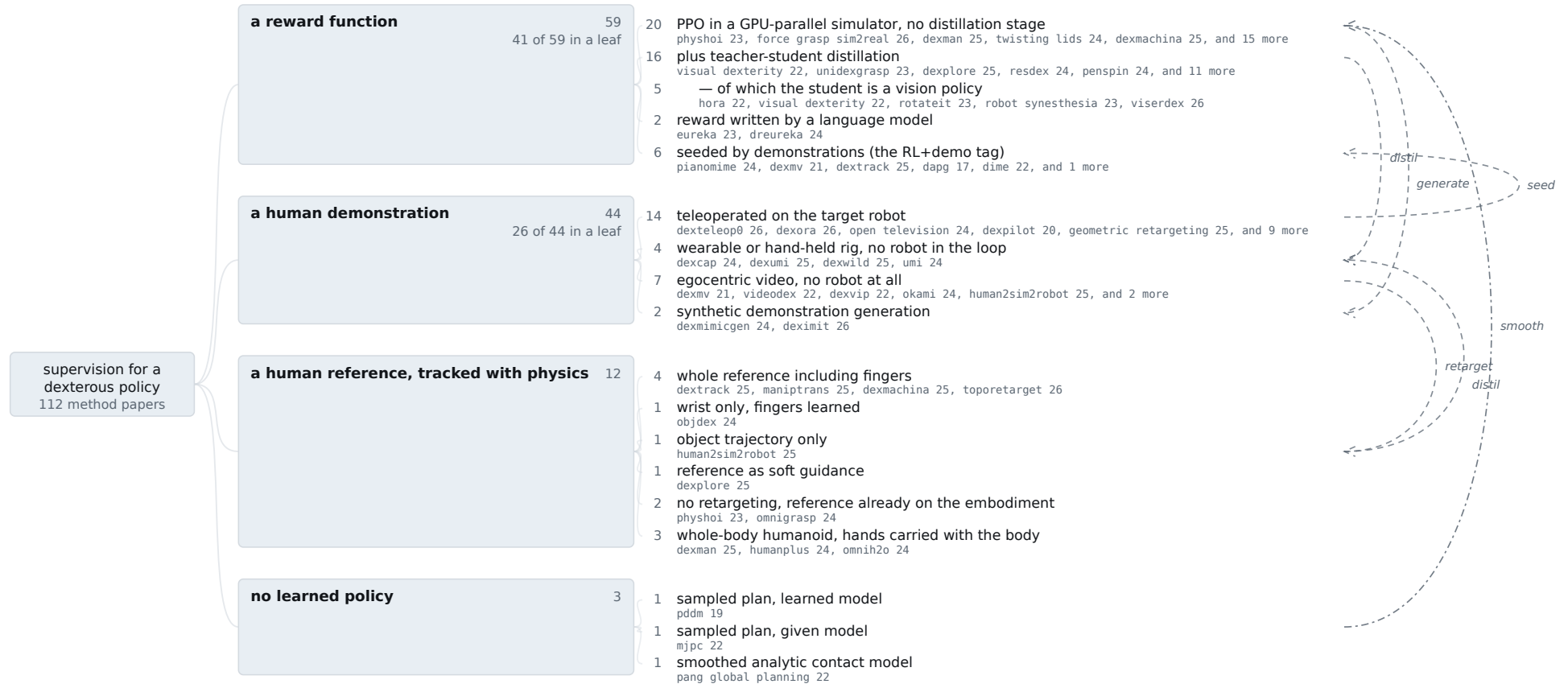
| method        | yr   | task     | paradigm         | algorithm                                                                                                                         | hand                                                             | DoF | bi | sim       | real | trials | unseen obj | penetration   | code |
|---------------|------|----------|------------------|-----------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|-----|----|-----------|------|--------|------------|---------------|------|
|               |      |          |                  | encoder and Gemma-based backbone, predicting actions in a unified Function-Actuator-Aligned Space (FAAS) across 8 dexterous hands | Dataset); real-robot eval uses Inspire, Wuji, and Oymotion hands |     |    |           |      |        |            |               |      |
| viserdex_2026 | 2026 | reorient | RL, distillation | PPO (RSL-RL)                                                                                                                      | Allegro Hand                                                     | 16  | no | Isaac Lab | yes  | 50     |            | not addressed | no   |

*112 rows; 224 of 1568 cells (14%) are values no source stated. No cell is truncated. A value wider than its column is wrapped at a word boundary, so a cell that runs to several rendered lines is one value and not several.*

## Figure 4. What supervises a dexterous policy

Four kinds of supervision, subdivided to the level at which the 112 method papers actually differ.

Every leaf count is the rows its stated predicate selects, recomputed from corpus/rows; a paper may appear more than once.



The branches are not exclusive, and the dashed curves on the right are the overlaps. Most work after 2024 trains with a reward in simulation and ships something shaped like an imitation policy, so it belongs to the first branch and the second at once. Eight further rows carry a trajectory-optimisation or MPC tag inside a learned pipeline and are counted on the branch that learns. The double-dashed smoothing edge is the one overlap that is a theorem rather than a pipeline: `pang\_global\_planning\_2022` proves that a policy gradient and an analytic log-barrier relaxation compute the same local model of contact.

## 5.2 Reinforcement learning

**The standard recipe.** Fifty-nine of the 112 papers learn from a reward. Forty-eight name PPO as the algorithm, and the next most frequent, DAPG, appears four times. Forty-three run their own experiments in a GPU-parallel simulator from the Isaac family or Genesis, and 36 do both. Only 31 rows state an environment count, with a median of 8192 and a maximum of 64000. Twenty-three rows distill a privileged teacher into a deployable student, and 16 combine all three of PPO, a GPU simulator and distillation. That 16-row intersection is the recipe as it is actually practised.

The hand is almost always an Allegro. It appears in 35 of them, and its hold on in-hand reorientation is tighter still: more than half of those rows use it, against roughly a fifth for the Shadow and an eighth for LEAP.

Privilege is handled two ways. `openai_dexterity_2018` gives the value network object and target orientation, joint angles, joint velocities and object velocities the policy never sees, with a footnote recording that current object orientation was left out of the policy inputs by accident, and `dextreme_2022` uses the same asymmetric critic with a 2048-unit LSTM against the actor's 1024. `hora_2022` instead compresses nine privileged object properties into an eight-dimensional vector regressed from 30 steps of proprioception, and `visual_dexterity_2022` distills twice, from a state teacher to a synthetic point cloud and then to a rendered one, for a fivefold speedup.

Domain randomisation is the part of the recipe with the least discipline. `openai_rubiks_cube_2019` made it adaptive, pushing each range boundary out when performance at that boundary exceeds 20 successes and pulling it in below 10, and `dextreme_2022` reproduced the mechanism with a 256-sample queue and 40 percent of environments dedicated to boundary evaluation. Both publish the discovered ranges. Below that standard, two shipped configurations disagree with their own papers about what was randomised: `hora_2022` states a joint-noise range of  $U(0, 0.005)$  against a shipped `jointNoiseScale` of 0.02, in a commit its own README says is not the one that reproduces the paper, and `penspin_2024`'s appendix states a disturbance force of 0.2 times the object's mass at probability 0.25 while the only shipped config that names one sets `forceScale` to zero. `dexpbt_2023`'s `randomize: False` is not a third case: it is the IsaacGymEnvs default and agrees with the paper's statement that randomisation was not used here.

**Reward engineering.** Table 5 puts the 21 in-hand reorientation methods against nine recurring term families and marks each cell by where the term was found: in the paper, in the code, in both, or in the code with every shipped configuration setting its weight to zero.

**Table 5. Reward terms across in-hand reorientation methods**

| method                       | yr   | goal or rotation tracking | object velocity | finger-object distance | hand-pose deviation | action rate or magnitude | torque, work or joint velocity | drop or failure | contact or force | success bonus | terms | code | paper/code mismatch |
|------------------------------|------|---------------------------|-----------------|------------------------|---------------------|--------------------------|--------------------------------|-----------------|------------------|---------------|-------|------|---------------------|
| openai_dexterity_2018        | 2018 | paper                     |                 |                        |                     |                          |                                | paper           |                  | paper         | 3     | no   |                     |
| openai_rubiks_cube_2019      | 2019 | paper                     |                 |                        |                     |                          |                                | paper           |                  | paper         | 3     | no   |                     |
| dextreme_2022                | 2022 | both                      |                 |                        |                     | both                     | both                           | code (0)        |                  | both          | 6     | yes  | yes                 |
| hora_2022                    | 2022 | both                      | both            |                        | both                |                          | both                           |                 |                  |               | 5     | yes  | yes                 |
| visual_dexterity_2022        | 2022 | both                      |                 | both                   | paper               |                          | both                           | both            | paper            | both          | 7     | yes  | yes                 |
| dexpbt_2023                  | 2023 | both                      |                 | both                   |                     | code                     | paper                          | code (0)        |                  | both          | 4     | yes  | yes                 |
| eureka_2023                  | 2023 | paper                     | paper           | paper                  |                     |                          |                                |                 |                  |               |       | yes  | yes                 |
| robot_synesthesia_2023       | 2023 | paper                     | paper           | paper                  |                     | paper                    | paper                          | paper           |                  |               | 6     | no   |                     |
| rotateit_2023                | 2023 | paper                     | paper           |                        | paper               |                          | paper                          |                 |                  |               | 6     |      |                     |
| rotating_without_seeing_2023 | 2023 | paper                     | paper           | paper                  |                     |                          | paper                          | paper           |                  |               | 6     |      |                     |
| anyrotate_2024               | 2024 | paper                     | paper           |                        | paper               |                          | paper                          | paper           | paper            | paper         | 10    | no   |                     |
| demostart_2024               | 2024 |                           |                 |                        |                     |                          |                                |                 |                  | paper         | 1     | no   |                     |
| dreureka_2024                | 2024 | paper                     | paper           |                        | paper               |                          |                                | paper           |                  |               | 4     | yes  | yes                 |
| penspin_2024                 | 2024 | both                      | both            |                        | both                | code (0)                 | both                           |                 |                  |               | 7     | yes  | yes                 |
| dexndm_2025                  | 2025 | paper                     | paper           |                        | paper               |                          | paper                          |                 |                  | paper         | 7     | no   |                     |
| dexremoe_2025                | 2025 | paper                     | paper           |                        |                     | paper                    |                                |                 |                  | paper         | 5     | no   |                     |
| dexteritygen_2025            | 2025 |                           |                 |                        |                     |                          |                                |                 |                  |               |       | no   |                     |
| force_grasp_sim2real_2026    | 2026 | paper                     |                 |                        | paper               | paper                    | paper                          | paper           | paper            | paper         | 7     | no   |                     |
| poise_2026                   | 2026 | paper                     |                 |                        |                     |                          | paper                          | paper           | paper            | paper         | 4     | no   |                     |
| teledexter_2026              | 2026 | paper                     |                 |                        |                     |                          |                                |                 |                  | paper         |       | no   |                     |
| viserdex_2026                | 2026 | paper                     | paper           |                        |                     | paper                    | paper                          | paper           |                  | paper         | 10    | no   |                     |

21 rows; 97 of 189 term cells (51%) are families the method does not use. `paper` means the term is in the paper and either no code was released or it is absent from the released reward code; `code` means it is in the released code and not in the paper's stated reward; `both` means it is in both. Marks are read from papers/notes/, term by term; the per-method source section is in corpus/reward\_matrix.json.

The families are not equally popular. Nineteen of 21 methods have a goal or rotation tracking term, which is the task. Thirteen penalize effort as torque, work or joint velocity, 13 pay a sparse success bonus, 11 penalize object velocity and 10 penalize a drop. Eight penalize deviation of the hand from a canonical grasp pose, a family the plan for this table did not anticipate and which had to be added. Six penalize action rate or magnitude, five reward closing the distance from fingertips to the object, and four carry a contact or force mark at all.

Three cells an earlier draft marked `code` print `code` (0) instead, because in each the term is in the released code with every shipped configuration setting its weight to zero: `dextreme_2022`'s timeout reward, `dexpbt_2023`'s fall penalty, and `penspin_2024`'s action penalty. Marking them `code` would tell a reader the code optimises something the paper does not state, and leaving them blank would hide a term that is in the file. The repository carries the config key and the zero beside each of the three marks.

Only three of the four contact marks are the finding. `anyrotate_2024` scores good and bad fingertip contacts, `poise_2026` rewards a friction-cone wrench margin, and `force_grasp_sim2real_2026` tracks a commanded grasp force. The fourth, `visual_dexterity_2022`, penalizes the *object* touching the table, a task-shaping term against using the table as a third finger rather than a hand-object term, and its table-contact flag defaults to off, with no shipped config setting it true. Three of 21 in-hand reorientation methods, then, put a hand-object contact or force quantity in the reward, and none puts interpenetration in it. `teledexter_2026` penalizes interpenetration with a differentiable signed-distance term, but during offline reference construction, not in the policy's reward. Across the corpus, 85 papers do not address penetration at all, 5 constrain it, 3 measure it, 3 penalize it and 16 say nothing either way. The physical quality of the contact is not something this literature optimises.

Term counts range from one to ten. `demostart_2024` gives a single binary terminal reward of 1 and lets a curriculum over demonstration states do the shaping. `anyrotate_2024` and `viserdex_2026` each publish ten weighted terms. `viserdex_2026` is the cleanest specification in the table, with every term, weight and equation in one appendix table, and a statement that the same weights are used for every object.

Weights are frequently unrecoverable. Three of the 21 rows give no numeric weight that survives PDF conversion, because the equation blocks are images, which is a limit of this survey and is recorded as such. `robot_synesthesia_2023` is a worse case. Its six coefficients are described only as "tuned hyper-parameters" and no number is printed anywhere in the paper, so the reward it reports cannot be reconstructed from the paper. `dexremoe_2025` names an angular-velocity penalty weight in prose with no entry in its hyperparameter table, and `dextrack_2025` names an affinity reward that is missing from its weight table.

**Curricula, populations and machine-written rewards.** Curricula in this literature relax physics or tighten tolerances. `dexpbt_2023` tightens a success tolerance from 0.075 to 0.01 by a factor of 0.9 every 3000 environment steps once three successes are logged, and `maniptrans_2025` starts at zero gravity and high friction and restores both while narrowing a fingertip threshold from 6 cm to 4 cm. `dexmachina_2025` attaches virtual spring-damper controllers to the object, lets them do the task at first, then decays their gains to zero. The two that ablate the curriculum report it is load-bearing: `rotateit_2023` sets its rotation-penalty weight to zero and ramps it to -0.1, because applying it from the start makes the policy learn only to hold the object still, and `viserdex_2026`, whose performance-based curriculum runs over regularisation weight, action latency and time between successes, fails completely without the regularisation component.

Population-based training appears once. `dexpbt_2023` trains populations of up to 32 agents, mutating the middle third of each population and replacing the bottom with mutated copies of the top, and reports 0.32 trillion environment steps for its largest run. The technique is routine outside robotics and nothing else in this corpus uses it.

A language model can write the reward outright, which removes the hand-tuning and replaces it with a search whose objective nobody has read. [eureka\\_2023](#) has GPT-4 write the reward function directly, constrained to return a total and a dictionary of named components, and feeds per-component statistics back between iterations. Its appendix gives an example whose reward correlates at  $-0.26$  with the human-written one and still scores 1.45 on the human-normalized metric, the strongest argument in the corpus that hand-tuned weights are not a ceiling. [dreureka\\_2024](#) moves the safety requirement into the prompt instead of a penalty term, asking in words for a cube rotating at about 0.25 radians per second with fingers penalized for leaving their initial pose. Neither can be checked against a file. Eureka's generated rewards are written at run time to a gitignored directory and never committed, so the reward behind any reported number cannot be recovered from the repository, though its README documents the paper's budget of five iterations and sixteen samples as the default and marks the shipped one-iteration block a fast-test preset. DrEureka's repository contains only the locomotion and globe-walking trees, so its LEAP-hand reward has no code at all.

**Where reinforcement learning stalls.** Reinforcement learning stalls on objectives that are not learnable as stated. [anyrotate\\_2024](#) found the angular-velocity objective unlearnable in the multi-axis setting and replaced it with a moving keypoint target. [rotateit\\_2023](#) reports that its multi-axis policy "does not converge when training with only reinforcement learning" and adds an imitation loss against single-axis oracles. [eureka\\_2023](#) cannot spin a pen from scratch and needs a pretrain-then-finetune split, with the from-scratch ablation failing to complete one cycle. Section 5.3's contact-graph term exists because of the general shape of the problem: contact moves the object off the reference, so return goes down, so the policy learns not to touch the object.

It stalls on geometry too. [hora\\_2022](#) fails on objects under 4 cm across because the fingers collide with each other. What it does not stall on is publication: the real-trial counts behind these headlines are small enough that section 7.1 treats them as the reporting problem they are.

### 5.3 Learning from human data

**Teleoperation and retargeting.** Table 6 lists the 30 corpus rows that describe a system for getting human motion onto a robot hand, with the retargeting objective compressed to one clause each.

**Table 6. Teleoperation and human-data systems**

| system         | operator interface                                                                                                                                                                      | robot hand                                                                                                                    | retargeting objective                                                                                                                                                                                                                                                                                                                 | latency                                                                  | rig USD | data collected |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|---------|----------------|
| dexpilot_2020  | four RealSense D415 RGB-D cameras tracking a bare hand, markerless; a coloured glove is worn only offline to train the neural hand-pose priors                                          | Wonik Robotics Allegro hand (16-DoF, 4x4-joint fingers), retrofitted with 4 SynTouch BioTac tactile sensors at the fingertips | minimise a weighted squared distance between human and Allegro fingertip task-space vectors, with switching weights that close primary-finger-to-thumb gaps and enforce a minimum finger separation, plus a regulariser toward the open hand, solved online with SLSQP                                                                | about one second, end to end (Sec. IV)                                   |         |                |
| dexmv_2021     | human demonstrator recorded on video (cubic capture rig, two RealSense D435 cameras), no live teleoperation; hand pose (MANO) and object pose extracted offline from the recorded video | Adroit Hand                                                                                                                   | match human (MANO forward-kinematics) and robot task-space vectors (palm-to-fingertip and palm-to-mid-phalanx pairs) per frame via SLSQP with a temporal-consistency term, then fit a minimum-jerk joint trajectory and compute torques via inverse dynamics                                                                          |                                                                          |         | 700 traj, 7 h  |
| dexvip_2022    | no live operator; a consensus hand pose is mined offline from curated HowTo100M video frames via FrankMocap 3D hand-pose estimation and k-medoid clustering                             | Adroit Hand                                                                                                                   | deterministic 4-stage geometric mapping (not an optimization) from FrankMocap's 21-joint hand pose to Adroit's 30-DoF joint space, applied once offline to produce a single static consensus target pose per object category, used only as a reward target                                                                            |                                                                          |         |                |
| dime_2022      | vision-based, single RGB camera, MediaPipe hand detector, no glove/headset/exoskeleton                                                                                                  | Allegro Hand                                                                                                                  | map human fingertip 2D image locations directly to fixed-height robot fingertip 3D targets (no depth estimation), then solve per-finger inverse kinematics for joint angles                                                                                                                                                           | not stated; targets at 30 Hz, PD loop at 300 Hz                          |         | 30 traj        |
| holo_dex_2022  | VR headset (Meta Quest 2), built-in 4-camera hand tracker, no glove/exoskeleton                                                                                                         | Allegro Hand                                                                                                                  | direct joint-angle copy for index/middle/ring fingers; thumb fingertip position matched via inverse kinematics; pinky ignored (no cost function or optimization)                                                                                                                                                                      | under 100 ms on a local network (Sec. IV-D)                              | 399     |                |
| videodex_2022  |                                                                                                                                                                                         | LEAP Hand                                                                                                                     | hand poses mapped via a distilled MLP replicating a Robotic-Telekinesis-style fingertip/palm keypoint-vector energy function; wrist trajectory computed via a PnP + monocular SLAM + heuristic gravity-alignment + workspace-rescaling pipeline, both applied offline to build pretraining trajectories, not run online at deployment |                                                                          |         | 965 traj       |
| anyteleop_2023 | vision-based, camera only (RGB or RGB-D, single or multi-camera), MediaPipe hand-keypoint detection, no glove/headset/exoskeleton                                                       | Allegro Hand                                                                                                                  | minimize the difference between human and robot fingertip keypoint vectors via forward kinematics, with joint-limit constraints and a temporal-smoothness penalty                                                                                                                                                                     | 26-35 ms hand pose, 9-10 ms retargeting (Table II); no end-to-end figure |         |                |
| pgdm_2023      |                                                                                                                                                                                         | ShadowHand                                                                                                                    | for pre-grasps only: inverse kinematics fits the robot hand's fingertip positions to a single human hand pose frame near first contact, used solely to initialize RL exploration; full object trajectories                                                                                                                            |                                                                          |         | 40 traj        |

| system               | operator interface                                                                                                                                                                                        | robot hand                                                                         | retargeting objective                                                                                                                                                                                                                                                                                                                | latency                                                                | rig USD | data collected  |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|---------|-----------------|
|                      |                                                                                                                                                                                                           |                                                                                    | are used directly as the RL goal with no hand-trajectory retargeting                                                                                                                                                                                                                                                                 |                                                                        |         |                 |
| ace_teleop_2024      | visual exoskeleton: 3D-printed bimanual 6-DoF-per-arm exoskeleton with wrist-mounted cameras for MediaPipe finger tracking; not a glove or VR headset                                                     | Ability Hand / Inspire Hand / parallel-jaw gripper (embodiment-dependent)          | wrist/end-effector pose mapped via a scale-and-recenter IK transform (Normal/Mirror/Bimanual modes) from exoskeleton forward kinematics; finger joints retargeted via AnyTeleop's fingertip-keypoint-vector optimization with joint limits and temporal smoothing                                                                    | 27 Hz hand tracking; no end-to-end figure                              | 600     |                 |
| bidex_teleop_2024    | Manus Meta motion-capture gloves for the fingers plus GELLO-style teacher arms for wrist and arm pose; no headset                                                                                         | LEAP Hand (16 DoF); LEAP Hand V2 (21 DoF) used for 'extreme dexterity' experiments | inverse kinematics from glove fingertip positions to LEAP fingertip targets, with the arm commanded directly in joint space from the teacher arm rather than in end-effector space                                                                                                                                                   | claimed low-latency, no number given                                   | 6000    | 50 traj         |
| bunny_visionpro_2024 | VR headset (Apple Vision Pro) hand/wrist tracking, plus a custom ERM haptic feedback device driven by fingertip FSR tactile sensors; no glove or exoskeleton                                              | Ability Hand                                                                       | minimize scaled human-robot fingertip keypoint-vector distance via forward kinematics with joint limits and temporal smoothness (hand), solved online via SQP with a reduced-dimension reformulation for loop joints; separately, a unified IK + singularity-avoidance + sphere-approximated self-collision objective for arm motion | 3.43 ms retargeting, 15.93 ms motion control, >60 Hz overall (Table 1) |         |                 |
| cyberdemo_2024       | vision-based teleoperation (single RealSense camera observing operator hand motion, real-time hand detection and retargeting via a cited third-party system), not a glove or headset                      | Allegro Hand                                                                       | delegated entirely to the cited third-party teleoperation system; CyberDemo does not describe or modify the human-to-robot mapping itself                                                                                                                                                                                            |                                                                        |         |                 |
| dexcap_2024          | wearable backpack rig: Rokoko EMF gloves, one SLAM camera per wrist and a chest-mounted RGB-D LiDAR; no robot in the loop during collection                                                               | LEAP Hand                                                                          | fingertip-position inverse kinematics into the 16-dim LEAP joint space anchored by the mocap wrist pose, with the little finger discarded                                                                                                                                                                                            |                                                                        | 4000    | 787 traj, 4.5 h |
| egomimic_2024        | human data: head-worn Project Aria glasses (egocentric RGB + onboard SLAM/hand-tracking, no glove or exoskeleton); robot demonstrations: standard bimanual ViperX teleoperation                           | parallel-jaw gripper (not a dexterous hand; see note scope flag)                   | no per-finger kinematic retargeting (end effector is a 2-finger gripper); domain alignment via a shared per-timestep camera-centered reference frame, per-embodiment Gaussian normalization of proprioception/actions, and SAM2-based visual masking with a directional line overlay applied identically to human and robot streams  |                                                                        | 1000    | 2,150 traj, 4 h |
| hudor_2024           | VR headset (Meta Quest 3) built-in hand tracking for fingertip 3D positions, ArUco-marker rigid-body calibration into the robot frame; demonstrator is in the same scene as the robot, not internet video | Allegro Hand                                                                       | direct Cartesian-space correspondence: human fingertip 3D positions are treated as target Cartesian positions for the corresponding robot fingertips with no kinematic-chain optimization, converted to joint commands via a custom Jacobian-based gradient-descent IK                                                               |                                                                        |         | 4 traj          |

| system               | operator interface                                                                                                                                                                                                                                                                                                                | robot hand                                                                                                        | retargeting objective                                                                                                                                                                                                                                                                                                                                                                                     | latency                                                        | rig USD | data collected     |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|---------|--------------------|
| okami_2024           | single human demonstrator recorded on a static RGB-D camera (Intel RealSense D435i), no live teleoperation and no glove/headset; single video per task                                                                                                                                                                            | Inspire Hand (x2)                                                                                                 | factorized retargeting run once per demonstration video: arm/shoulder-elbow-wrist IK (via the Pink library, wrist position dominating the objective) plus hand joint angles computed via dex-retargeting from SMPL-H hand keypoints; the resulting trajectory is affinely SE(3)-warped at test time to match newly localized object start/end poses, with fingers retargeted independently of arm warping |                                                                |         |                    |
| open_television_2024 | VR headset (Apple Vision Pro or Meta Quest 3) with active stereo video streamed back from a 2-3 DoF camera gimbal on the robot's neck                                                                                                                                                                                             | Inspire Robots hand (6 actuated DoF, 12 total); Fourier GR-1 embodiment instead uses a 1-DoF parallel-jaw gripper | dex-retargeting keypoint-vector optimisation over five wrist-to-fingertip and two thumb-to-finger vectors, solved with SLSQP, with end-effector position taken relative to the head and orientation absolute                                                                                                                                                                                              | 60 Hz stereo round trip and 60 Hz control                      |         |                    |
| dexteritygen_2025    | human teleoperator tracked via a Manus Glove (retargeted to the Allegro hand at 300Hz via an unreleased, confidential fast retargeting method) plus a Vive tracker for 6-DoF wrist pose driving the arm                                                                                                                           | Allegro Hand                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                |         |                    |
| dexumi_2025          | hand-specific 3D-printed wearable exoskeleton (per-robot-hand optimized joint-to-fingertip kinematics), joint encoders reading exoskeleton angles directly (no visual fingertip retargeting), a wrist-mounted camera (ARKit-tracked wrist pose), and matching tactile sensors installed on both exoskeleton and target robot hand | Inspire Hand (12 DoF, 6 active) and XHand (12 active DoF)                                                         | no visual/kinematic-chain retargeting at inference; a learned per-joint regression model maps exoskeleton encoder values directly to robot motor values (calibrated by visually overlaying exoskeleton and robot at matched motor values), with the exoskeleton's mechanical linkage itself optimized offline so its fingertip workspace matches the target robot hand's forward-kinematics workspace     | per-sensor latency measured and compensated offline, no figure |         | 1,355 traj         |
| dexwild_2025         | wearable, calibration-free rig: motion-capture glove + two palm-mounted stereo cameras (pinky-side and thumb-side) + wrist ArUco marker tracking, worn by untrained operators; no VR headset                                                                                                                                      | LEAP Hand / LEAP Hand V2 Advanced                                                                                 | robot hand kinematics optimized to match observed human fingertip positions (fixed hyperparameters, no per-user tuning); a relative (delta) end-effector state-action representation removes the need for a global coordinate frame, and identical palm-camera mounting on human and robot hands makes RGB observations look the same across embodiments                                                  |                                                                |         | 9,290 traj, 46.2 h |
| doglove_2025         | haptic force-feedback exoskeleton glove (21-DoF encoder-based motion capture, 5-DoF cable-driven force feedback, per-fingertip LRA haptic feedback), plus an HTC Vive Tracker for wrist localization; not vision-based                                                                                                            | LEAP Hand                                                                                                         | map glove-FK fingertip positions to robot fingertip targets via the third-party Mink differential-1K solver, with a single scalar hand-size scaling factor; no paper-original cost function                                                                                                                                                                                                               | 30 Hz minimum system rate (120 Hz mocap, 30 Hz haptics)        | 600     |                    |
| egozero_2025         | Project Aria smart glasses (egocentric fisheye RGB + SLAM                                                                                                                                                                                                                                                                         | Franka Panda parallel-jaw gripper (not a dexterous hand; see note                                                 | no kinematic hand-to-robot retargeting; both human and robot                                                                                                                                                                                                                                                                                                                                              |                                                                |         | 700 traj           |

| system                            | operator interface                                                                                                                                                                                                                                                       | robot hand                                                                                                                                                                                             | retargeting objective                                                                                                                                                                                                                                                                                                                                                                | latency                                                                   | rig USD | data collected      |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|---------|---------------------|
|                                   | cameras + onboard hand/camera pose tracking), no glove, no live teleoperation of a robot during data collection; at inference an iPhone provides the egocentric view instead of Aria                                                                                     | scope flag)                                                                                                                                                                                            | are represented in a shared morphology-agnostic 3D-point space so the policy transfers zero-shot without any cross-embodiment pose-mapping step                                                                                                                                                                                                                                      |                                                                           |         |                     |
| geometric_retargeting_2025        | Manus glove for fingertip keypoints + HTC Vive tracker for wrist pose (real-world evaluation); the retargeting method itself is tracking-source-agnostic                                                                                                                 | Allegro Hand                                                                                                                                                                                           | learned per-finger neural network trained offline (3-5 minutes) to jointly satisfy motion-direction preservation, C-space coverage, flatness, pinch correspondence, and self-collision avoidance, replacing per-frame online optimization with a single 1kHz forward pass at inference                                                                                               | 1 kHz retargeting inference, against 60-100 Hz for optimisation baselines |         |                     |
| h_rdt_2025                        | for pretraining: none (consumes EgoDex's pre-existing released 3D hand-pose annotations, no live capture by this paper); for fine-tuning: standard teleoperation of each robot platform, not described in detail                                                         | parallel-jaw / 2-jaw grippers at deployment (Aloha-Agilex, ARX5, Franka-Panda, UR5+UMI); no dexterous hand deployed, though pretraining action space includes bimanual fingertip positions from EgoDex | no explicit kinematic retargeting; the 48-D bimanual wrist-fingertip human action space is treated as a superset of most end-effector action spaces, and the human-to-robot correspondence is instead learned implicitly by reinitializing and fine-tuning the state/action adaptors and action decoder on paired robot data while keeping the pretrained vision/language backbone   |                                                                           |         | 338,000 traj, 829 h |
| humanoid_policy_human_policy_2025 | consumer VR headsets (Apple Vision Pro built-in camera with ARKit hand/head tracking, or Meta Quest 3/Vision Pro with a 3D-printed ZED Mini stereo mount via an OpenTelevision web app), <\$700; robot-side demonstrations collected via standard humanoid teleoperation | Inspire Hand (x2, 5-fingered)                                                                                                                                                                          | no offline kinematic retargeting network; human and robot share an identical 54-D state-action space via a bijective fingertip-to-fingertip correspondence (both are 5-fingered hands), bridged only by action-speed interpolation (human trajectories slowed ~4x) and instructing operators to minimize torso movement                                                              |                                                                           | 700     |                     |
| wm_dex_human_videos_2025          |                                                                                                                                                                                                                                                                          | Allegro Hand                                                                                                                                                                                           | no explicit robot retargeting network; both human hands (MANO 21-keypoint) and robot end-effectors (Allegro fingertips via forward kinematics; parallel-jaw grippers approximated with dummy circular keypoints) are mapped into the same fingertip-keypoint-difference action representation, with the missing pinky filled by reusing the ring-finger-equivalent Allegro keypoints |                                                                           |         | 829 h               |
| dexteleop0_2026                   | VR headset (Meta Quest 3), egocentric hand/wrist tracking (26 tracked hand-joint transforms per hand), no external motion capture                                                                                                                                        | Sharpa Wave (x2)                                                                                                                                                                                       | DexPilot-style vector matching of critical inter-joint/finger-to-palm target vectors for the hand, plus analytical IK on wrist pose for the arm, producing a raw joint command that a real-time QP then corrects to keep fingertip contact forces in a safe window and balance net object force/torque                                                                               | 30 Hz QP control cycle                                                    |         |                     |
| egoscale_2026                     | large-scale pretraining: in-the-wild egocentric video with off-the-shelf                                                                                                                                                                                                 | Sharpa Wave (22-DoF); cross-embodiment target Unitree G1 (7-                                                                                                                                           | human 21-keypoint (MANO-style) hand pose retargeted via an                                                                                                                                                                                                                                                                                                                           |                                                                           |         | 20854 h             |

| system          | operator interface                                                                                                                                                                                                                     | robot hand                                                                                                                                         | retargeting objective                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | latency | rig USD | data collected |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|---------|----------------|
|                 | SLAM + hand-pose estimation, no dedicated capture rig; aligned mid-training: Vive trackers for 3D wrist pose + Manus gloves for 25-joint in-hand pose, same camera rig as the robot                                                    | DoF tri-finger hand)                                                                                                                               | optimization-based procedure enforcing joint limits and kinematic constraints into the 22-DoF Sharpa Wave hand joint space; wrist motion is represented as a shared relative SE(3) transform identical across human and robot data                                                                                                                                                                                                                                                                               |         |         |                |
| teledexter_2026 | NOKOV motion-capture system tracking operator hand pose and object 6D pose in real time (not vision-based, not a glove)                                                                                                                | SharpaWave (22-DoF, headline); also LeapHand (16-DoF)                                                                                              | two-stage geometry-aware retargeting: Stage 1 fingertip-vector alignment to human hand geometry; Stage 2 object-mesh-aware refinement combining a surface-contact term, a differentiable-SDF interpenetration penalty, a self-collision sphere term, and a temporal-smoothness term, producing the fingertip+object-pose subgoals the RL controller tracks                                                                                                                                                       |         |         |                |
| unidex_2026     | human-video side: no live capture rig, converts existing egocentric RGB-D datasets (H2O, HOI4D, HOT3D, TACO) via offline retargeting; fine-tuning demonstrations collected via OpenTeleVision + dex-retargeting on an Apple Vision Pro | Inspire, Leap, Shadow, Allegro, Ability, Oymotion, XHand, Wuji (8 hands in UniDex-Dataset); real-robot eval uses Inspire, Wuji, and Oymotion hands | two-stage human-in-the-loop kinematic retargeting: an automatic PyBullet multi-end-effector IK stage aligns a 6-DoF offset between a virtual human-hand-fixed base and the real robot base to match fingertip positions (respecting joint limits, with iterative mimic-joint correction), followed by an interactive GUI stage where a human manually corrects alignment; a separate visual-alignment step masks the human hand from the pointcloud and re-renders the retargeted robot-hand mesh into the scene |         |         | 52,000 traj    |

30 rows; 60 of 210 cells (28%) are values no source stated. No cell is truncated. A value wider than its column is wrapped at a word boundary, so a cell that runs to several rendered lines is one value and not several.

Retargeting splits four ways. The dominant family matches task-space vectors between the human and robot hands. `dexpilot_2020` set the pattern with a weighted squared distance between fingertip vectors, switching weights of 1, 200 and 400 by whether a pair is within threshold, a minimum-separation constant of 3 cm between primary fingers, and a regulariser toward the open hand, solved online with SLSQP; `anyteleop_2023`, `open_television_2024`, `bunny_visionpro_2024` and `ace_teleop_2024` all use the same formulation. The second copies joint angles where the kinematics allow it, as `holo_dex_2022` does for index, middle and ring while solving the thumb by inverse kinematics and discarding the pinky the Allegro does not have. The third learns the map offline: `geometric_retargeting_2025` trains a per-finger network in three to five minutes against motion-direction, coverage, flatness, pinch and collision criteria, then runs at 1 kHz against the 60 to 100 Hz of online optimization, raising configuration-space coverage from 38 to 90 percent and one-time grasp success from 55 to 87.5 percent. The fourth removes retargeting from the loop: `dexumi_2025` builds an exoskeleton whose fingertip workspace is optimized to match the target hand, then regresses encoder values straight to motor values.

The table's empty cells are the point. Only 12 of the 30 papers carry a latency cell at all, three of those give no number, and only 7 state a rig cost. Where costs are stated they are low and falling, from `holo_dex_2022`'s \$399 headset through the \$600 exoskeleton of `ace_teleop_2024` and the \$600 haptic glove of `doglove_2025` to the \$4,000 backpack of `dexcap_2024` and the \$6,000 rig of `bidex_teleop_2024`. Latency is not one quantity: `dexpilot_2020`'s "about one second" and `holo_dex_2022`'s "under 100 milliseconds" are end to end, while `geometric_retargeting_2025`'s 1 kHz and `anyteleop_2023`'s 9 to 10 ms retargeting are single stages and several cells give a control rate. The column mixes the two, so its spread is not a range. Only 13 rows report how many trajectories were collected, and only 7 report hours.

Where throughput is measured, human collection beats teleoperation by a wide margin. `humanoid_policy_human_policy_2025` times a grasp demonstration at 3.79 seconds for a bare human and 19.72 seconds through a teleoperated humanoid, and a pour at 4.81 against 37.31 seconds. It attributes the gap to retargeting latency and to the workspace of a 7-DoF arm rather than to anything about the hand.

**Imitation architectures.** Four architectures cover the corpus. Action chunking came from `aloha_act_2023`, which predicts a chunk of joint targets with a CVAE and combines overlapping chunks by exponentially weighted ensembling, reaching 80 to 90 percent on fine bimanual tasks from about 50 demonstrations each. Diffusion came from `diffusion_policy_2023`, which denoises an action sequence and reports a 46.9 percent average success improvement over LSTM-GMM, IBC and BET across 15 tasks. `dp3_2024` swaps the image encoder for a small point-cloud encoder and reports 74.4 percent against 59.8 percent across 72 simulated tasks, and 85.0 against 35.0 percent on four real tasks from 40 demonstrations each. Flow matching is the 2024-onward default for large models and carries eight papers. Discrete action tokens are the fourth, used by `openvla_2024` and `metis_2025`, and are the only one of the four that needs no continuous head at all.

Data generation is a separate lever and is underrated. `dexmimicgen_2024` turns 60 human source demonstrations into 21,000 simulated ones across 9 tasks and 3 embodiments, and a real Fourier GR1 with two Inspire hands reaches 90 percent from 40 generated demonstrations against 0 percent from the 4 source ones. `dex1b_2025` iterates optimization, a CVAE proposal model and a simulator filter into roughly a billion synthetic grasps, and its CVAE baseline beats the prior best by 22 points.

**Human video without a robot.** Seven rows train a dexterous-hand policy from video with no teleoperation at any stage. `dexmv_2021` retargets 700 self-recorded demonstrations by matching palm-to-fingertip task-space vectors and estimating actions by inverse dynamics, `videodex_2022` mines 965 trajectories from EpicKitchens to pretrain a neural dynamic policy, and `dexvip_2022` goes further into the reward, clustering 715 curated HowTo100M frames into one consensus grasp pose per object category and adding it as a reward term. `okami_2024`, `human2sim2robot_2025` and `hudor_2024` work from a single video each, and `wm_dex_human_videos_2025` pretrains a world model on 829 hours of EgoDex.

**The embodiment gap.** Three papers measure the gap directly rather than asserting it. `okami_2024` runs the same reference-plan and warping pipeline with and without human body and hand retargeting, and the version without it loses 75.0, 42.1, 82.0 and 74.0 points across four task and setting pairs. That is the largest clean embodiment-gap number in the corpus.

`humanoid_policy_human_policy_2025` ablates two mitigations separately on the same vertical-grasp task over 10 trials. The full method scores 4 out of 10. Removing the interpolation that slows human trajectories by a factor of four drops it to 1 out of 10. Removing the unified 54-dimensional state space while keeping the slowdown drops it to 0 out of 10, because the policy is handed a shortcut for telling the two embodiments apart. `egozero_2025` reports the sharpest version of the same effect. Removing 3D augmentation drops every one of its seven tasks to 0 out of 15, and substituting monocular depth for triangulated depth does the same, because the best metric depth models it tested carry more than 5 cm of error. It also quantifies what its hand-pose pipeline costs, at 1 to 2 cm of action-label error.

More human correspondence is not reliably better, and one paper here tested it directly. `objdex_2024` supplies the counterexample. It retargets only the wrist and lets reinforcement learning discover finger motion, and its ablation that adds a fingertip-matching reward "does not yield benefits and even leads to lower performance". Richer human correspondence is not uniformly better, and this is the one result in the corpus that says so with an ablation.

**Tracking a human reference with physics.** Twelve papers sit in the human-reference tracking family. Eight track a human hand on an object and are covered here; the other four sit at the edges, `human2sim2robot_2025` tracking only the object's trajectory, `dexman_2025` retargeting bimanual video onto a full humanoid, and `humanplus_2024` and `omnih2o_2024` tracking whole-body motion. A reference is retargeted onto the robot and a policy trained to make the simulated body follow it: the reference supplies the shaping reward engineering would otherwise have to invent, and the simulator the physical consistency pure imitation lacks.

The multiplicative reward that most human-reference methods now use has a single source. `physhoi_2023` is the origin of the form. It multiplies a body term, an object term, an interaction-graph term and a contact-graph term, and reaches 95.4 percent success on GRAB against 27.0 percent for a DeepMimic baseline. The contact-graph term exists to stop the policy learning not to touch the object. `omnigrasp_2024` replaces the action space with a pretrained latent motion prior and reports 94.6 percent grasp success and 84.8 percent trajectory success on GRAB, while stating outright that it omits penetration metrics.

The next step was to stop treating demonstrations as fixed and let the method generate its own. `dextrack_2025` adds an imitation loss to PPO and mines its own demonstrations through a homotopy search over easier neighbouring trajectories, reaching 46.70 and 65.48 percent on GRAB at loose and strict thresholds against 38.58 and 54.82 for the best baseline. It has a penetration-depth formula and applies it only to input references, never to its own rollouts, and presents tolerance of "severe hand-object penetrations" as robustness. `maniptrans_2025` freezes a generalist hand imitator and trains a per-task residual on top, reaching 58.1 percent single-hand and 39.5 percent bimanual success on OakInk-V2. Its stance on contact is to raise the friction coefficient above the real value rather than model skin deformation, and its real deployment is open-loop replay.

Two papers in this corpus ran the same curriculum and disagreed about whether it helps. `dexmachina_2025` moves to Genesis with 12,000 environments and six hand URDFs, and drives the object with virtual controllers that decay to zero. Its re-implementation of `maniptrans_2025`'s curriculum does not beat no curriculum on its own setup, which is a direct disagreement between two papers worth quoting in both directions. `dexplore_2025` drops the explicit retargeting stage and treats the reference as soft guidance, with termination thresholds derived from the failure rate, reaching 87.7 percent on GRAB with an Inspire hand.

`toporetaret_2026` is the only corpus method that quantifies penetration as an evaluation quantity, reporting maximum depth and the fraction of frames past 2 mm, and constraining it during retargeting with a 1 mm soft tolerance and a 30 mm hard bound. It reaches 87.5 percent on its own pen-spinning set against 46.9 for the best baseline. It does not re-measure penetration after the tracking policy runs, so the property it constrains is a property of the reference and not of the behavior.

That is the pattern across all eight, and it is the reference-versus-rollout split in its sharpest form. Penetration is handled at the reference, if at all, and never at the rollout. `objdex_2024` completes the set with the only real-robot numbers among them, from 100 percent on a microwave and a laptop down to 41.2 percent on a ketchup bottle over 20 trials each.

**The retargeting map, and what would close it.** Human data does not port across hands, and the map is usually left unstated. Fifty-three papers use human data and name 44 distinct hand strings between them; twenty of those appear in Table 6 with a

stated retargeting objective. The other 33 never say how the human motion reached the hand, and the objectives that are stated do not converge, running from a fingertip keypoint-vector energy in `anyteleop_2023` through a per-joint regression from exoskeleton encoders in `dexumi_2025` to no retargeting at all in `egozero_2025`. What would close it is a retargeting benchmark in TopoRetarget's form `toporetaret_2026`, reporting contact precision, contact alignment, maximum penetration and the share of frames past 2 mm, per hand and per dataset.

## 5.4 Generalist and vision-language-action policies

The finding is the size of the hand. Eighteen rows carry the generalist tag. Fourteen of them settle whether the reported evaluation ran on a multi-fingered hand. Eight of those fourteen did, six report no hand result at all, and the remaining four never say, `groot_n16_2025` naming no end effector anywhere on its page.

Five of the eight state a hand size, and that comparison has to be made in actuated degrees of freedom rather than joints, for the reason section 3 opens with. Four are six, the Inspire RH56DFX among them actuating six of its twelve joints, and one is twelve, `dexora_2026`'s XHAND. Their median is 6. Two of the four rows that never settle the hand question do state a size, and both are large: 21 for `gr_dexter_2025`'s ByteDexter V2 and 22 for `egoscale_2026`'s Sharpa Wave. Of the 59 reward-learning rows, 48 state a count, 38 of those are 16 or above, and their median is 16: an Allegro and a LEAP actuate 16, and a Shadow actuates 20 of its 24 joints. So no row that settles the question reaches the band the reinforcement-learning literature of section 5.2 works in, and the two rows that reach it on paper are the two that never say whether the hand was in the evaluation. An earlier version of this claim counted eleven rows as evaluating on a hand, which mixed the settled eight with rows the notes leave open.

The six with no hand result are parallel-jaw throughout (`pi0_2024`, whose released code encodes each ALOHA gripper as one scalar, `pi05_2025`, `pistar06_2025`, `openvla_2024`, `helix_2025`, which claims individual-finger control of a 35-DoF upper body with no success rate or trial count for any task, and `gemini_robotics_2025`, whose Apollo hand appears in a figure caption with no number attached). Its successor `gemini_robotics_15_2025` closed that gap with Apollo progress scores, not success rates, of 0.74 down to 0.62 across five generalisation axes. `dexvla_2025` is the borderline case the other way: a hand in one real-robot task family, grippers in the other five and in all 91 pretraining tasks.

The released code also drifts. `groot_n1_2025`'s repository is a later N1.7 generation with a different backbone, a different action horizon and embodiments the paper does not describe, and `pi05_2025`'s supports only the flow-matching head, not the hybrid recipe the paper describes.

**The size of the gap, and what would close it.** The gap is size rather than absence. A generalist policy that does touch a hand runs it at roughly a third of the actuation the reinforcement-learning literature assumes, and the rows that never settle the question are the smaller half of the same gap. What would close it is one dexterous-hand task inside the standard generalist suite, with the hand's actuated degrees of freedom and its vendor printed beside the number.

## 5.5 Model-based baselines and hybrids

Three corpus methods solve dexterous tasks without learning a policy, and they are the control group the rest of this section lacks.

Not every method here learns a policy. A smaller line plans through a model instead, and it reaches dexterous tasks on a fraction of the data. `pddm_2019` learns an ensemble of dynamics models and plans through it with an MPPI-style optimiser, replanning every step. It needs 1 to 2 hours of data in simulation and 2 to 4 hours on a real 24-DoF Shadow Hand. `mjpc_2022` removes the learned model too and samples ten rollouts per step through MuJoCo itself, planning in 1 to 20 milliseconds. It reorients a cube with a Shadow Hand in real time from scratch, and reports no success criterion, no trial count and no real-robot result.

One of the three does more than plan: it explains why the learned and analytic routes arrive at the same place. `pang_global_planning_2022` is the most substantial. It proves that the randomised smoothing implicit in reinforcement learning and an analytic log-barrier relaxation compute the same local linear model of contact, then uses the analytic version inside a trajectory optimiser and an RRT. Allegro in-hand rotation takes 19.59 seconds to optimize and its plate-pickup task

117.16 seconds of planning on a 16-core CPU, against the GPU-days of section 5.2, and it is the only method here that imposes non-penetration as a hard constraint. Its own limitation is the honest part: its 3D systems transfer to hardware far worse than its 2D ones, because the quasi-dynamic assumption breaks and planned grasps miss contacts under a second-order solver.

**Hybrids, and where the field has converged.** The recurring shape is reinforcement learning in simulation distilled into a policy that looks like an imitation policy, and it appears in four variants. The first distills a privileged teacher into a vision student inside one paper, which is `hora_2022`, `visual_dexterity_2022`, `rotateit_2023`, `robot_synesthesia_2023` and `viserdex_2026`. The second uses reinforcement learning as a demonstration factory. `dextrack_2025` mines demonstrations with per-trajectory RL and trains a generalist on them, and `maniptrans_2025` does the same to build a 3.3K-episode dataset. `dexteritygen_2025` goes further, training a diffusion controller on ten billion simulated grasp-to-grasp transitions and projecting a human teleoperator's coarse command onto it, taking four reorientation tasks from 0 out of 20 under raw teleoperation to 12, 13, 10 and 9 out of 20. The third generates data without reinforcement learning at all, which is `dexmimicgen_2024`, `dex1b_2025` and `deximit_2026`; `dex1b_2025`'s row is classed as a dataset, so figure 4's leaf, which counts papers, holds the other two. The fourth wraps an RL policy in a residual, which is `resdex_2024` at 88.8 percent over 3,200 objects in 12 GPU-hours.

In the variants that dominate 2025 and 2026 work, the reward is no longer the objective of the deployed policy. It is the objective of the process that produced the deployed policy's training data. Reward engineering has moved upstream into data curation, and every reward-shaping pathology in section 5.2 now reaches the shipped policy through a dataset rather than a gradient.

## 5.6 Paper against released code

Thirty-eight of the 112 papers record a disagreement between a paper and the code it released, and all 38 released code, so they sit inside the 62 rows that released anything. They are not one kind of thing. Eight are contradictions, where paper and code state different values or different terms. Fourteen are limits of this survey's own parse, where the body or config that would settle the question was never recovered and the row says so. Eight released code without the described component in it, four are version skew against a later repository, and four are a paper disagreeing with itself. The fourth version skew is `groot_n16_2025`, which ships a main branch one generation later than the checkpoint its page describes.

Eight is the number to quote, seven at high confidence and one, `penspin_2024`, held at medium. Eight of 62 is 13 percent, and it is a floor rather than a rate: the census covers method rows only, so `robopianist_2023`, whose row is a benchmark, does not enter it although it sums five reward terms against the three its Table 2 documents. It is also a screened count rather than a first pass. Sixteen rows were drawn at first and eight have since left the class. Six went under adversarial re-reading. One, `omnih2o_2024`, went when writing to its authors sent someone back to the evidence. The eighth went because its corresponding author replied: the term this survey called a departure from the paper's formula was the paper's formula, and the line defining it had never been in the parse. Two more rows were narrowed and stayed. Each revision is recorded in the row it concerns, beside the comparison it revises. That screening is evidence that the eight were checked, not a bound on how many a fuller reading would find. Among the 21 reorientation methods of Table 5, seven released a repository and four of those state something different from their paper, the other three being a missing environment, a later generation and a default-value question their own READMEs settle.

Table 11 is the whole of the finding, in the form the finding is made: a public repository, the commit `tools/fetch_code.py` cloned, the file inside it, and the two values. The repository and the commit are the ones `corpus/code_manifest.json` records; the file is in the parsed copy under `code/md/`, which is the same snapshot every other claim in this survey about that repository is made from; `what the paper prints` names the table or equation the value was read from. No cell states a cause, and none is a claim about what the work's authors did.

### Table 11. Paper and released repository, the nine rows that state different values

| method                | confidence | repository, fetched commit               | file in it, and where                                                                        | what the paper prints                                                     | what that file contains                                                                              |
|-----------------------|------------|------------------------------------------|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| pddm_2019             | high       | google-research/pddm 06b88cdbaf          | pddm/envs/cube/cube_env.py<br>_get_obs                                                       | observation dimension 46 for in-hand reorientation, Table 2               | the six fields _get_obs concatenates sum to 39 on the code's own inline comments                     |
| dextreme_2022         | high       | isaac-sim/IsaacGymEnvs aeed298638        | isaacgymenvs/cfg/task/AllegroHandDextremeADR.yaml<br>actionDeltaPenaltyScale                 | action-delta penalty weight -0.25, Table 2                                | actionDeltaPenaltyScale: -0.2, and -0.01 in the ManualDR yaml beside it                              |
| visual_dexterity_2022 | high       | Improbable-AI/dexenv ad9634e9d2          | dexenv/conf/dclaw.yaml<br>alg.num_envs                                                       | 32,000 teacher training environments, Table S1                            | num_envs: 8000, and 16384 in the parent hand_default.yaml                                            |
| dexpbt_2023           | high       | NVIDIA-Omniverse/IsaacGymEnvs aeed298638 | isaacgymenvs/tasks/allegro_kuka/allegro_kuka_base.py<br>compute_kuka_reward                  | four staged reward terms, Sec. III-C                                      | eight summed components, hand_delta_penalty multiplied by 0 with the comment currently <br>disabled  |
| physhoi_2023          | high       | wyhuai/PhysHOI 6095c605e2                | physhoi/env/tasks/physhoi.py<br>compute_humanoid_reward                                      | object-rotation weights 0.1 and 0.01 for GRAB, Table 4                    | eor and eorv set to torch.zeros_like(ep), the computation commented out on the same lines            |
| unidexgrasp_2023      | high       | PKU-EPIC/UniDexGrasp 36c9bfcf7c          | dexgrasp_policy/dexgrasp/tasks/shadow_hand_grasp.py<br>compute_hand_reward, goal_cond branch | seven named weights from 0.1 to 10, Table 7                               | a torch.where cascade on literals -0.5, -1.0, 0.9, 0.1, 0.2, none of the seven present               |
| penspin_2024          | medium     | HaozhiQi/penspin 5035c52dc9              | configs/task/AllegroHandHora.yaml<br>forceScale                                              | a disturbance force of 0.2 times object mass at probability 0.25, Table 8 | forceScale: 0.0 and randomForceProbScalar: 0.0, the only shipped config naming either                |
| pianomime_2024        | high       | sNiper-Qian/pianomime c4abefac8d         | single_task/piano_with_shadow_hands_res.py_set_rewards                                       | two weighted terms, 2/3 and 1/3, Table 3                                  | five summed terms; _compute_energy_reward ends return 0 and _compute_fingering_reward return <br>0.0 |

8 rows. The repository and the commit are the ones `corpus/code_manifest.json` records, and the file is in the parsed copy at `code/md/<key>.md`. what the paper prints names the table or equation the value was read from. `confidence` is the row's own `mismatch_confidence` field, 7 high, 1 medium; `penspin_2024` is held below high pending a direct code read this survey has not made, and Appendix C prints the review note. No cell states a cause, and none is a claim about what the work's authors did: a reader with a browser settles every line of this table without asking anyone.

The case with the most at stake is `physhoi_2023`, and it is four items. The repository is `wyhuai/PhysH0I`, the commit is `6095c605e2`, the file is `physhoi/env/tasks/physhoi.py`, and inside `compute_humanoid_reward` the object rotation error and the object rotation-velocity error are set to `torch.zeros_like(ep)`, with the computation that would produce them commented out on the same two lines. The paper's Table 4 gives those two terms weights of 0.1 and 0.01 for GRAB. The paper does say that both are zero for BallPlay, which supplies no ball rotation; the zeroing in the file is not conditioned on the dataset. In that file the object's orientation error is the constant zero and its weight cannot change the reward, and the success criterion the paper scores its 95.4 percent with is itself position-only, so a run of that file would not report the difference either. Two documents say different things, and that is the whole of the claim.

Two more read the same way and are quicker. `dextreme_2022`'s Table 2 prints an action-delta penalty weight of -0.25; at commit `aead298638` of `isaac-sim/IsaacGymEnvs`, `AllegroHandDextremeADR.yaml` sets `actionDeltaPenaltyScale: -0.2` and the `ManualDR.yaml` beside it sets -0.01. `pianomime_2024`'s Table 3 prints two weighted terms, at two thirds and one third; at commit `c4abefac8d` of `sNiper-Qian/pianomime`, `_set_rewards` in `single_task/piano_with_shadow_hands_res.py` sums five, and two of the five, `_compute_energy_reward` and `_compute_fingering_reward`, compute a value and then end `return 0` and `return 0.0`. The remaining six are in Table 11 in the same four parts, and every one of the nine can be checked by opening the repository at the commit in that table.

**Everyone was written to first.** The authors of these nine works were written to on 19 September 2026, before this survey was posted. Each letter quotes the claim, its evidence and the sentences the survey would print, and asks whether the reading is right, whether there is a reason the released configuration differs, and whether the authors want the wording changed; each gave until 10 October 2026 to reply, and said that silence would be recorded as silence and not as agreement. What a reply can add is what the artifacts cannot give: which shipped configuration belongs to which stage of a pipeline, whether the fetched commit is the one behind the reported numbers, and whether an untagged branch holds the code that was run. The letters are in `outreach/` in the corpus that accompanies this survey, so a reader can see exactly what every author was asked, and `outreach/RECIPIENTS.md` records the address each went to and where that address came from. A tenth letter was drafted and not sent, because drafting it broke its own claim; it is kept with the nine. This version was written between those two dates: no reply had been received at the time of writing, no row here has been changed by one, and a window that closes in silence will be recorded in the next version as silence and not as assent. Any reply, and its effect on the row it concerns, is recorded in `mismatch_review` in `corpus/rows/`. What this section says is still written to the narrow form: a repository, a commit, a file, and two values. It attributes nothing to intent, and a reader with a browser can confirm or refute any line of Table 11 at the repository and commit that line names, without anyone's agreement and without this survey's own corpus. Two limits come with that, and neither is a hedge. A repository at a fetched commit is not the code that produced a paper's numbers: it may postdate that code, precede it, or have diverged from it on a branch nobody tagged, and a snapshot cannot say which, so each line compares a published document with one public artifact and claims nothing beyond the two. One work in this corpus says exactly that about itself. `hora_2022`'s README sends a reader to tag `v0.0.1` rather than to the default branch to reproduce the paper's numbers, which is why its row is classed version skew and is not one of the eight: told which commit to read, this survey read it, and nobody else was in a position to tell us, because nobody else was asked. The other limit is the remedy. Every one of the eight is correctable in public. An author who shows that the file says something other than what Table 11 prints, or that the fetched commit is not the one behind their numbers, changes the row: `mismatch_class` and `mismatch_review` in `corpus/rows/`, the counts that follow from them, and the sentence in the next version, with the correction printed beside the original comparison as the withdrawals already are. The route is the address in the author block of this paper. It is the only route this survey can offer today, because the corpus is not deposited yet: it is available from the author on request, and will be deposited with a persistent identifier. A correction asked for that way is a commit and a replacement version rather than a negotiation. Eight of the sixteen rows an earlier draft drew have already gone that way on this survey's own evidence and two more were narrowed; a further correction would cost it nothing.

Zeroed terms recur, and they are not the same thing as a term that is missing. A term present and zeroed survives a reader's check of the file, which is why Table 5 marks it separately, and it is only worth marking where the paper claims the term. Weights differ as well: a penalty printed at one value in a table and set to another in a config, an equation's term that is not in the released reward file, a term in the file that the table does not list. And in 14 rows the repository does not settle the question at all, which is this survey's limit and not a finding against the paper. Appendix C prints all 38 row by row in their five classes, each with the file, the value on both sides, and the review note where a comparison was narrowed or withdrawn.

A reward table is a claim about a training run and the code is a claim about a repository. Here the two state different things in nine cases, in the other 29 the released artifacts do not settle the question, and in exactly one, `hora_2022`, the repository says so itself. Read the reward function before the reward table, and treat a printed weight as a hypothesis about the code.

**The withdrawals, and what would close it.** This is a result about publishing practice in robot learning more than about dexterous manipulation: the dexterous corpus is its sample, not its subject. Eight is also a floor, since 46 method rows released nothing to check and four more are unsettled. The first count was sixteen. An adversarial re-reading withdrew six outright, `maniptrans_2025`, `eureka_2023`, `open_television_2024`, `dexmachina_2025`, `artigrasp_2023` and `graspxl_2024`: two refuted by the repository's own README, two resting on reward code that was never in the parse, one reading as a difference where the code carries the structure the paper prints, one against a paper with no reward function. `dexpbt_2023` was narrowed rather than dropped, its domain-randomisation half withdrawn and its zeroed reward term left standing, so that row is still a contradiction and the narrowing takes nothing off the count. That left ten, and ten held until the letters to the authors were drafted. Writing to `omnih2o_2024` meant reading the comparison again before the letter went out, and reading it again is what broke it. Four of its five reward-weight comparisons match the paper's own table to the digit, once a systematic x1.25 curriculum factor is applied. Only the stumble weight differs, and it differs by a factor of about a million, which reads as a typo in a table rather than as a policy trained on a different objective. Its hands are driven open-loop from a VR pose as well, outside the policy and outside the reward, which made the work a poor fit for a reward census in a dexterous-manipulation survey whatever the weight said. The comparison is withdrawn and the row moves to an internal inconsistency, which is where the count above sits it. `penspin_2024` was narrowed the same way and before the same deadline: half of it, that the released code turns off the paper's tactile channel, is withdrawn, because the config that was read carries 96 observation dimensions and `enable_tactile: False`, which is what the paper's proprioception-only student should carry and not a claim the paper makes about that stage. What is left is the line in Table 11, and it is held at medium because this survey could not establish from the parse whether a second task config exists elsewhere in that repository.

**Eight of the sixteen rows first drawn have left the contradiction count: six under adversarial re-reading, one at the point of writing to the authors, because someone sat down to write the letter and looked at the evidence again, and one because a reply came back and the claim did not survive it. Two more were narrowed and remain in it.** Each retraction and narrowing is recorded in the row it concerns, beside the comparison it revises: a survey that names people should carry its corrections beside its findings, in public and not just in the corpus. What would close the finding itself is a reward table generated from the released config at a named commit, so a reviewer diffs two artifacts instead of reading two documents.

## 6. Bimanual dexterous manipulation

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Two hands are not two instances of the same problem. They share one object, they can collide with each other, and the action space doubles, so neither the hardware result nor the policy carries over from one hand. Counting the work is the first difficulty. Fifty-three corpus papers carry the two-hand flag, and that flag says only that the robot has two end effectors. Thirteen of the 53 put no dexterous hand on the robot at all. `aloha_act_2023`, `rdt1b_2024`, `egomimic_2024`, `h_rdt_2025` and `umi_2024` are parallel-jaw throughout; `pi0_2024`, `pi05_2025`, `pistar06_2025` and `diffusion_policy_2023` name no hand and run grippers on every embodiment; `gemini_robotics_2025` and `gemini_robotics_15_2025` report every bimanual number on grippers and show five-fingered hands only qualitatively; and `helix_2025` and `groot_n16_2025` name no end effector anywhere on the page. Three more run a gripper on one embodiment and a hand on another: `ace_teleop_2024`, `dp3_2024` and `open_television_2024`, whose Unitree H1 carries 6-DoF Inspire hands and whose Fourier GR-1 carries a 1-DoF jaw.

**The denominator for this section is 28:** the method rows whose notes place a learned closed-loop controller on two multi-fingered hands. It is the 53 less those 16; less two static grasp-synthesis methods, `bimangrasp_2024` and `bidexgrasp_2026`; less three systems with no learned policy, `castro_sap_contact_2021`, `dexteleop0_2026` and `pang_global_planning_2022`; and less four rows where no learned policy holds both hands: `omnih2o_2024`, whose fingers are mapped open-loop from the Vision Pro and sit outside its 19-DoF policy; `okami_2024`, whose headline pipeline is open-loop retargeting with a learned policy only in a side experiment; `dexdeform_2023`, a skill model refined by trajectory optimization rather than a closed-loop controller; and `omnigrasp_2024`, a simulated human body with no bimanual task. `dexterous_handover_2025` never enters, because its row records one hand. Benchmarks and datasets are outside the 28 by class, `bidexhands_2022`, `bench2dex_2026` and `robopianist_2023` being benchmarks and `rplm_2024` and `humanoidgen_2025` datasets; they are quoted here as evidence and never counted. Every paper below runs two multi-fingered hands unless said otherwise.

### 6.1 Coupling, roles and collision

When both hands hold the same object, the object closes a kinematic loop between them. Neither hand can move without changing what the other must do. `bidexgrasp_2026` reports that the coupled objective "often yields imbalanced solutions, where one hand dominates stability while the other contributes marginally." `artigrasp_2023` reports simulation speed scaling "roughly quadratically with the number of contacts," and trains each hand alone before pairing them.

Role asymmetry is an assumption most of these systems make silently. `bidexhd_2024` states it outright: "we assume the robot to be right-handed by default, i.e., the left hand handles the target object and the right hand handles the tool," and `twisting_lids_2024` bakes the same split into its reward, putting reference contact keypoints on the bottle base for the left fingertips and on the lid for the right. The bias is in the data first: `taco_2024` recruited 14 right-handed subjects and measures the right hand moving consistently faster.

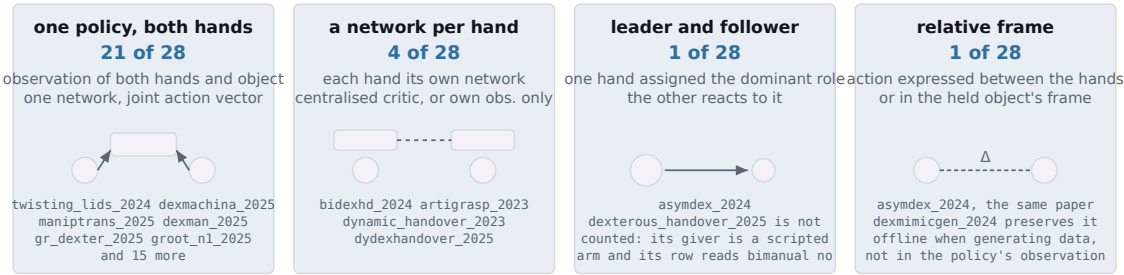
The arms collide, and the corpus handles this by construction rather than by control. `robopianist_2023` ships a forearm-forearm collision term that its own reward table never lists. Two papers name arm collision as unsolved. `dexmicgen_2024` "does not explicitly handle collisions," and `dexman_2025` says its control parameterisation "overlooks full arm posture," which it needs to avoid them.

The action space doubles and the observation grows faster. `bidexhands_2022` uses a 52-dimensional action for 18 of its 20 tasks against observations of 398 to 446 dimensions. `gr_dexter_2025` emits an 88-dimensional chunk of arm joints, end-effector poses, hand joints and fingertips. What the larger vector does not carry is any statement of who does what. The usual answer is one scalar reward summed over both sides, and the cost is a compounding failure rate. `maniptrans_2025` scores success only if both hands succeed, and its OakInk-V2 success rate falls from 58.1 percent on single-hand sequences to 39.5 percent on bimanual ones.

## 6.2 Coordination architectures

**Figure 5. Four ways to control two dexterous hands**

Counts are the 28 corpus rows that put a learned closed-loop controller on two dexterous hands. The panels are not a partition: `asymdex_2024` is drawn in two of them, and the two rows whose notes never say which they use are in none.



The field is concentrated in the first panel. The two on the right are where the structure of the problem is actually used, and `asymdex_2024` is the only corpus paper that puts either one inside a policy.

Figure 5 sets the four mechanisms side by side. Every row of the 28 carries exactly one architecture, read from its note: one policy over both hands, a network per hand, leader and follower, or a note that does not say. The fourth panel is not a fourth architecture but the relative frame, and it is the same single row as the third.

Twenty-one of the 28 put one policy over both hands, which is three quarters of the set: the trackers and tracking-adjacent methods `dextrmachina_2025`, `dexman_2025`, `maniptrans_2025`, `objdex_2024` and `humanplus_2024`; the reinforcement-learning tasks `dexpbt_2023`, `twisting_lids_2024`, `eureka_2023`, `pianomime_2024` and `humanoid_sim2real_recipe_2025`; the demonstration pipelines `bidex_teleop_2024`, `dexcap_2024`, `dexwild_2025`, `dexmimicgen_2024`, `hato_visuotactile_2024` and `humanoid_policy_human_policy_2025`; and the generalist policies `gr_dexter_2025`, `groot_n1_2025`, `dexora_2026`, `metis_2025` and `egoscale_2026`. In every one of them a single network takes a concatenated two-hand observation and emits a two-hand action. Two rows do not say which they are, `bunny_visionpro_2024` and `deximit_2026`, whose notes describe the rig and the data pipeline but never the policy's own decomposition. The concentration is not the outcome of a comparison that was won.

Four of the 28 give each hand its own network, and the two papers that compare the choice disagree. `bidexhands_2022`, a benchmark row and so outside the 28, ships the MARL baselines and finds PPO over the full observation beats HAPPO and MAPPO "in most cases," with the gap narrowing on tasks that need both hands, because PPO "can use all observations" where MARL sees only part. `bidexhd_2024` concludes the opposite. Its independent PPO teachers score 74.59 percent stage-two tracking rate on trained tasks against 53.88 for a centralised policy over both observations. Both are simulation only, on different tasks and hands, so neither settles it. `artigrasp_2023` trains one PPO policy per hand, and `dynamic_handover_2023` and `dydexhandover_2025` use MAPPO with one agent per arm-hand system.

One of the 28 assigns explicit leader and follower roles, and the same one is the only policy expressed in a relative frame. `asymdex_2024` is that paper, and the only corpus paper that puts either mechanism inside a policy. It gives the dominant hand full finger and wrist control, restricts the facilitating hand to a 6-DoF base pose, and writes the dominant hand and the object in a frame attached to the object the facilitating hand holds. That cuts the observation from 176 dimensions to 88 and the action from 52 to 26. It ablates the two mechanisms separately, which no other corpus paper does. On Block in cup the full method scores 0.7701 over five seeds, against 0.1086 for relative frames without asymmetry and 0.0164 for asymmetry without them. Twist Lid transfers zero-shot at 18 of 20 real trials. The real rig pairs a 16-DoF Allegro with a 6-DoF Ability Hand because only one Allegro was available, so the roles are confounded with the hardware. Two other rows use a relative quantity without giving either hand a role: `dexwild_2025` appends the inter-hand pose to its observation, and `dexmimicgen_2024` applies one shared SE(3) transform to both arms' source segments when generating data, which is the mechanism offline rather than in a policy.

**What would close it.** One architecture carries three quarters of the field, and the choice behind it has been compared twice, with opposite outcomes, and ablated once. What would close it is a single handover task run on three architectures with the same hand and the same seeds, reporting giver and receiver returns separately rather than one shared number.

## 6.3 Benchmarks, datasets and the shared axes

Two purpose-built bimanual dexterous suites exist in the corpus, four years apart.

The two bimanual benchmarks in the corpus trade coverage against measurement, in opposite directions. `bidexhands_2022` is 20 tasks on two Shadow Hands in Isaac Gym, ordered by the infant age at which humans acquire the skill, at 2048 environments and a reported 30,000-plus FPS. Its measurement discipline is weaker than its coverage. It reports reward and normalized score, never a success rate. Its only success flag in code tests the object-to-goal distance against 3 cm, which ignores orientation and exists only in the four catching tasks. Any success rate later work attributes to Bi-DexHands comes from that flag or its own definition.

`bench2dex_2026` is the more instrumented of the two. It runs 26 long-horizon tasks in Isaac Lab across 12 arm-and-hand embodiments, with roughly 1.3K teleoperated demonstrations in eight modalities, including a ray-cast tactile image that maps all 12 hand geometries into one representation. Success needs the terminal predicate held for 0.5 s. Matched, GR00T N1.5 leads at 631 of 1300 rollouts, 48.5 percent, against ACT at 29.5,  $\pi 0.5$  at 27.3 and Diffusion Policy at 12.9. Under combined perturbation those first two compress to 19.8 and 19.7. Tasks are not crossed with embodiments, so no contrast isolates the hand, and the authors state the hand and tactile models are not calibrated against matching physical hardware.

Ten corpus rows are bimanual datasets, and five are two-hand interaction capture. `arctic_2022` gives 339 mocap sequences of 11 one-DoF articulated objects held in both hands, and deliberately labels over-shooting vertices as in contact, so its ground truth records interpenetration as contact. `taco_2024` gives 2.5K bimanual tool-use sequences fitted markerlessly under an attraction loss and a penetration loss the paper says trade against each other. `oakink2_2024` is the only corpus dataset publishing a penetration number for its own annotations, a mean depth of 0.25 cm over its 627 long-horizon sequences. `hot3d_2024` gives 833 egocentric minutes with mocap-grade poses for both hands and up to six objects, and `gigahands_2024` gives 2,034 markerless minutes from 56 subjects, neither with a penetration metric. These five feed the tracking work here: `dexmachina_2025` and `artigrasp_2023` from ARCTIC, `bidexhd_2024` from TACO, `maniptrans_2025` from OakInk-V2 and `dexman_2025` from both.

**Methods on the same axes.** Table 7 carries the bimanual papers on the same columns as every other method here, and three of those columns are worth reading together.

Real trials. `asymdex_2024` reports 20 per task, `hato_visuotactile_2024` 10 per condition, `dexmimicgen_2024` 20 on a Fourier GR1 and `bidexgrasp_2026` 260 across 30 objects. `maniptrans_2025` reaches real Inspire hands by open-loop replay with no trial count and no success rate. Seven of the 28 have no real robot at all: `artigrasp_2023`, `bidexhd_2024`, `dexmachina_2025`, `dexman_2025`, `dexpbt_2023`, `dydexhandover_2025` and `pianomime_2024`. The two piano rows outside the denominator, `robopianist_2023` and `rp1m_2024`, are simulation-only as well.

Penetration. Most bimanual rows read "not addressed." `bimangrasp_2024` gates on it, failing any grasp whose total penetration across hand-object, self and inter-hand exceeds 1.5 mm, and reports that "penetration remains the primary cause of grasp failure." `bidexgrasp_2026` reports it as a number, 0.15 to 0.20 cm maximum depth for its own grasps. `deximit_2026` carries a named hand-hand penetration term in its grasp-synthesis objective. `dexmachina_2025` resolves it once, replaying retargeted joints against a fixed object "to eliminate object penetrations" and measuring nothing during rollout. `artigrasp_2023` declines the metric outright, because its baselines "include a physics simulation which exhibits no interpenetration."

Comparability. `bimangrasp_2024` reports 54.03 percent success in Isaac Gym at friction 3. `bidexgrasp_2026` re-runs the same grasps in MuJoCo at friction 0.6 and gets 26.80 percent, at 1.52 cm penetration depth. Neither number transfers, and section 5.3's re-implementation result is the same lesson on the training side.

## 6.4 Handover, and what transfers from one hand

Handover is the one bimanual task where the hands are unambiguously asymmetric, because one gives and one receives. All three corpus handover papers use a single shared reward across giver and receiver. None defines separate objectives for the two roles.

Whether the two hands optimize the same objective is usually left implicit. [dynamic\\_handover\\_2023](#) states one formula,  $r = r_{\text{dis}} + r_{\text{linvel}} + r_{\text{torque}}$ , and never says whether thrower and catcher receive different decompositions of it. [dydexhandover\\_2025](#) is explicit that they do not: "Both hands share aligned objectives, forming a fully cooperative relationship," with one weighted sum feeding both agents. Only the thrower gets an extra KL regulariser toward a policy pretrained on human throws, which constrains style and not role.

The third case is weaker still. In [dexterous\\_handover\\_2025](#) only the receiver is learned. The giver is a UR5e with an Allegro hand that "holds the object without moving during the whole episode." There is one agent, one reward and no second policy, its row in Table 7 accordingly records one hand, and it is one of the rows the 28 excludes. The 94 percent often attached to this paper needs its conditions. It is Total Success, which counts "Indetermination" cases where the simulator failed to resolve collisions and the object clipped through the giver's hand, on the short prism, in simulation, over 100 episodes, with no real robot in the paper.

Nothing in the handover literature measures contact quality. Contact appears only as a positive signal, a boolean per-phalange touch in [dexterous\\_handover\\_2025](#) and a boolean contact reward in [dydexhandover\\_2025](#). The two systems that do handover well sidestep the problem. [hato\\_visuotactile\\_2024](#) reaches 10 of 10 on a real slippery handover by imitating teleoperation with fingertip touch sensing and no physics model. [humanoid\\_sim2real\\_recipe\\_2025](#) stages its reward with a discrete variable switching which hand's fingertips are scored, and still calls handover its hardest task at 52.5 percent real success.

**What transfers from single-hand work.** The training recipe transfers intact. PPO with thousands of parallel environments, an asymmetric critic on privileged state, domain randomisation and distillation into a vision policy is the same in [twisting\\_lids\\_2024](#) and [humanoid\\_sim2real\\_recipe\\_2025](#) as in section 5.2. Reward forms transfer literally, and the contact-goal reward  $r_{\text{contact}} = \sum 1/(1+\alpha d)$  appears in both.

Multi-task learning does not transfer, and the bimanual benchmarks show it clearest. [bidexhands\\_2022](#) finds multi-task PPO and ProMP fail on MT4, MT20, ML4 and ML20. [robopianist\\_2023](#) finds F1 on the shared training song drops "from roughly 0.7 F1 for 1 song to almost 0 F1 for 16 songs," with no positive transfer "regardless of the size of the pre-training tasks."

Single-hand evaluation does not transfer either. Success criteria compound, because [maniptrans\\_2025](#) fails a trajectory "if either hand fails to meet these conditions." And object-centric metrics cannot say which hand caused the failure. [bidexhd\\_2024](#) scores the steps where both objects track their references, and [dexmachina\\_2025](#) averages ADD across object parts before an AUC. Both say whether the task worked and nothing about which hand lost it.

One failure mode has no single-hand counterpart. Two hands can penetrate each other, and the corpus almost never looks. Three papers carry an inter-hand penetration term, and all three are static grasp synthesis: [bimangrasp\\_2024](#), [bidexgrasp\\_2026](#) and [deximit\\_2026](#). Not one learned bimanual controller in the corpus measures or penalizes hand-hand penetration during a rollout. [pianomime\\_2024](#) ships a flag that turns inter-hand collision off in the physics.

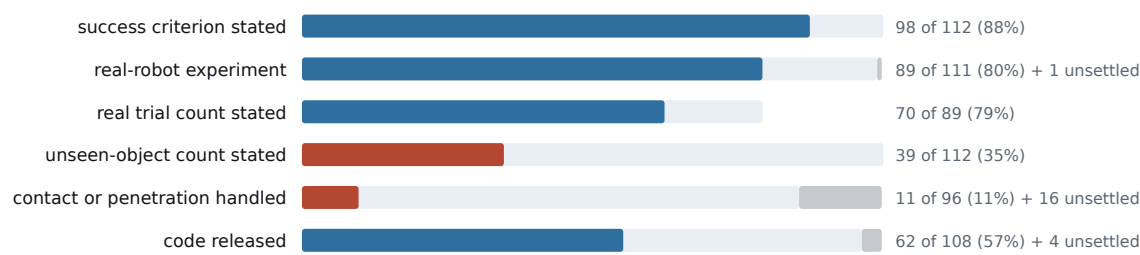
# 7. Evaluation, and a protocol for comparison

## 7.1 Reporting practice

Most of this literature does test on hardware, which is the check that matters and the one a simulator cannot substitute for. Of the 112 papers in the corpus, 89 report a real-robot experiment, 22 do not and one row is unsettled, which is 80 percent of the 111 the note settled. Among those 89, 70 state how many real trials produced the headline number, 79 percent of them. The 89 is the denominator that belongs to this statistic: the 22 rows with no real robot cannot state a real trial count, and counting them as silent turns a definitional impossibility into a reporting failure. Thirty-nine rows state a count of unseen test objects, 35 percent. Ninety-eight state how a rollout is scored, 88 percent. Sixty-two released code and 46 did not, with four rows unsettled, 57 percent of the 108 the note settled. Figure 6 draws these six shares, each against the denominator that belongs to it. Every bar is a lower bound, for the reason the next section gives. The two items flagged red there, unseen-object count and contact or penetration handling, are the pair a reader actually needs to compare two methods: no unseen-object count means no generalisation denominator for a success rate, and no contact handling means no way to tell whether the hand passed through the object. They are also the two the field states least often.

**Figure 6. What the method literature reports**

Share of the 112 surveyed method papers whose row records each item, each against its own denominator: the pale track is the rows the item can apply to, the grey tail the rows whose note did not settle it. A bar is evidence the item was stated, not that the work did it well.



Red marks an item fewer than a third of the rows it applies to state. Every bar is a floor: a hand audit of the nulls recovered 15 trial counts, 19 criteria and 7 object counts (§7.1).

**These are counts of what this survey captured, not of what papers reported, and every one is a floor.** A structured row holds a scalar. A paper that reports ten trials on each of nine tasks, or a scoring rubric instead of a threshold, or a count spread over four tables, has nothing the extraction can reduce to one integer, so it produces a null, and a null is then indistinguishable from a paper that said nothing. The bias runs one way: every miss converts a reporting paper into a silent one, and the survey's argument is that the field reports badly, so the artifact flatters the argument. Section 5.6 makes the same disclosure about the paper/code count, subtracting the 13 disagreements that are limitations of this survey's own parsing before declaring which number to quote, and the coverage statistics above need it more.

The nulls behind four of those six shares were therefore audited by hand: the trial counts, the success criteria and the unseen-object counts against the notes they came from, and the penetration field against the parsed sources themselves. Those four numbers above are post-audit. Appendix A gives what the audits recovered and what they could not: the counts moved by tens of rows, 19 trial counts and 4 criteria remain genuinely unsettled, and every bar in Figure 6 is still a lower bound. The other two shares are not post-audit. The code-release field has never been audited this way, so its share is a floor of unmeasured depth, and the real-robot field's one unsettled row is left out of its denominator rather than counted as silence.

The remaining gap is the one that matters. Nineteen of the 89 papers with a real robot never say how many times they ran it. A percentage with no denominator cannot be given an interval, so it cannot be compared with anything.

Where the denominator is stated it is small, and it is not one quantity. Some stored counts are per-cell, meaning ten trials on each task, or twenty per condition, or five per object. Others are grand totals over every cell. The rows now record which of the two each value is, and the two distributions are quoted separately. The 39 per-cell counts run from 5 to 100 with a median

of 15 and quartiles at 10 and 20. The modal cell is 10 trials, in 16 rows, then 20, in 12. The 24 grand totals run from 12 to 750 with a median of 110. Pooling the two gives a median of 20 and a range of 5 to 1287, and that pooled figure is the one an earlier draft of this section quoted. It describes nothing, because `hora_2022`'s 240 and `visual_dexterity_2022`'s 20 are experiments of comparable size recorded on different bases. Seven further counts could not be assigned a basis at all.

The per-cell count is the one a Wilson interval attaches to. At 20 trials a reported 60 percent carries a 95 percent Wilson interval of 39 to 78 percent, and a reported 80 percent an interval of 58 to 92 percent. Overlapping intervals are not a test, so the point is made with one: 12 of 20 against 16 of 20 is  $z = 1.38$ ,  $p = 0.17$ , and the difference is not established. Two methods separated by 20 points at 20 trials, the larger of the two modal cell sizes, are not separated at all.

Reporting has improved a little and the experiments have not. About two thirds of papers from 2025 onward state a trial count, against three fifths before, but the counts themselves have not moved: the median has sat near twenty trials per cell every year since 2024, and earlier years rest on too few observations to read as a trend. A field can get better at disclosing a number without the number itself getting any better.

The denominators also sit on different hardware. The 103 papers that name their own hand give 78 distinct hand strings between them, and this survey applies no normalization to those strings, so 78 is a count of strings and not of hand designs. Matching on the string, Allegro appears in 35, Shadow in 21, Inspire in 19 and LEAP in 12. A success rate on a 16-degree-of-freedom Allegro and a success rate on a 6-actuator Inspire hand are not measurements of the same thing.

Nor are the criteria. `dexverse_2026` counts PickCube a success when the cube is "lifted at least 0.20 m above its resetting height". `bench2dex_2026` requires its terminal predicate to hold for a continuous dwell time of 0.5 s, to reject transient contacts. `colosseum_2024` counts an episode successful "if the model completes the task fully". `dextrack_2025` reports every success rate as a pair under two threshold sets, which on GRAB gives 46.70 and 65.48 percent for the same rollouts. Of the 14 papers that still state no criterion, ten have no success predicate at all. They report radians rotated or time-to-fall and never define a success, which is a fact about the paper rather than a gap in this survey. And four are unsettled by the note. Of the 19 criteria the audit recovered, all but one are a rubric, a staged partial credit, a judgment by eye or a deferral to a benchmark's own definition rather than a threshold, and exactly one, `pistar06_2025`, states a verbatim numeric one; Appendix A counts the four kinds. A rubric is a milder failure than silence and a worse one than a threshold, because it is reproducible inside a lab and not across two.

**The axes that matter.** Seven quantities dissociate in the published data, so they have to be reported separately.

**Task success.** Binary success discards the difference between near-misses and inaction. `beyond_binary_success_2026` puts it plainly: "a policy that completes 90% of the task is clearly better than a policy that is frozen the whole time, yet their success rates would be identically 0%". In `kress_gazit_policy_eval_2024` policy C scores 17 percent overall on the pancake task while picking up the spatula and flipping the pancake in 23 of 23 attempts.

**Robustness to perturbation.** How much a reported rate moves when the setting changes has been measured, though not on a hand. `colosseum_2024` measures a 30 to 50 percent success drop under single perturbation factors and at least 75 percent under all 14 together. In `bench2dex_2026` GR00T N1.5 leads the matched condition at 48.5 percent and falls to 19.8 percent under combined shift, while  $\pi 0.5$  goes from 27.3 to 19.7 and retains the most at 72.1 percent. The ranking at the anchor is not the ranking under shift.

**Generalisation to unseen objects.** Only 39 rows state a count and the median is 11 objects.

**Physical plausibility of the contact.** Eleven of the 96 papers whose contact handling the note settled address it, 11 percent, with 16 rows unknown. Section 7.2 takes them apart.

**Sample and wall-clock cost.** Thirty-one of the 112 papers state a parallel environment count and 18 a simulated episode count. `robopianist_2023` is the exception, at 5 million samples per song and roughly 5 hours per run on four Tesla K80 GPUs.

**Real-robot transfer.** Simulated rank order is not real rank order. `autoeval_2025` scores Open- $\pi 0$  on put-eggplant-in-sink at 6 of 50 in SIMPLER and 47 of 50 on the real WidowX. `suresim_2025` states the limit directly: "the simulation-to-real gap

precludes rigorous statistical inferences about real-world outcomes from simulation results alone".

**Reproducibility.** Releasing code is not the same as releasing something that settles a question. 62 rows released code that could be parsed against the paper, and 38 of the 112 papers record a disagreement of some kind between the paper and that code. All 38 released code, so the raw rate among code-releasing rows is 61 percent. That raw rate is not the finding, because the 38 are not one thing. Section 5.6 classifies them: 8 contradictions, 14 limitations of this survey's own parsing, 8 components never released, 4 version skews and 4 inconsistencies internal to a paper. Only the contradictions are a finding about the work itself, so 9 of 62 code-releasing rows, which is 15 percent, is the figure this section and Table 8 use. `physhoi_2023` is the clearest of the 9. It lists a non-zero object-orientation weight for GRAB in Table 4, and the reward function in the repository at the commit this survey fetched sets that orientation error to zero, so in that file the object is tracked in position only.

## 7.2 Physical plausibility

One quantity is specific to hands, and it is the one closed-loop policies do not record: how far the hand passes into the object it is holding. Eleven of the 96 papers whose notes settle the question handle interpenetration at all, and every one of them measures it at a reference rather than at a rollout. That is a narrower claim than it sounds, and reading it more widely would be wrong. Outside closed-loop control the quantity is a standard comparative column and has been for years: `bimangrasp_2024` fails any grasp past 1.5 mm, `toporetaret_2026` reports a maximum depth and a share of frames past 2 mm against a baseline retargeter, and `oakink_2022` scores a dataset split on depth, intersection volume and simulation displacement. The gap is specific to learned closed-loop control: seven of the eleven score a pose, a trajectory or a contact model before anything executes, the four that are closed-loop policies score the references they were given, and none reports the measurement for rollouts of its own trained policy. Appendix A gives the audit behind that null, which sampled 25 of the 85 papers recorded as not addressing penetration and recovered nothing.

Where in the pipeline those eleven act is the reference-versus-rollout split that section 1 takes from `zhao_dexhand_survey_2026`. A reference is a pose or a trajectory scored before execution, and a rollout is what the trained policy actually did. What this section supplies on that axis is a measurement method and a count, and it does not supply a threshold. The count is the eleven of 96 above, with four closed-loop policies inside it and none we found reporting a number for its own rollouts. The method is the plausibility row of Table 8: maximum and mean penetration depth over the evaluation rollouts, on a dense surface sample, computed by code that never entered the reward or the termination rule. The threshold is borrowed, and section 7.4 says from where and why it does not bind. Six of the eleven are grasp synthesisers or trajectory optimisers, namely `bidexgrasp_2026`, `bimangrasp_2024`, `deximit_2026`, `pang_global_planning_2022`, `toporetaret_2026` and `unidexgrasp_2023`, and `castro_sap_contact_2021` is a contact model rather than a controller. That leaves four closed-loop policies in the whole corpus: `clutterdexgrasp_2025`, `dexmachina_2025`, `dextrack_2025` and `teledexter_2026`.

The reason only four are closed-loop is a measurement trap: a quantity a policy optimizes cannot also judge it, because the policy learns the measure rather than the property the measure stands for. `physhoi_2023` documents the failure in the clean direction, its kinematic imitation reward maximized by not touching the object at all because contact perturbed the reference trajectory. `dextrack_2025` shows the same trap from the other side: it defines a maximum hand-object penetration depth over all frames, applies it only to its input kinematic references, and then reports tolerance of "severe hand-object penetrations" as a result. `toporetaret_2026` is the strongest case in the corpus and still stops one step short on the same line. It constrains penetration during retargeting with a 1 mm soft tolerance and a 30 mm hard bound, and reports a maximum penetration of 1.07 mm and 0.00 percent of frames above 2 mm against 22.22 mm and 96 percent for its baseline retargeter, measured on retargeted references rather than on a rollout.

Definitions are not shared either. `grab_2020` estimates contact by proximity, because "contact cannot be directly observed", and reports that "'Use' grasps have  $3.25 \pm 0.68$  mm average penetration" without saying whether that is a maximum, a mean or a median. `oakink_2022` supplies the fullest published vocabulary: penetration depth, solid intersection volume and simulation displacement. The analytic tradition scored a grasp without simulating it at all, in the wrench-space terms of

`ferrari_canny_1992` and `roa_suarez_grasp_quality_2015`; neither could be obtained for this survey, and no definition here rests on their contents.

**What would close it.** The gap is specific to learned closed-loop control, and it is a choice. Nothing prevents the measurement. What would close it is a maximum and a mean penetration depth over the evaluation rollouts, on a dense surface sample, computed by a measure the policy never optimized.

### 7.3 Statistical practice, and a proposed protocol

A protocol already exists for evaluating learned robot policies, and what it does not supply is the part a reader needs. `kress_gazit_policy_eval_2024` prescribes process, not numbers. Write the success criteria before the run and have someone other than their author score the runs. Match initial conditions with image overlays. Interleave the policies blind within one session. Report counts rather than percentages, alongside the initial conditions and the failure modes. Use a posterior over the Bernoulli parameter instead of a point estimate. It prescribes no minimum trial count anywhere and no frequentist confidence-interval width anywhere. Its own example report uses 10 initial conditions with two runs each, 20 evaluations per policy, and its worked case shows what 20 buys. Pancake success of 15 of 18 against 11 of 17, nominally 83 against 65 percent, leaves a 0.11 posterior probability that the worse policy is actually better. At 150 of 180 against 110 of 170 the same rates separate.

One study does put numbers on it, without saying where they come from. `lbm_careful_examination_2025` supplies them by fiat rather than derivation: 50 rollouts per task per policy per condition on hardware, 200 in simulation, blind, with randomised policy order inside per-initial-condition bundles. It replaces confidence intervals with Beta-posterior violins and corrects its pairwise tests under Bonferroni, "unless otherwise noted". Its own warning is the strongest sentence in this literature: "there is significant risk that many robotics papers are measuring statistical noise due to insufficient statistical power".

The rest fall short of their own advice. `roboarena_2025` runs 612 double-blind pairwise comparisons across seven institutions and 4284 rollouts, and reports no confidence intervals and no per-policy trial counts. `colosseum_2024` evaluates 235 test sets at 25 episodes each with "one training seed and one evaluation seed" per baseline and gives no intervals in simulation. `autoeval_2025` runs 50 trials per policy per task and calls  $\pm 10$  percent "the natural variance of robot evaluations", without giving the formula behind the intervals it plots.

Two papers do give usable numbers. `suresim_2025` pairs real and simulated trials and de-biases the simulation with a rectifier, saving 20 to 25 percent of hardware trials at paired correlations of roughly 0.6 to 0.7, and nothing at all at a correlation near zero. The 25 percent figure is the better of its two reported settings, the DP case at  $\rho = 0.702$ . Its decision rule is exact: combining helps only when the rectifier variance is below the variance of the real evaluations. `beyond_binary_success_2026` gives the largest saving. On the LBM rubrics its sequential test on graded scores cuts simulated evaluation by about 70 percent and hardware by about 45 percent, 286 trials against a nominal 500, with per-task decisions landing in 12 to 36 paired trials. On RoboArena's data a 30-point gap on continuous progress scores reaches significance in 18 trials, while a 20-point gap on binary success needs about 80.

**A proposed protocol.** Every count below is printed by `tools/make_eval_tables.py` in its derivation mode, and each axis is derived for the statistic that axis actually reports: a single rate takes a Wilson half-width, a matched comparison takes McNemar, a ratio takes the standard error of the log ratio, a correlation takes the Fisher-z interval.

Appendix E gives the full derivation; what it concludes is the following, and every count in Table 8 is one of these. A single reported rate takes 100 real rollouts per cell, because 93 is where the worst-case Wilson half-width falls to 10 points and 10 points is the coarsest width at which a rate is worth printing. A comparison is a different question and a harder one. The design is paired, so the count follows McNemar and depends on the discordance rate rather than on the two rates alone, and separating 50 from 70 percent at 80 percent power takes 57 matched pairs per arm at the assumed discordance of 0.3, which is 114 rollouts against the 186 two independent arms would need. Simulated cells take 200 episodes for a 6.9-point half-width. Perturbation axes are screened at 40 per axis, which ranks the axes and cannot establish that any one of them hurt, and certified at 101 per arm. Unseen objects resample the object rather than the trial, at 20 objects and 5 trials each, and a 10-point

claim about an object distribution needs about 93 objects. The transfer axis takes the 100 real trials paired to 100 simulated, and about 457 pairs to separate two correlations 0.1 apart. One hundred is a cap and not a bill, because a sequential test on a graded score reached its decision in 12 to 36 paired trials in `beyond_binary_success_2026`, and a cell that stops early reports a confidence sequence rather than a Wilson interval, never both. The appendix also keeps the counts an earlier draft of this section quoted and this one withdrew, since the test they were computed for was the wrong one.

**Table 8. The proposed evaluation protocol**

| axis                       | what is measured                                                                                                                                                 | how                                                                                                                                                                                                            | minimum trial count                                                                                                                                                                                     | reported alongside it                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | why                                                                                                                                                                                                                                                                              |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Task success               | fraction of episodes meeting a criterion, and a graded rubric score in [0,1] over equally weighted milestones                                                    | criterion written before the run by the designer and scored by someone else; terminal predicate held for a 0.5 s dwell; initial conditions matched by image overlay; policies interleaved blind in one session | 100 real and 200 sim per policy per task per condition for an absolute rate at a 10-point interval; a comparison alone needs 57 matched pairs per arm (McNemar, 50 against 70 percent, discordance 0.3) | raw counts, the criterion verbatim, the rubric, the initial-condition protocol, and an interval whose kind is named: Wilson for a cell run to a fixed n, an anytime-valid confidence sequence (WSR, as in <code>beyond_binary_success_2026</code> ) for a cell stopped early by a sequential test, because a Wilson interval computed at a data-dependent stopping time is not a 95 percent interval. Intervals are marginal, not simultaneous: a paper comparing k policies corrects its $k(k-1)/2$ pairwise tests to a global 95 percent level, as <code>lbm_careful_examination_2025</code> does, or says it has not | a rate without a denominator and an interval cannot be compared; the rubric separates a near-miss from inaction; the design is paired, so the count that governs it is paired                                                                                                    |
| Robustness                 | success under each perturbation axis as an absolute rate; the ratio to the unperturbed anchor is reported descriptively and is not tested at the screening count | axes that must not change the action and axes that must be scored separately; ranges stated; anchor replayed exactly                                                                                           | 40 per axis per policy in sim to rank the axes; 101 per axis and 101 at the anchor before any claim that a named axis hurt                                                                              | both absolute rates with their Wilson intervals, the ratio without an interval unless the certifying count was run, the axis ranges, and which split each axis is in                                                                                                                                                                                                                                                                                                                                                                                                                                                    | the ranking at the anchor is not the ranking under shift; at 40 per arm a fall from 0.50 to 0.30 is a ratio of 0.60 with a 95 percent interval of 0.34 to 1.06, which contains 1                                                                                                 |
| Unseen objects             | mean over held-out objects of per-object success                                                                                                                 | objects drawn from a stated distribution disjoint from training; bootstrap over objects, not trials                                                                                                            | 20 objects at 5 trials for a screening claim; 93 objects for a 10-point claim                                                                                                                           | the object list, per-object rates, the bootstrap interval, the object-count caveat, and the within-object trial count, since the 93 is a Wilson width that treats one object as one Bernoulli draw                                                                                                                                                                                                                                                                                                                                                                                                                      | the resampling unit is the object, so trial counts overstate the precision                                                                                                                                                                                                       |
| Physical plausibility      | maximum and mean hand-object penetration depth per rollout, and the fraction of frames above 2 mm                                                                | dense surface sample against the object mesh or SDF, computed on the policy's own rollouts by code that never entered the reward or the termination rule                                                       | 100 rollouts, simulation only                                                                                                                                                                           | interval over rollouts, sample density, threshold, solver depenetration settings, and at least one rendered rollout                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | a measure the policy optimised is not evidence about the policy                                                                                                                                                                                                                  |
| Sample and wall-clock cost | environment steps and wall-clock to the checkpoint that produced the headline number                                                                             | counted to that checkpoint, not to the end of training; GPU model and count stated                                                                                                                             | 3 training seeds                                                                                                                                                                                        | the range over seeds, and the statement that 3 seeds is a range and not an interval                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | training variance is not rollout variance and the two are routinely conflated                                                                                                                                                                                                    |
| Real-robot transfer        | paired real and simulated outcomes on matched initial conditions, their correlation, and the rectifier variance against the variance of the real evaluation      | the same 100 initial conditions run in both; correlation on the pairs; variance ratio bootstrapped over the same pairs                                                                                         | the 100 real trials paired to 100 sim, which fixes the correlation to about $\pm 0.10$ ; about 457 pairs for a claim that separates two correlations 0.1 apart                                          | the correlation with its Fisher-z interval, the variance ratio with its bootstrap interval, and which of the two decisions the count supports                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | a simulator earns a real-world claim only through its measured paired correlation; at 100 pairs $\hat{\rho} = 0.70$ carries 0.58 to 0.79, enough to say a simulator tracks reality at all and not enough to separate <code>suressim_2025</code> 's 0.70 regime from its 0.59 one |
| Reproducibility            | code, checkpoints, and the exact config that produced the headline run                                                                                           | diff the paper's stated objective against the released config and name the file and line of every disagreement                                                                                                 | not a trial count                                                                                                                                                                                       | the named file and line for each disagreement, or an explicit statement that none was found                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 16 of the 62 code-releasing method rows carry a true contradiction between the paper's stated objective and the released code, and 38 carry a recorded disagreement of any kind (\$5.8)                                                                                          |

## 7.4 Feasibility and limitations

The proposed protocol has not been run. In the twelve-method comparison assembled for this survey, no method supplied all the measurements needed for a complete protocol row. DeXTreme `dextreme_2022`, for example, reports a success criterion and a confidence interval, but uses ten trials. This limits retrospective comparison; it does not imply that the reported experiments failed their own stated objectives.

**What would have to be true.** The bill comes first. Per policy and per task the protocol asks for 57 matched trials on the anchor set, which is the paired count from §7.3, and 100 on the unseen-object set, at 20 objects and 5 trials each, with robustness and plausibility absorbed by simulation. A two-policy comparison on three tasks is then 342 real rollouts on the matched set and 600 on the unseen-object set, 942 in all, and at one minute per rollout including the reset that is about 16 hours of robot time, before failed resets, repairs and scoring. The same bill computed from the independent-arm count, which an earlier draft used, was 1200 rollouts and 20 hours. The pairing removes about a fifth of it rather than the four fifths the pair counts suggest, because a pair is two rollouts and only the anchor set is paired. `autoeval_2025` ran about 850 episodes in 24 hours on a WidowX with three human interventions, and had to pause 20 minutes every 6 hours once the motors overheated. A tendon-driven multi-finger hand is more fragile, so 16 hours of rollouts is most of a week of calendar time.

That bill is large but not unprecedented. `autoeval_2025` records that evaluating OpenVLA against its baselines took more than 2500 rollouts and more than 100 hours of human labour across three institutions, and `lbm_careful_examination_2025` analyzed about 1800 real rollouts across nine hardware stations. What is unprecedented is paying it for a single dexterous-hand paper, where the modal per-cell count is 10 trials and the median per-cell count 15, and where the median stated count has been 20 or 22.5 in each of the last three years.

Four things would have to change. Reviewers would have to reward 57 matched trials on one task over 20 unmatched trials on five, and nothing in the corpus suggests that is happening. The loop would have to be automated, and `autoeval_2025` shows it is buildable for a gripper at one to three hours of setup, while stating that it supports binary success only and no robustness axes. Somebody would have to run the penetration measure on their own rollouts, which is a choice rather than a capability: section 4.2 shows the depth is computable from the poses and the meshes in a few lines of Warp, and IsaacGymEnvs already ships a task that does it every step. An earlier draft of this section made the engine the barrier, and that claim is withdrawn, because the released code refutes it. And the comparison would have to be sequential, because the savings in `beyond_binary_success_2026` are the only reason 100 is a cap rather than a cost.

Four limits apply to the proposal itself. This survey re-ran no method, so every count in Table 8 is derived from an interval width, a power calculation or another paper's measurement. The counts are worst-case at  $p = 0.5$ , so a method near 90 percent needs fewer trials for the same width and a method near 50 percent needs all 100, and the paired count additionally rests on an assumed discordance of 0.3, which no paper in this corpus reports. The perturbation axes are borrowed from `colosseum_2024` and `simpler_2024`, which run parallel-jaw grippers on rigid objects, where `simpler_2024` found physical parameters moved success rates by at most 15 percent. That is the sensitivity expected to grow with multi-finger contact, and no work in this corpus measures it. The 2 mm penetration threshold is taken from `toporetaret_2026` with no independent justification, and the captured human grasps in `grab_2020` sit above it at 3.25 mm, which makes 2 mm a simulator convention rather than a physical bound. So the three things this survey adds to the reference-versus-rollout axis are a count, a measurement method and a protocol slot for them, and a threshold is not among them. It is borrowed from one paper and reported as borrowed, and it will stay a convention until somebody measures penetration on rollouts across hands and engines and finds a value that separates behavior a physicist would accept from behavior they would not.

**The methodology literature has no dexterous hand in it.** Seven corpus rows are evaluation protocols, a small denominator, and not one of them uses a dexterous hand: `suressim_2025` runs a parallel-jaw gripper and the other six state no hand. Everything proposed above is therefore assembled from work on grippers and on whole-arm tasks. What would close that, and close it cheaply, is to run the Table 8 protocol once, on one in-hand reorientation task with one 16-DoF hand, and release the rollouts and per-trial outcomes for independent analysis.

## 8. Conclusion: the seven claims, and what to do next

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This corpus describes a field that has converged. Most of it trains with reinforcement learning on privileged state in a GPU simulator and distills the result into a vision-only student; a second branch, growing faster, learns from teleoperated demonstration instead; and almost all of it runs on four hands. What the field cannot currently do is compare its own results, and most of the seven claims below follow from that. Each is one sentence, with the section that carries its evidence and the experiment that would close it. Nothing is argued here: the denominator, the evidence and the prescription sit in the section named, so that a finding and its consequence are read together and stated once.

1. Eight hands that can be bought today or built from published designs take zero papers between them, while the corpus's own experiments concentrate on four designs (section 3.4, Figure 2).
2. Human data does not port across hands and the map is usually unstated: 33 of the 53 method rows that use human data never say how the human motion reached the robot hand (section 5.3, Table 6).
3. Twenty-one of the 28 papers that put a learned closed-loop controller on two multi-fingered hands run one policy over a concatenated two-hand observation, a choice that has been compared twice with opposite outcomes and ablated once (section 6.2, Figure 5).
4. The generalist and vision-language-action policies that do evaluate on a multi-fingered hand run it at a median of 6 actuated degrees of freedom, against 16 across the reinforcement-learning rows that state a count, and the two rows that reach the larger band on paper never say whether the hand was in the evaluation (section 5.4).
5. The evaluation-methodology literature the protocol of section 7.3 is assembled from contains no dexterous hand at all: of its seven corpus rows, one runs a parallel-jaw gripper and the other six state no hand (section 7.4, Table 8).
6. Eight of the 62 papers that released parseable code state, in a named file at a named commit, something other than the value their paper prints, and eight further rows have been withdrawn from that count since its first draft, one of them because an author replied, with two more narrowed. Table 11 gives each of the eight as a repository, a commit, a file and two values, and all of their authors were written to before this was posted: section 5.6 says that beside the finding, with the letters in `outreach/` and the route by which a disputed case is corrected (section 5.6 and Appendix C).
7. No closed-loop policy in the corpus reports interpenetration for the rollouts of its own trained policy, and all eleven rows that handle penetration at all sit on the reference side of the reference-versus-rollout split (section 7.2, and the contact-handling bar of Figure 6).

Six of the seven are gaps in the literature. The sixth is a result about publishing practice, and this survey's own corrections to it are printed beside it in section 5.6 rather than kept in the repository.

Three of those claims carry a case worth remembering. Hardware and software have come apart on a narrower claim than the hand count first suggests: 11 of the 19 hands that take no paper are neither sold nor open and take none for that reason, which is why the first claim is eight hands and not nineteen. A reward table in a paper is a claim about a document rather than about a run, and `physhoi_2023` is the instance to keep in mind, since the term its table weights at 0.1 is set to zero in the file at the commit this survey fetched and the success criterion the paper reports is itself position-only. And the obstacle to measuring contact is not the engines: IsaacGymEnvs ships a task that computes a per-environment maximum interpenetration depth and gates the policy update on a one-millimetre threshold, so the tooling sits in the field's own benchmark repository while the number goes unreported.

What this survey cannot establish is which method is better than which. It re-runs nothing, and Section 7 argues that the published numbers do not compare. Six works are cited by metadata only, and a seventh, Ma and Dollar 2011, is on disk but unread; no claim rests on any of them. Every coverage statistic here counts what this survey's extraction captured rather than what the literature reported. Each is a floor and not a rate, because every miss converts a reporting paper into a silent one.

Three things to do next week, cheapest first.

If you are publishing, generate the reward table from the config that trained the reported run, print it, and cite the commit. State the trial count, state the success predicate, and release the per-trial outcomes. None of that needs a GPU.

If you are running experiments, measure penetration depth over your evaluation rollouts on a dense surface sample, and never let that measure become a reward. `toporetaret_2026` shows what the number looks like when it is reported carefully, and what the widely used retargeters look like when it is not.

If you are choosing hardware, the corpus names four hands and no more. The Allegro carries 35 papers, the Shadow 21, the Inspire 19 and LEAP 12, and every other hand in Tables 2 and 3 carries eight rows or fewer. An announced hand has no URDF, no datasheet that can be checked, and no paper in this corpus that used it.

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## Appendix A. Method

The survey is written from a corpus that lives on disk. No claim about a paper is made from memory. Each claim traces to a note, each note to a parsed markdown file, and each markdown file to a PDF or repository with a recorded hash or commit.

### Selection

Six topic-specific bibliographies were assembled in parallel (simulators and physics, hands and vendors, reinforcement learning, imitation and human data, bimanual, benchmarks and evaluation), seeded with the canonical works in each area and extended by search up to 2026-09-17. Every arXiv identifier was checked by fetching the abstract page and matching the title, and entries that could not be checked that way are marked. The six lists were merged with deduplication on arXiv identifier and normalized title, giving 221 entries.

Seven further bibliography entries are not part of that corpus and are not counted anywhere in this survey. They are the prior audits and case studies section 1 positions this survey against, they carry the topic `related` in `corpus/bib_related.json`, and none of them carries a structured row. No source for them was parsed either, so each is quoted only from its abstract and its stated method.

### Acquisition

PDFs were downloaded from arXiv or, for work without a preprint, from the publisher or vendor page recorded in the bibliography. Vendor pages for hands without any paper were fetched as HTML and converted to markdown, so that a specification quoted in this survey is quoted from a stored copy of the page and dated. Repositories were shallow-cloned and parsed, then deleted. What remains is one markdown per repository holding the README, a pruned file tree, the task and reward configuration files, and the bodies of reward and observation functions.

### Parsing

PDFs were converted with `pymupdf4llm`, falling back to raw text extraction when the layout parse returned too little. The corpus manifest records, per entry, the source URL, page count, sha256 of the PDF and the converter used. Two consequences matter for reading this survey. Equations rendered as images do not survive conversion, so where a reward weight exists only inside a figure it is recorded as unreadable rather than guessed. Tables are flattened, so a number taken from a table is quoted with the table it came from and, where the flattening is ambiguous, the ambiguity is stated.

### Notes Each entry was read into a structured note under a fixed template: embodiment,

learning method, objective, contact handling, evaluation, reproducibility, stated limitations, and quotable claims. Notes were written only from the parsed files. Where the source is silent the note says "not stated", and nothing was inferred from the reviewer's prior knowledge of the work. For method papers the reward or loss was quoted from the paper and, separately, from the released code, so that disagreements between the two are visible rather than smoothed over. Those disagreements turned out to be common enough to become a finding in their own right. The reading into notes was done with language-model assistance, from the parsed files only. The template's conventions are the safeguard against what that can get wrong: a

quotation rather than a paraphrase, "not stated" rather than an inference, and a named source beside every value, so a note can be checked against the file it came from. Four fields carrying a headline claim were re-read against their sources afterwards, the trial counts, the success criteria, the unseen-object counts and the penetration field, and all four audits are below; the coverage statistics are reported as floors for the same reason.

## Sources that could not be obtained

Six works are cited by metadata only and no claim in this survey rests on their contents. Five are behind publisher paywalls with no author-hosted copy found on 2026-09-18: Okamura et al. 2000, Bicchi 2000 (IEEE T-RO), Piazza et al. 2019, Roa and Suarez 2015, and Butterfass et al. 2001 on DLR-Hand II. The sixth is Hwangbo et al. 2018 on RaiSim, whose hosted PDF returned no body. The freely circulating PDF often taken for Bicchi 2000 is a different work, a book chapter, and is listed separately.

Ma and Dollar 2011 is a seventh case with a different cause. The fetch that failed when its note was written succeeded on 2026-09-18, so a seven-page PDF and its hash are in the manifest, but no note has been read from it. It is cited by metadata only for that reason and not because the source is unavailable.

Three of the 221 bibliography entries carry no structured row. Two are the paywalled Bicchi 2000 and DLR-Hand II entries. The third is `bicchi_grasping_chapter_2001`, which was read into a note and quoted throughout but is a book chapter rather than a work with an embodiment, a method or a result to record in a row.

The reference list of the typeset edition is shorter than the corpus, and the two numbers are different quantities. It prints 208 entries: the 201 corpus entries that some sentence, table or figure of this paper cites, plus the 7 prior-work entries from outside the corpus. The other 20 corpus entries carry a row and a note and are counted in every statistic here, but no passage in the paper names them, so they have nothing to be cited from and do not appear in the list. This edition cites by key rather than by number and prints no list, so the place to count all 221 is `corpus/bib.json`. A reader counting the typeset reference list should get 208, and a reader counting the corpus should get 221.

## What this method cannot do

Mention counts over the corpus are counts of mentions, not of use: a related-work sentence counts the same as an experiment. Vendor specifications are manufacturer claims and are labeled as such throughout, and where a page has since gone offline the note says so. The corpus is large but not exhaustive, and selection by search favours work that is indexed, in English, and posted as a preprint. It also favours recent work. Of the 221 bibliography entries, 138 are dated 2024 or later and 16 predate 2018, so this is a corpus of the learned era and any claim here about a trend over time is a claim about 2022 onward.

Every coverage statistic in this survey measures what this survey's extraction captured, not what the literature reported. A structured row holds a scalar. A paper that reports a per-task count, a rubric, or a total spread across several tables produces a null, and a null is then counted as silence. The bias runs one way. Every miss converts a reporting paper into a silent one, so the field is made to look worse at reporting than it is. The size of the effect was measured on the statistic the survey leads with. Of the 34 method rows that had a real robot and no recorded trial count, 15 carried a count in plain text in their own note, dropped because the paper reports it per task and the field takes a single integer. Those 15 have since been re-extracted, which moved the stated-trial-count row from 55 to 70. The success criterion and the unseen-object count were audited the same way, rising from 79 to 98 and from 32 to 39. What those 19 recovered criteria say is mostly not a threshold: eleven score by rubric or staged partial credit, five judge binary completion against a task description by eye, two defer to a benchmark's own definition, and exactly one, `pistar06_2025`, states a verbatim numeric threshold. The audit counted only cases where the note itself carried the number, so a count the note also missed is still uncounted, and 19 trial counts, 8 unseen-object evaluations whose object count the note never gives, and 4 criteria remain genuinely unsettled. Fifteen of the recovered trial counts are named in the rows: `pi0_2024` at ten trials per task, `rdt1b_2024` at 139 across seven tasks, `umi_2024` at 260, `pistar06_2025` at 750, `gemini_robotics_2025` at twenty per task, and ten more, which is 44 percent of the audited nulls and moved the "never says" figure from 34 of 89 down to 19. Read every coverage statistic in this survey as a floor rather than as a rate, and read the bars in Figure 6 the same way.

**The penetration null, sampled.** A null is worth what the search behind it is worth, and this survey's extraction under-counted every other field it was audited against, by twenty to forty-five percent. The 85 method rows whose contact-handling field records that the work does not address penetration were therefore sampled: 25 of the 85, drawn at random with a fixed seed, each read again in its own parsed source and its released repository rather than in the note the field was written from, since the note is the artifact under suspicion. Not one of the 25 reports a measurement of penetration depth, intersection volume or physical plausibility on its own rollouts. Zero recoveries in 25 bounds the rows that could be hiding one at 8 of the 85 at 95 percent confidence, so the eleven is a floor and nineteen a ceiling; had this field under-counted at even the mildest rate the other audits found, a sample of 25 would have missed every recoverable row with probability 0.001. The nearest miss is worth naming, because it is the one a reader might count differently: `graspxl_2024` puts hand-object interpenetration to 35 human raters as one of four dimensions of a single realism score, which is a judgment of its own rollouts rather than a measurement of one, and counting it would give one recovery in 25 and a bound of 13 rows. Three near misses recurred across the sample and none of them is a measurement of a rollout: self-collision avoidance in a retargeter, a binary self-collision penalty in a released reward, and an engine's de-penetration velocity left at its default. The draw, the seed and the per-row results are in `reviews/penetration_audit.md`, and the two fields that remain unaudited are named above.

**The penetration field, audited the same way.** The same mechanism reaches the penetration, code-release and failure-mode fields. The penetration field has since been audited by hand, because this survey's second finding is a null over it and a null is worth what the search behind it is worth. The population is the 85 method rows whose contact-handling field records that the work does not address penetration. It excludes the 16 nulls, which are already counted as unsettled rather than as silence, so recovering one would not move the finding. Twenty-five of the 85, 29 percent, were drawn with `random.Random(20260919).sample` over the sorted population, so the draw is fixed and can be redrawn. Each sampled row was read again in `papers/md/`, in its OCR recovery where one exists, and in `code/md/` where a repository was parsed, but not in `papers/notes/`, because the note is the artifact under suspicion: a field is wrong exactly when the source says something the note did not carry. A regular expression over the whole source collected every occurrence of penetration, interpenetration, intersection, intersection volume, solid intersection, simulation displacement, contact consistency, physical plausibility, signed distance and contact depth, and of the words that are mistaken for them, and every hit was read in its context. A row counted as a recovery if its own source reported a measurement of any of those quantities on that method's own rollouts. A statement that penetration occurs, a citation whose title says physically plausible, a collision-avoidance constraint on a reference trajectory and a solver setting did not count, and the three recurring near misses of that kind are listed in section 7.2. Recoveries: none of the 25, against the 20 to 45 percent the three earlier audits recovered on the fields above. The one-sided 95 percent bound, computed hypergeometrically over the finite population, is therefore 8 of the 85, and section 7.2 states it beside the claim and names the one borderline case that a reader might count differently. Why this field held where the others did not is visible in the per-row results: the recovered trial counts and criteria were numbers present in the source and dropped by a field shaped to hold a scalar, whereas a penetration number is absent from the source altogether. The earlier audits measured a defect in this schema; this one looked for an absence in the literature. `reviews/penetration_audit.md` holds the seed, the sample and a result per row beside the sentence it rests on, and `tools/audit_penetration.py` redraws the sample and re-runs the sweep. The code-release and failure-mode fields have still not been audited this way, and their counts stay floors.

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## Appendix B. The full hand table with its sources

Table 2 and Table 3 split these rows by whether the hand can be bought, and truncate their cells to stay readable. This appendix prints every hand row once and at full width. Nothing here is a measurement by this survey. Every specification is a manufacturer claim, a figure in a paper, or a figure in a press article, and the `source quality` column in B.2 says which. Cells reproduce the text stored in the row, with the source's own punctuation and wording.

### B.1 Specifications as the source states them

| hand                        | maker                                                   | DoF | act. DoF | actuation                                                                                                                                                                                                                                                                                   | weight g | tip force N | payload | tactile                                                                                                                                           | control rate | price USD | BOM USD | open HW | licence | sim model                                                     | status        |
|-----------------------------|---------------------------------------------------------|-----|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------|---------|---------|---------|---------------------------------------------------------------|---------------|
| agibot_omni_hand_2025       |                                                         |     |          |                                                                                                                                                                                                                                                                                             |          |             |         |                                                                                                                                                   |              |           |         |         |         |                                                               |               |
| onex_neo_hand_2026          | 1X Technologies                                         | 25  | 25       | quasi-direct-drive tendons via the '1X Tendon Drive' at low gear ratios (~5:1-15:1); motors in the forearm, tendons pulled through the wrist                                                                                                                                                |          | 45          |         | high-resolution tactile sensing across fingertips and finger surfaces (normal force, contact location, shear); sensor count/technology not stated | not stated   |           |         | no      |         |                                                               | announced     |
| leap_hand_v2_adv_2025       | Kenneth Shaw, Deepak Pathak, Carnegie Mellon University | 21  | 17       | tendon (PIP/DIP coupled by a single tendon per finger); motor type not stated                                                                                                                                                                                                               |          |             |         | none mentioned                                                                                                                                    | not stated   | 3000      | 3000    | yes     |         | URDF, loaded into PyBullet by the released teleoperation node | announced     |
| paxini_dexh13_2024          | PaXini Tech                                             |     |          |                                                                                                                                                                                                                                                                                             |          |             |         | 1,140 ITPU multidimensional tactile processing units (press-release claim; unclear if PX-6AX-GEN3 sensor family)                                  |              |           |         |         |         |                                                               | announced     |
| clone_robotics_hand_2024    | Clone Robotics                                          | 27  |          | hydraulic artificial muscle ('Myofiber'; small water-filled tubes that contract when pressurized), carbon-fiber bones and ligament-style tethers, driven by a 500 W water pump and 36 electro-hydraulic valves; each Myofiber generates up to 1 kg grip force, survived 650,000 test cycles | <910     |             |         | pressure pads in the palm detect grip firmness (count not given); 70 inertial sensors track angle/speed (proprioception, not tactile)             |              |           |         | no      |         |                                                               | internal-only |
| sanctuary_phoenix_hand_2024 | Sanctuary AI (Sanctuary Cognitive)                      | 21  |          | hydraulic, 'unique miniaturized                                                                                                                                                                                                                                                             |          |             |         | not in the press release text; sidebar-                                                                                                           |              |           |         | no      |         |                                                               | internal-only |

| hand           | maker                                                                                                                                                                                                                 | DoF | act. DoF | actuation                                                                                                                                                                   | weight g | tip force N | payload                                                                                              | tactile                                                                                                                | control rate                                                        | price USD | BOM USD | open HW | licence                                                                                                                                | sim model                                                                        | status      |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------------|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|-----------|---------|---------|----------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|-------------|
|                | Systems Corporation)                                                                                                                                                                                                  |     |          | hydraulic valves'                                                                                                                                                           |          |             |                                                                                                      | only headlines mention new touch/tactile sensors as a separate later announcement , not confirmed as part of this hand |                                                                     |           |         |         |                                                                                                                                        |                                                                                  |             |
| bidexhand_2025 | Zhengyang Kris Weng, Center for Robotics and Biosystems, Northwestern University                                                                                                                                      | 21  | 16       | cable-driven, N-configuration tendon routing (FeeTech servos, endless-loop antagonistic pull-pull); thumb CMC flexion driven via a 4-bar linkage instead of a direct tendon |          | 2.14        | 4.54 kg (10 lb) lifted                                                                               | none mentioned                                                                                                         | serial to an onboard ESP32, ROS 2 position control; rate not stated |           |         | yes     | MIT (code, per README); CAD/hardware licence unconfirmed (bib's claimed CC BY-NC-SA is not verified in the parsed README or file tree) | URDF, consumed by the ROS 2 and MoveIt stack; no physics simulator               | open-source |
| ruka_v2_2026   | Xinqi (Lucas) Liu, Ruoxi Hu, Alejandro Ojeda Olarte, Zhuoran Chen, Kenny Ma, Charles Cheng Ji, Lerrel Pinto, Raunaq Bhirangi, Irmak Guzey, New York University and NYU Shanghai                                       | 20  | 16       | tendon-driven, forearm-mounted actuators; decoupled parallel 2-DoF wrist via a passive spherical ball joint; single-tendon-plus-spring abduction/adduction                  |          |             |                                                                                                      | optional e-flesh fingertips (magnetic touch-sensing form factor); not part of the base design                          | Dynamixel over USB serial; no rate stated                           | 1500      | 1500    | yes     |                                                                                                                                        | MuJoCo XML in the shared v1 repo; not confirmed to carry the v2 wrist            | open-source |
| orca_hand_2025 | Clemens C. Christoph, Maximilian Eberlein, Filippas Katsimalis, Arturo Roberti, Aristotelis Sympetheros, Michel R. Vogt, Davide Liconti, Chenyu Yang, Barnabas Gavin Cangan, Ronan J. Hinchet, Robert K. Katzschmann, | 17  | 17       | tendon-driven (antagonistic fishing-line tendon pairs per joint); wrist uses a GT2 timing belt drive                                                                        | 1200     |             | 10.5 kg on all four fingers and 2 kg on the index finger alone, both at a fixed 600 mA motor current | yes: FSR-based binary tactile sensing on all 5 fingertips (RP-C7.6-ST), absolute detection threshold as low as 0.05 N  | serial to Dynamixel or Feetech motors; rate not stated              |           |         | yes     |                                                                                                                                        | URDF-derived kinematic constants only; no URDF or MJCF file in the released tree | open-source |

| hand                     | maker                                                                                                                     | DoF | act. DoF | actuation                                                                                                                                                       | weight g | tip force N | payload                                                                                                          | tactile                                                                                                        | control rate                                                                            | price USD | BOM USD | open HW | licence                                                                   | sim model                                                                                  | status      |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------|-----|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------------|------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|-----------|---------|---------|---------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|-------------|
|                          | Soft Robotics Lab, ETH Zurich                                                                                             |     |          |                                                                                                                                                                 |          |             |                                                                                                                  |                                                                                                                |                                                                                         |           |         |         |                                                                           |                                                                                            |             |
| leap_hand_2023           | Kenneth Shaw, Ananye Agarwal, Deepak Pathak, Carnegie Mellon University                                                   | 16  | 16       | direct-drive (Dynamixel servos, e.g. XC330-M288), joint velocity ~8 rad/s                                                                                       | 595      | 19.5        |                                                                                                                  | none (future work only: 'we plan to develop and integrate LEAP Hand with low-cost touch sensors')              | up to 500 Hz querying over USB serial; the paper's own sim-to-real policy runs at 20 Hz | 2000      | 2000    | yes     | Code: MIT License; CAD: CC BY-NC-SA (non-commercial use with attribution) | URDF claimed in the paper; the released LEAP_Hand_API repo contains no URDF, xacro or MJCF | open-source |
| ruka_2025                | Anya Zorin, Irmak Guzey, Billy Yan, Aadithya Iyer, Lisa Kondrich, Nikhil X. Bhattasali, Lerrel Pinto, New York University | 15  | 11       | tendon-driven (11 Dynamixel actuators in the forearm: XM430-W210T for the thumb, XL330-M288-T for the other fingers)                                            |          | 2.74        | 6.0 kg: weight added to a curled cloth bag grip until joint-angle error exceeds 15 degrees, best of three trials | none (stated limitation: 'lacks tactile sensing')                                                              | Dynamixel over a USB-to-serial bridge; rate not stated, data collection ran at 15 Hz    | 1300      | 1300    | yes     |                                                                           | MJCF                                                                                       | open-source |
| dexhand_open_source_2023 | Rob Knight, The Robot Studio (design); electronics/firmware/ROS 2 by Trent Shumay, IoT Design Shop                        |     |          | tendon (fishing line, Sufix 832 80lb / 0.8mm kilteline); Emax ES3301/ES3302/ES3351/ES3352 micro-servos for fingers, Feetech SCS2332 or PWM servos for the wrist |          |             |                                                                                                                  | none mentioned                                                                                                 | Arduino SCServo or SBUS firmware, BLE; rate not stated                                  | 300       | 300     | yes     | CC BY-NC-SA 4.0 (README); Onshape CAD freely available                    | URDF, in the IoT Design Shop ROS 2 packages                                                | open-source |
| ilda_hand_2021           | Ajou University, Korea Institute of Machinery & Materials, Korea University                                               | 20  | 15       | linkage-driven, direct linear drive (3 Maxon DCX8M motors per finger with GPX8 16:1 gearboxes driving ball screws via parallel+serial four-bar linkages)        | 1100     | 34          | 18 kg                                                                                                            | 6-axis F/T sensor per fingertip (5 total), force resolution 62 mN, range +/-35 N; not a distributed taxel skin | CAN to a desktop; loop rate not stated                                                  |           |         | no      |                                                                           |                                                                                            | prototype   |
| pisa_iit_softhand_2014   | Centro E. Piaggio, University of Pisa and IIT (Catalano, Grioli, Farnioli, Serio, Piazza, Bicchi)                         | 19  | 1        | tendon-driven adaptive synergy, single motor via differential gears (6 W Maxon RE-max21, 84:1 reduction)                                                        |          |             | holding force about 20 N along z and holding torque 2 N m, 28 N and 3.5 N m with a stronger motor                | none (motor encoder only; a padded work glove supplied contact compliance during experiments)                  | not stated                                                                              |           |         | no      |                                                                           | URDF and Gazebo models in the ROS repo (BSD 3-Clause)                                      | prototype   |

| hand                            | maker                                                                                                                                                          | DoF | act. DoF | actuation                                                                                                                                                           | weight g | tip force N | payload                                              | tactile                                                                                                                                                         | control rate                                                                             | price USD | BOM USD | open HW | licence | sim model                                                   | status    |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------------|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|-----------|---------|---------|---------|-------------------------------------------------------------|-----------|
| faive_hand_2023                 | Yasunori Toshimitsu, Benedek Forrai, Barnabas Gavin Cangan, Ulrich Steger, Manuel Knecht, Stefan Weirich, Robert K. Katzschmann, Soft Robotics Lab, ETH Zurich | 16  | 11       | tendon-driven, rolling-contact joints (16 Dynamixel XC330-T288-T servos, 6 antagonistic pairs)                                                                      | 1100     |             | 10 kg, whole-hand downward power grasp on a dumbbell | none on the physical hand; a simulated 'fingertip force' (15-dim, critic-only/privileged) exists only in the RL training observation, not real tactile hardware | Dynamixel over serial; joint angles estimated from tendon length by EKF; rate not stated |           |         |         |         | MJCF, used directly by the released Isaac Gym training repo | prototype |
| boston_dynamics_atlas_hand_2026 | Boston Dynamics                                                                                                                                                |     |          | electric (robot is fully electric); specific hand mechanism not stated                                                                                              |          |             |                                                      | tactile sensing in the fingers and palms, confirmed present; type and count not stated                                                                          |                                                                                          |           |         | no      |         |                                                             | prototype |
| daxo_muscle_v0_2025             | Daxo Robotics                                                                                                                                                  |     |          | ultra-redundant tendon-driven; compliant structure with no rigid joints, flexible materials and tendon routing                                                      | 750      |             |                                                      |                                                                                                                                                                 |                                                                                          |           |         |         |         |                                                             | prototype |
| figure_03_hand_2025             | Figure AI                                                                                                                                                      |     |          |                                                                                                                                                                     |          |             |                                                      | not stated on page for the hand itself; palm cameras in each hand are vision, not tactile                                                                       |                                                                                          |           |         | no      |         |                                                             | prototype |
| tesla_optimus_hand_2025         | Tesla                                                                                                                                                          |     |          | V3: tendon-driven from forearm actuators, three tendons per finger through a crosstalk-managing wrist. Gen 2: in-hand actuators/sensors (unlocated in the article). |          |             |                                                      | Gen 2: tactile sensing on all fingers, demonstrated matching contact location and force in an egg-pickup video; type/count not stated. V3: not stated.          |                                                                                          |           |         | no      |         |                                                             | prototype |
| xiaomi_cyberone_hand_2026       | Xiaomi                                                                                                                                                         |     |          | motors located in the hand (compact motors generate heat                                                                                                            |          |             |                                                      | full-palm tactile sensing, area ~8,200 sq mm, detects                                                                                                           |                                                                                          |           |         | no      |         |                                                             | prototype |

| hand                       | maker                            | DoF | act. DoF | actuation                                                                                                                                | weight g | tip force N | payload                                                  | tactile                                                                                                                                                                                                                              | control rate                                                                            | price USD | BOM USD | open HW | licence | sim model                                                        | status |
|----------------------------|----------------------------------|-----|----------|------------------------------------------------------------------------------------------------------------------------------------------|----------|-------------|----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|-----------|---------|---------|---------|------------------------------------------------------------------|--------|
|                            |                                  |     |          | requiring liquid cooling); no tendon/linkage statement, though contrast with tendon-driven hands' cycle life implies a non-tendon design |          |             |                                                          | pressure and contact across the whole palm not just fingertips; taxel count and type not stated                                                                                                                                      |                                                                                         |           |         |         |         |                                                                  |        |
| shadow_dexterous_hand_2005 | Shadow Robot Company             | 24  | 20       | tendon-driven, motors in the forearm (20 Smart Motor nodes, Maxon motors, PWM)                                                           | 4300     |             | 4 kg in a power grasp, vendor claim, protocol not stated | Shadow Tactile Fingertips (STF): 17x3 DoF Hall-effect taxels, 1000 Hz, uncalibrated; up to 5 fingertips; software also supports BioTac/MST/U BIO/PST fingertip options                                                               | EitherCAT at 1 kHz to the host, with a 5 kHz tendon-force loop inside each motor module |           |         | no      |         | MuJoCo models for Hand E and variants, plus Gazebo, in sr_common | sold   |
| proception_prohand_2026    | Proception Inc (YC W25)          | 22  |          | tendon-driven, motors pull cables to move the fingers                                                                                    |          |             |                                                          | integrated skin-like sensors detecting contact, supporting grip control; same 'sensor skin' used on the companion ProGlove; type/count not stated                                                                                    |                                                                                         |           |         | no      |         |                                                                  | sold   |
| sharpa_wave_2026           | Sharpa Robotics (Sharpa Pte Ltd) | 22  | 22       |                                                                                                                                          | 1300     | 20 / 12     | 40 kg / 24 kg (the same two unlabelled columns)          | 'Dynamic Tactile Array' (DTA), camera-type array at the fingertip: resolution 240x240 / 60x60, force resolution 0.02 N / 0.05 N, spatial resolution 1 mm / 2 mm, frame rate 180 fps / 30 fps, force range 0-30 N, latency 20 ms, 6-D | 500 Hz over 1000BASE-T Ethernet                                                         |           |         | no      |         | MuJoCo and Isaac Sim models                                      | sold   |

| hand                 | maker                                                                                          | DoF | act. DoF | actuation                                                                                                                                                              | weight g | tip force N | payload                                                                          | tactile                                                                                                                                     | control rate                                                                    | price USD | BOM USD | open HW | licence | sim model                                              | status |
|----------------------|------------------------------------------------------------------------------------------------|-----|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------------|----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|-----------|---------|---------|---------|--------------------------------------------------------|--------|
|                      |                                                                                                |     |          |                                                                                                                                                                        |          |             |                                                                                  | F/T; number of sensors per hand not stated                                                                                                  |                                                                                 |           |         |         |         |                                                        |        |
| tesollo_dg5f_2024    | Tesollo Inc. (Incheon HQ, Gwangmyeong R&D/factory)                                             | 20  | 20       | one integrated actuator per joint ('high-torque actuation', absolute encoder); page does not say direct-drive                                                          | 1763     |             | pinching 2.5 kg rated and 5 kg maximum; enveloping 10 kg rated and 20 kg maximum | none standard; optional fingertip sensors (6-axis F/T, 3-axis force, or tactile) available; count/type not given                            | 250 Hz over Modbus RTU or TCP and Ethernet TCP/IP                               |           |         | no      |         |                                                        | sold   |
| unitree_dex5_2025    | Unitree Robotics (Yushu Technology Co., Ltd.)                                                  | 20  | 16       | in-joint geared motor ('hollow-cup motor' + high-precision encoder + low-damping small-clearance reducer), backdrivable; page does not use the words tendon or linkage | 1100     | 10          | 3.5 kg palm down and 4.5 kg palm left, on a 5 cm round hard object               | Dex5-1: none. Dex5-1P: 94 pressure sensors per hand (2x5 palm + 2x3x5 finger pad + 2x3x5 fingertip + 2x3x4 finger root), range 10 g-2500 g. | 1000 Hz over USB 2.0, with per-joint stiffness and damping commands             |           |         | no      |         |                                                        | sold   |
| wuji_hand_2025       | Wuji Technology (founded 2019)                                                                 | 20  | 20       | direct-drive rotary, backdrivable                                                                                                                                      |          |             |                                                                                  | 'Multi-Axis Force/Torque Fingertip Sensing' confirmed present; no type, count or range given                                                | 1000 Hz across 20 axes over 100BASE-TX Ethernet, MIT force-position hybrid mode |           |         | no      |         | MuJoCo and Isaac Sim models                            | sold   |
| allegro_hand_v4_2016 | Wonik Robotics Co. Ltd. (Seoul, South Korea); earlier versions 1.0/2.0 made by SimLab Co. Ltd. | 16  | 16       |                                                                                                                                                                        |          |             |                                                                                  | none                                                                                                                                        | 333 Hz, CAN, the hand's own real-time clock (driver README)                     |           |         | no      |         | URDF and xacro, left and right, in the ROS driver repo | sold   |
| inspire_rh56dfx_2023 | Beijing Inspire Robots Technology Co., Ltd.                                                    | 12  | 6        |                                                                                                                                                                        | 540      | 10          |                                                                                  | none (this variant is the 'without tactile sensors' table; a tactile FTP variant exists per bib but is not on this page)                    | RS485; rate not stated                                                          |           |         | no      |         |                                                        | sold   |
| robotera_xhand1_2024 | ROBOTERA                                                                                       | 12  | 12       | gear-driven force-controlled joint                                                                                                                                     | 1100     |             | 25 kg, tracker's 'Strength'                                                      | tactile/force sensors on every fingertip                                                                                                    | not stated                                                                      | 14000     |         | no      |         | URDF exists but is licensed; maniptrans_               | sold   |

| hand                       | maker                                                       | DoF | act. DoF | actuation                                                                                                   | weight g | tip force N | payload                             | tactile                                                                                                                                                                                                    | control rate                                                                   | price USD | BOM USD | open HW | licence | sim model                                                                                   | status |
|----------------------------|-------------------------------------------------------------|-----|----------|-------------------------------------------------------------------------------------------------------------|----------|-------------|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|-----------|---------|---------|---------|---------------------------------------------------------------------------------------------|--------|
|                            |                                                             |     |          | modules per finger segment, back-drivable                                                                   |          |             | figure, over 25 kg gripping palm-up | confirmed present (senses contact, force, temperature); count and array size not stated                                                                                                                    |                                                                                |           |         |         |         | 2025 withholds it                                                                           |        |
| shadow_dex_ee_2024         | Shadow Robot Company, in collaboration with Google DeepMind | 12  |          |                                                                                                             | 4100     |             |                                     | stereo camera-based fingertip tactile sensors (hundreds of taxels each); multi-taxel 3-DoF arrays on middle and proximal phalanges; counts not stated                                                      | not stated; the page claims 'high bandwidth torque and position control loops' |           |         | no      |         |                                                                                             | sold   |
| brainco_rev_o2_2025        | BrainCo Inc.                                                | 11  | 6        |                                                                                                             | 383      | 15          | >=20 kg                             | Touch variant only: multi-dimensional fingertip tactile module with 'Tactile Adaptive Control'; count/type not stated. Basic and Pro variants: none.                                                       | RS485 and CAN FD; EtherCAT on Pro and Touch; rate not stated                   |           |         | no      |         |                                                                                             | sold   |
| linkerbot_l20_2025         | LinkerBot (SDK author 'CHIUS INC')                          |     |          |                                                                                                             |          |             |                                     | L10/L20: per finger 4 channels (normal force, tangential force, tangential direction, proximity), 5 sites x 4 channels, 0-255 range, sensor type not stated. O6 (reseller): 16 capacitive tactile regions. | USB-to-CAN at 1 Mbit/s, optional Modbus or RS485; rate not stated              |           |         | no      |         | URDF, with PyBullet and Isaac Gym examples in the SDK                                       | sold   |
| psyonic_abi_lity_hand_2021 | PSYONIC (San Diego, CA, USA)                                |     | 6        | linkage (repo file/function names indicate a four-bar finger linkage; not stated on the vendor page itself) |          |             |                                     | 30 touch sensor values streamed (FSR-based per API and URDF 'no_fsr' variant); per-finger split not stated                                                                                                 | BLE, I2C, UART or RS485; rate not stated                                       |           |         | no      |         | URDF (left, right, large, small, no-FSR) plus MuJoCo and Isaac Sim paths in the vendor repo | sold   |

33 rows. 216 of 528 cells (40 percent) are values no source stated. A blank price is a hand with no public list price rather than a free hand.

## B.2 Where each specification came from

| hand                        | source quality                 | claim date                                                            | caveat recorded in the row                                                                                                                                                            | source URL                                                                                                        | parsed source in the manifest       | note                                        | corpus method rows running it                                                                                                                                                                                                  |
|-----------------------------|--------------------------------|-----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|-------------------------------------|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| agibot_omnihand_2025        | unavailable                    | no source: both store.agibot.com URLs 404, fetched 2026-09-17         |                                                                                                                                                                                       |                                                                                                                   | no manifest entry                   | papers/notes/agibot_omnihand_2025.md        | 1: clutterdexgrasp_2025                                                                                                                                                                                                        |
| onex_neo_hand_2026          | vendor page                    | vendor page, 2026-07-09                                               |                                                                                                                                                                                       |                                                                                                                   | no manifest entry                   | papers/notes/onex_neo_hand_2026.md          | 0                                                                                                                                                                                                                              |
| leap_hand_v2_adv_2025       | vendor page                    | undated project page; API repo at commit a0936196, fetched 2026-09-17 | the page claims 21 DoF with 17 powered motors and the README calls the same hand 17-DOF                                                                                               | <a href="https://github.com/leap-hand/LEAP_Hand_V2_Adv_API">https://github.com/leap-hand/LEAP_Hand_V2_Adv_API</a> | no manifest entry                   | papers/notes/leap_hand_v2_adv_2025.md       | 2: bidex_teleop_2024, dexwild_2025                                                                                                                                                                                             |
| paxini_dexh13_2024          | press                          | press, CES 2026-01-07                                                 |                                                                                                                                                                                       |                                                                                                                   | no manifest entry                   | papers/notes/paxini_dexh13_2024.md          | 0                                                                                                                                                                                                                              |
| clone_robotics_hand_2024    | press                          | press, 2025-11-24, on a video of 2025-11-15                           | 'under 2 pounds' is an upper bound, and the source does not say whether the 500 W pump and 36 valves are inside it                                                                    |                                                                                                                   | no manifest entry                   | papers/notes/clone_robotics_hand_2024.md    | 0                                                                                                                                                                                                                              |
| sanctuary_phoenix_hand_2024 | press                          | undated press release; trade reprint 2024-12-17                       |                                                                                                                                                                                       |                                                                                                                   | no manifest entry                   | papers/notes/sanctuary_phoenix_hand_2024.md | 0                                                                                                                                                                                                                              |
| bidexhand_2025              | paper                          | paper, 2025 (ICRA Dexterity workshop abstract)                        | the paper states 16 independently actuated DoF and 21 joints; the README states 15 servos driving 15 joints                                                                           | <a href="https://github.com/wengmister/BiDexHand">https://github.com/wengmister/BiDexHand</a>                     | 5 pp, 89f39715, fetched 2026-09-17  | papers/notes/bidexhand_2025.md              | 0                                                                                                                                                                                                                              |
| ruka_v2_2026                | paper                          | paper, 2026                                                           | the AS5600 encoders are attachable and detachable and are read for calibration and measurement only, not in the control loop                                                          | <a href="https://github.com/ruka-hand/RUKA">https://github.com/ruka-hand/RUKA</a>                                 | 20 pp, d71d6f90, fetched 2026-09-17 | papers/notes/ruka_v2_2026.md                | 0                                                                                                                                                                                                                              |
| orca_hand_2025              | paper                          | paper, 2025                                                           | the 19.6 N previously tabulated as a fingertip force is the newton equivalent of the 2 kg index-finger payload at a control-imposed current limit, and no source states it as a force | <a href="https://github.com/orcahand/orca_core">https://github.com/orcahand/orca_core</a>                         | 8 pp, f3967f2e, fetched 2026-09-17  | papers/notes/orca_hand_2025.md              | 0                                                                                                                                                                                                                              |
| leap_hand_2023              | paper                          | paper, 2023                                                           |                                                                                                                                                                                       | <a href="https://github.com/leap-hand/LEAP_Hand_API">https://github.com/leap-hand/LEAP_Hand_API</a>               | 11 pp, ac08a57e, fetched 2026-09-17 | papers/notes/leap_hand_2023.md              | 12: bidex_teleop_2024, bidexhd_2024, cross_embodiment_world_models_2025, dexcap_2024, dexndm_2025, dexterous_functional_grasping_2023, dextrack_2025, dexwild_2025, dreureka_2024, teledexter_2026, unidex_2026, videodex_2022 |
| ruka_2025                   | paper                          | paper, 2025                                                           |                                                                                                                                                                                       | <a href="https://github.com/ruka-hand/RUKA">https://github.com/ruka-hand/RUKA</a>                                 | 13 pp, 3a19161f, fetched 2026-09-17 | papers/notes/ruka_2025.md                   | 0                                                                                                                                                                                                                              |
| dexhand_open_source_2023    | project page and GitHub README | page posts 2023-08-08 to 2023-10-01; no hardware release date         | no joint count is given anywhere, so no DoF figure can be quoted; the servo count is 16 finger and thumb micro-servos plus 2 wrist servos, with a third                               | <a href="https://github.com/TheRobotStudio/V1.0-Dexhand">https://github.com/TheRobotStudio/V1.0-Dexhand</a>       | no manifest entry                   | papers/notes/dexhand_open_source_2023.md    | 0                                                                                                                                                                                                                              |

| hand                            | source quality      | claim date                                                                                   | caveat recorded in the row                                                                                                                                                                                                                                | source URL                                                                                                                                                                                                                                    | parsed source in the manifest       | note                                            | corpus method rows running it                                                                                                                                                                      |
|---------------------------------|---------------------|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                 |                     |                                                                                              | optional. The \$300 is 'additional total cost of components', excluding printing and the wrist servos                                                                                                                                                     |                                                                                                                                                                                                                                               |                                     |                                                 |                                                                                                                                                                                                    |
| ilda_hand_2021                  | paper               | paper, accepted 2021-11-05                                                                   |                                                                                                                                                                                                                                                           | <a href="https://www.nature.com/articles/s41467-021-27261-0.pdf">https://www.nature.com/articles/s41467-021-27261-0.pdf</a>                                                                                                                   | 13 pp, ed0e19c2, fetched 2026-09-17 | papers/notes/ilda_hand_2021.md                  | 0                                                                                                                                                                                                  |
| pisa_iit_softhand_2014          | paper               | paper, IJRR 2014                                                                             |                                                                                                                                                                                                                                                           | <a href="https://www.centropiaggio.unipi.it/sites/default/files/PisaIIT_Soft_Hand_0.pdf">https://www.centropiaggio.unipi.it/sites/default/files/PisaIIT_Soft_Hand_0.pdf</a>                                                                   | 14 pp, 9c9cd17f, fetched 2026-09-17 | papers/notes/pisa_iit_softhand_2014.md          | 0                                                                                                                                                                                                  |
| faive_hand_2023                 | paper               | paper, 2023                                                                                  | the row's open-hardware status is unconfirmed: the repo has LICENSE and LICENSE-NVIDIA files whose contents were not captured, and no open-hardware claim appears in the sources                                                                          | <a href="https://github.com/srl-ethz/faive_gym_oss">https://github.com/srl-ethz/faive_gym_oss</a>                                                                                                                                             | 7 pp, 13feb753, fetched 2026-09-17  | papers/notes/faive_hand_2023.md                 | 1: graspxl_2024                                                                                                                                                                                    |
| boston_dynamics_atlas_hand_2026 | press               | trade press, 2026-01-05 (CES); the Boston Dynamics blog of the same date has no hand content |                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                               | no manifest entry                   | papers/notes/boston_dynamics_atlas_hand_2026.md | 0                                                                                                                                                                                                  |
| daxo_muscle_v0_2025             | third-party tracker | third-party catalogue post, 2026-09-16; daxo-robotics.com returned HTTP 404                  | 120 actuators in 750 g is about 6 g per actuator including structure, tendons, routing and skin; no DoF number is stated, and a redundant tendon architecture with no rigid joints has fewer kinematic DoF than actuators                                 |                                                                                                                                                                                                                                               | no manifest entry                   | papers/notes/daxo_muscle_v0_2025.md             | 0                                                                                                                                                                                                  |
| figure_03_hand_2025             | third-party tracker | undated tracker page, fetched 2026-09-17; figure.ai launch page returned HTTP 404            | the tracker's 'Degrees of freedom, hands / 20' sits beside 'Number of fingers / 10', so the 20 is most likely the pair and about 10 per hand; the 16-per-hand figure in circulation has no reachable source                                               |                                                                                                                                                                                                                                               | no manifest entry                   | papers/notes/figure_03_hand_2025.md             | 0                                                                                                                                                                                                  |
| tesla_optimus_hand_2025         | press               | Gen 2 video 2023-12-13; V3 patents filed 2024-10, relayed by press 2026-04                   | the 22 DoF in circulation is a Teslarati article's arithmetic of four DoF on each of five fingers plus two at the wrist, not a Tesla statement; Gen 2's own figure is 11 DoF. This row's actuation cell describes V3 and its tactile cell describes Gen 2 | <a href="https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/9505134">https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/9505134</a>                                                                               | 20 pp, 93664ecb, fetched 2026-09-17 | papers/notes/tesla_optimus_hand_2025.md         | 0                                                                                                                                                                                                  |
| xiaomi_cyberone_hand_2026       | press               | press, 2026-03-30                                                                            |                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                               | no manifest entry                   | papers/notes/xiaomi_cyberone_hand_2026.md       | 0                                                                                                                                                                                                  |
| shadow_dexterous_hand_2005      | datasheet           | specification dated 2024-12-04                                                               | the four distal joints are coupled and not independently controllable, so the '24 DoF' quoted in learning papers is the joint count                                                                                                                       | <a href="https://shadowrobot.com/wp-content/uploads/2024/12/shadow_dexterous_hand_e_technical_specification_20241204.pdf">https://shadowrobot.com/wp-content/uploads/2024/12/shadow_dexterous_hand_e_technical_specification_20241204.pdf</a> | 11 pp, 66541d02, fetched 2026-09-17 | papers/notes/shadow_dexterous_hand_2005.md      | 21: asymdex_2024, bidexgrasp_2026, bimangrasp_2024, cross_embodiment_world_models_2025, dexdeform_2023, dexman_2025, dp3_2024, eureka_2023, graspxl_2024, maniptrans_2025, mjpc_2022, objdex_2024, |

| hand                    | source quality | claim date                                                 | caveat recorded in the row                                                                                                                                                                    | source URL                                                                                                                                                                          | parsed source in the manifest | note                                    | corpus method rows running it                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------|----------------|------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         |                |                                                            |                                                                                                                                                                                               |                                                                                                                                                                                     |                               |                                         | openai_dexterity_2018,<br>openai_rubiks_cube_2019,<br>pddm_2019, pgdm_2023,<br>pianomime_2024,<br>resdex_2024, unidex_2026,<br>unidexgrasp_2023,<br>unidexgrasp_pp_2023                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| proception_prohand_2026 | press          | launch post and press, 2026-06-29                          | the evidence is a shipping announcement, not a datasheet: no weight, force, actuated-DoF count or control rate is stated anywhere                                                             |                                                                                                                                                                                     | no manifest entry             | papers/notes/proception_prohand_2026.md | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| sharpa_wave_2026        | vendor page    | undated page, copyright 2026, fetched 2026-09-17           | the page footnotes 'Specifications may vary between products', and no column of the spec table is labelled                                                                                    |                                                                                                                                                                                     | no manifest entry             | papers/notes/sharpa_wave_2026.md        | 6: dexteleop0_2026,<br>egoscale_2026, metis_2025,<br>poise_2026,<br>simtoolreal_2026,<br>teledexter_2026                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| tesollo_dg5f_2024       | vendor page    | undated page, copyright 2026, download links dated 2026-08 |                                                                                                                                                                                               | <a href="https://en.tesollo.com/wp-content/uploads/2025/02/TESOLLO-DG-5F-catalogue-En-2.pdf">https://en.tesollo.com/wp-content/uploads/2025/02/TESOLLO-DG-5F-catalogue-En-2.pdf</a> | no manifest entry             | papers/notes/tesollo_dg5f_2024.md       | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| unitree_dex5_2025       | datasheet      | undated page, copyright 2016-2025, fetched 2026-09-17      |                                                                                                                                                                                               | <a href="https://github.com/unitreerobotics/unitree_sdk2">https://github.com/unitreerobotics/unitree_sdk2</a>                                                                       | no manifest entry             | papers/notes/unitree_dex5_2025.md       | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| wuji_hand_2025          | vendor page    | undated page, copyright 2026, fetched 2026-09-17           | the page footnotes that these figures 'reflect the current Beta1 product and the SPEC will continue to iterate'                                                                               |                                                                                                                                                                                     | no manifest entry             | papers/notes/wuji_hand_2025.md          | 2: toporetargert_2026,<br>unidex_2026                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| allegro_hand_v4_2016    | driver repo    | driver repo, versions 1.0-4.0 undated, fetched 2026-09-17  | allegrohand.com/v4 returned HTTP 404 and the Wonik wiki timed out; every cell here comes from the ROS driver repository, and weight, joint torque, payload and price have no reachable source | <a href="https://github.com/simlabrobotics/allegro_hand_ros_v4">https://github.com/simlabrobotics/allegro_hand_ros_v4</a>                                                           | no manifest entry             | papers/notes/allegro_hand_v4_2016.md    | 35: anyrotate_2024,<br>anyteleop_2023,<br>asymdex_2024,<br>castro_sap_contact_2021,<br>cross_embodiment_world_models_2025,<br>cyberdemo_2024,<br>dexmachina_2025,<br>dexpbt_2023,<br>dexpilot_2020,<br>dexplore_2025,<br>dexpoint_2022,<br>dexteritygen_2025,<br>dexterous_handover_2025,<br>dextrack_2025,<br>dextreme_2022, dime_2022,<br>dp3_2024,<br>dynamic_handover_2023,<br>eureka_2023,<br>geometric_retargeting_2025, grasppl_2024,<br>holo_dex_2022, hora_2022,<br>hudor_2024,<br>human2sim2robot_2025,<br>maniptrans_2025,<br>pang_global_planning_2022, penspin_2024,<br>robot_synesthesia_2023,<br>rotateit_2023,<br>rotating_without_seeing_2023, twisting_lids_2024, |

| hand                      | source quality      | claim date                                                                | caveat recorded in the row                                                                                                                                                     | source URL                                                                                                  | parsed source in the manifest | note                                      | corpus method rows running it                                                                                                                                                                                                                                                                                                                                                                        |
|---------------------------|---------------------|---------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|-------------------------------|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           |                     |                                                                           |                                                                                                                                                                                |                                                                                                             |                               |                                           | unidex_2026,<br>viserdex_2026,<br>wm_dex_human_videos_2025                                                                                                                                                                                                                                                                                                                                           |
| inspire_rh56dfx_2023      | vendor page         | undated page, fetched 2026-09-17                                          | the vendor's own page reads 'Degrees of freedom 6, Numbers of joints 12'; the joints column here prints the joint count, so the vendor's DoF figure is the actuated-DoF column |                                                                                                             | no manifest entry             | papers/notes/inspire_rh56dfx_2023.md      | 19: ace_teleop_2024,<br>articulated_tools_inhand_2025, being_h05_2026,<br>being_h0_2025,<br>bidexgrasp_2026,<br>dexmachina_2025,<br>dexmimicgen_2024,<br>dexplore_2025,<br>dexumi_2025, dexvla_2025,<br>humanoid_policy_human_policy_2025,<br>humanoid_sim2real_recipe_2025, humanplus_2024,<br>maniptrans_2025,<br>metis_2025, okami_2024,<br>omnih2o_2024,<br>open_television_2024,<br>unidex_2026 |
| robotera_xhand1_2024      | third-party tracker | undated tracker page, fetched 2026-09-17; the vendor page was not fetched | the 15 N in circulation comes from a survey table, not from the tracker or the vendor; the tracker states no fingertip force                                                   |                                                                                                             | no manifest entry             | papers/notes/robotera_xhand1_2024.md      | 8:<br>cross_embodiment_world_models_2025, deximit_2026,<br>dexmachina_2025,<br>dexora_2026, dexumi_2025,<br>dp3_2024,<br>force_grasp_sim2real_2026, unidex_2026                                                                                                                                                                                                                                      |
| shadow_dex_ee_2024        | vendor page         | undated page, copyright 2026, fetched 2026-09-17                          |                                                                                                                                                                                |                                                                                                             | no manifest entry             | papers/notes/shadow_dex_ee_2024.md        | 1: demostart_2024                                                                                                                                                                                                                                                                                                                                                                                    |
| brainco_revo2_2025        | datasheet           | undated page, fetched 2026-09-17                                          |                                                                                                                                                                                |                                                                                                             | no manifest entry             | papers/notes/brainco_revo2_2025.md        | 1: bidexgrasp_2026                                                                                                                                                                                                                                                                                                                                                                                   |
| linkerbot_l20_2025        | vendor page         | undated page, fetched 2026-09-17                                          |                                                                                                                                                                                | <a href="https://github.com/linkerbotai/linker_hand_sdk">https://github.com/linkerbotai/linker_hand_sdk</a> | no manifest entry             | papers/notes/linkerbot_l20_2025.md        | 1: being_h05_2026                                                                                                                                                                                                                                                                                                                                                                                    |
| psyonic_ability_hand_2021 | vendor page         | undated page, fetched 2026-09-17                                          | the four-bar finger linkage rests on repository file names and the five fingers on URDF meshes; the vendor page states neither                                                 | <a href="https://github.com/psyonicinc/ability-hand-api">https://github.com/psyonicinc/ability-hand-api</a> | no manifest entry             | papers/notes/psyonic_ability_hand_2021.md | 7: ace_teleop_2024,<br>asymdex_2024,<br>bunny_visionpro_2024,<br>cross_embodiment_world_models_2025,<br>dexmachina_2025,<br>hato_visuotactile_2024,<br>unidex_2026                                                                                                                                                                                                                                   |

33 rows. 31 of 264 cells (11 percent) are values no source stated. `parsed source in the manifest` gives page count, the first eight characters of the sha256 of the stored copy, and the date it was fetched. A hand with no manifest entry was recorded from a page that could not be stored, and its row should be read as a claim with no retrievable source.

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## Appendix C. Per-paper reward term extraction

Table 5 marks nine recurring term families across the in-hand reorientation methods. This appendix is the extraction underneath it, over every method row that records a reward at all. `terms stated` counts the terms the paper itself names. `where the reward was read` gives the section of the paper or the file in the released code that the note quotes. A row with a `paper/code class` is one where the two disagree, and C.2 prints the disagreement in full. Cells and entries reproduce the text stored in the row, with the source's own punctuation and wording.

### C.1 The extraction

| method                             | yr   | paradigm         | terms stated | term names as the paper names them                                                                                    | where the reward was read                                                           | in reward matrix | code released | paper/code class       | confidence |
|------------------------------------|------|------------------|--------------|-----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|------------------|---------------|------------------------|------------|
| openai_dexterity_2018              | 2018 | RL               | 3            |                                                                                                                       | Method, reward (Sec 4.2 / App C.1)                                                  | yes              | no            |                        |            |
| openai_rubiks_cube_2019            | 2019 | RL, distillation | 3            |                                                                                                                       | Method, reward (Sec. 6.1)                                                           | yes              | no            |                        |            |
| pddm_2019                          | 2019 | MPC              |              |                                                                                                                       |                                                                                     |                  | yes           | contradiction          | high       |
| dexpoint_2022                      | 2022 | RL               | 4            | reach, contact, lift, action penalty                                                                                  |                                                                                     |                  | yes           | parse-limitation       | low        |
| dextreme_2022                      | 2022 | RL               | 6            |                                                                                                                       | Reward block, paper Table 2 against allegro_hand_dextreme.py and the two DR yamls   | yes              | yes           | contradiction          | high       |
| dexvip_2022                        | 2022 | RL               | 4            |                                                                                                                       |                                                                                     |                  | no            |                        |            |
| hora_2022                          | 2022 | RL, distillation | 5            |                                                                                                                       | Block C, paper Sec 3.1/App. C against configs/task/AllegroHand Hora.yaml            | yes              | yes           | version-skew           | high       |
| mjpc_2022                          | 2022 | MPC              | 3            |                                                                                                                       |                                                                                     |                  | yes           |                        |            |
| visual_dexterity_2022              | 2022 | RL, distillation | 7            | c1_sparse_task, c2_dense_task, c3_fingertip_dist, c4_energy, c5_push_away_penalty, c6_table_contact, c7_finger_height | Method, Table S1 against dexenv/envs/rewards.py and dexenv/conf/task/dclaw.yaml     | yes              | yes           | contradiction          | high       |
| aloha_act_2023                     | 2023 | BC               |              |                                                                                                                       |                                                                                     |                  | yes           | internal-inconsistency | high       |
| artigrasp_2023                     | 2023 | RL               | 4            | r_p, r_c, r_reg, r_task                                                                                               |                                                                                     |                  | yes           | parse-limitation       | low        |
| dexpbt_2023                        | 2023 | RL               | 4            | r_reach, r_pick, r_targ, r_vel                                                                                        | Block C, paper Sec III-C/Table II against allegro_kuka_base.py and AllegroKuka.yaml | yes              | yes           | contradiction          | high       |
| dexterous_functional_grasping_2023 | 2023 | RL               | 2            | r_threshold, r_hand-obj                                                                                               |                                                                                     |                  | no            |                        |            |
| diffusion_policy_2023              | 2023 | BC, diffusion    |              |                                                                                                                       |                                                                                     |                  | yes           | parse-limitation       | high       |
| dynamic_handover_2023              | 2023 | RL               | 3            |                                                                                                                       |                                                                                     |                  | no            |                        |            |
| eureka_2023                        | 2023 | RL               |              |                                                                                                                       | Method, App. G.1/G.2 LLM-authored rewards; generated code not checked into the repo | yes              | yes           | parse-limitation       | low        |
| pgdm_2023                          | 2023 | RL               | 2            |                                                                                                                       |                                                                                     |                  | no            |                        |            |
| physhoi_2023                       | 2023 | RL               | 4            | body motion reward, object motion reward, ig reward, cg reward                                                        |                                                                                     |                  | yes           | contradiction          | high       |
| robot_synesthesia_2023             | 2023 | RL, distillation | 6            |                                                                                                                       | Method, reward (Sec IV.A.3 eq. 1); no code released                                 | yes              | no            |                        |            |

| method                       | yr   | paradigm                          | terms stated | term names as the paper names them                                                                                                                                                                | where the reward was read                                                   | in reward matrix | code released | paper/code class       | confidence |
|------------------------------|------|-----------------------------------|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|------------------|---------------|------------------------|------------|
| rotateit_2023                | 2023 | RL, distillation                  | 6            |                                                                                                                                                                                                   | Block C (paper Sec 3.1, App. B); no code released                           | yes              |               |                        |            |
| rotating_without_seeing_2023 | 2023 | RL                                | 6            | r_rot, r_vel, r_fall, r_work, r_torque, r_dist                                                                                                                                                    | Reward block (Sec IV.A.3, App. D Eq. 9); no code released                   | yes              |               |                        |            |
| unidexgrasp_2023             | 2023 | grasp-synthesis, RL, distillation | 4            | goal, reach, lift, move                                                                                                                                                                           |                                                                             |                  | yes           | contradiction          | high       |
| unidexgrasp_pp_2023          | 2023 | RL, distillation                  | 4            | r_reach, r_lift, r_move, r_bonus                                                                                                                                                                  |                                                                             |                  | yes           | parse-limitation       | low        |
| anyrotate_2024               | 2024 | RL, distillation                  | 10           | keypoint distance reward, rotation reward, goal bonus reward, good contact reward, bad contact penalty, angular velocity penalty, pose penalty, work penalty, torque penalty, termination penalty | Block C (App. B.1 weights); no code released                                | yes              | no            |                        |            |
| asymdex_2024                 | 2024 | RL                                | 4            |                                                                                                                                                                                                   |                                                                             |                  | yes           | code-absent            | medium     |
| bidexhd_2024                 | 2024 | RL, distillation                  | 4            | r_appro, r_lift, r_bonus, r_track                                                                                                                                                                 |                                                                             |                  | yes           | parse-limitation       | high       |
| bimangrasp_2024              | 2024 | grasp-synthesis, diffusion        | 7            |                                                                                                                                                                                                   |                                                                             |                  | no            |                        |            |
| demostart_2024               | 2024 | RL, distillation                  | 1            |                                                                                                                                                                                                   | Method, reward (App. VII-A.1); no code released                             | yes              | no            |                        |            |
| dexmimicgen_2024             | 2024 | data-collection, BC               | 1            |                                                                                                                                                                                                   |                                                                             |                  | yes           | code-absent            | high       |
| dp3_2024                     | 2024 | BC, diffusion                     |              |                                                                                                                                                                                                   |                                                                             |                  | yes           | internal-inconsistency | medium     |
| dreureka_2024                | 2024 | RL                                | 4            | ang_z_vel_reward, lin_vel_penalty, object_fall_penalty, deviation_penalty                                                                                                                         | Method, App. B1 Prompt 13; cube-rotation code absent from the released repo | yes              | yes           | code-absent            | high       |
| graspxl_2024                 | 2024 | RL                                | 8            | r_dis, r_v, r_ω, r_m, r_c, r_f, r_reg, r_anatomy                                                                                                                                                  |                                                                             |                  | yes           | parse-limitation       | low        |
| hudor_2024                   | 2024 | RL                                | 1            |                                                                                                                                                                                                   |                                                                             |                  | no            |                        |            |
| humanplus_2024               | 2024 | RL, BC                            | 8            |                                                                                                                                                                                                   |                                                                             |                  | yes           | parse-limitation       | medium     |
| objdex_2024                  | 2024 | BC, RL, distillation              | 3            | rotation, translation, joint angle                                                                                                                                                                |                                                                             |                  | no            |                        |            |
| omnigrasp_2024               | 2024 | RL, distillation                  | 3            | r_approach, r_pre-grasp, r_obj                                                                                                                                                                    |                                                                             |                  | yes           | parse-limitation       | medium     |
| omnih2o_2024                 | 2024 | RL, distillation                  | 24           |                                                                                                                                                                                                   |                                                                             |                  | yes           | internal-inconsistency | medium     |
| open_television_2024         | 2024 | teleop-system, BC                 |              |                                                                                                                                                                                                   |                                                                             |                  | yes           | code-absent            | low        |
| penspin_2024                 | 2024 | RL, distillation                  | 7            |                                                                                                                                                                                                   | Method, paper Table 4 against penspin/tasks/allegro_hands_hora.py and       | yes              | yes           | contradiction          | medium     |

| method                             | yr   | paradigm                    | terms stated | term names as the paper names them                              | where the reward was read                                                                            | in reward matrix | code released | paper/code class       | confidence |
|------------------------------------|------|-----------------------------|--------------|-----------------------------------------------------------------|------------------------------------------------------------------------------------------------------|------------------|---------------|------------------------|------------|
|                                    |      |                             |              |                                                                 | configs/task/AllegroHand<br>Hora.yaml                                                                |                  |               |                        |            |
| pi0_2024                           | 2024 | VLA, flow                   |              |                                                                 |                                                                                                      |                  | yes           | version-skew           | high       |
| pianotime_2024                     | 2024 | RL+demo, distillation       | 2            |                                                                 |                                                                                                      |                  | yes           | contradiction          | high       |
| resdex_2024                        | 2024 | RL, distillation            | 7            | task, proposal, pose, reach, lift, move, bonus                  |                                                                                                      |                  | yes           | parse-limitation       | medium     |
| twisting_lids_2024                 | 2024 | RL                          | 5            |                                                                 |                                                                                                      |                  | no            |                        |            |
| articulated_tools_inhand_2025      | 2025 | RL, BC, distillation        | 8            | r_pos, r_quat, r_goal, r_timer, r_inc, r_contact, r_slip, r_act |                                                                                                      |                  | no            |                        |            |
| being_h0_2025                      | 2025 | VLA, BC                     |              |                                                                 |                                                                                                      |                  | yes           | code-absent            | high       |
| clutterdexgrasp_2025               | 2025 | RL, diffusion, distillation | 4            | r_pos, r_neg, r_grasp, r_force                                  |                                                                                                      |                  | no            |                        |            |
| cross_embodiment_world_models_2025 | 2025 | world-model, MPC            | 2            |                                                                 |                                                                                                      |                  | no            |                        |            |
| dexgraspvla_2025                   | 2025 | VLA, diffusion              | 1            |                                                                 |                                                                                                      |                  | yes           | parse-limitation       | high       |
| dexmachina_2025                    | 2025 | RL                          | 4            |                                                                 |                                                                                                      |                  | yes           | internal-inconsistency | low        |
| dexman_2025                        | 2025 | RL, trajopt                 | 3            |                                                                 |                                                                                                      |                  | no            |                        |            |
| dexndm_2025                        | 2025 | RL, distillation            | 7            |                                                                 | Reward block (Sec. 3.1, App. A.1); no code released                                                  | yes              | no            |                        |            |
| dexplore_2025                      | 2025 | RL, distillation            | 7            | r_j^h, r_r^h, r_p^o, r_r^o, r_d, r_c, r_energy                  |                                                                                                      |                  | yes           |                        |            |
| dexremoe_2025                      | 2025 | RL                          | 5            | csuccess, cdist, crot, cω, ca                                   | Method, reward (Sec. III-B, Table II); no code released                                              | yes              | no            |                        |            |
| dexteritygen_2025                  | 2025 | RL, diffusion               |              |                                                                 | Method, reward/objective: the RL reward behind the pretrained controller is not quoted in the source | yes              | no            |                        |            |
| dexterous_handover_2025            | 2025 | RL                          | 3            | r_BASE, contact_term (c-w_c), r_MANIP                           |                                                                                                      |                  |               |                        |            |
| dextrack_2025                      | 2025 | RL+demo                     | 5            | r_o,p, r_o,q, r_wrist, r_finger, r_affinity                     |                                                                                                      |                  | yes           | parse-limitation       | low        |
| dexvla_2025                        | 2025 | VLA, diffusion              | 2            |                                                                 |                                                                                                      |                  | yes           | parse-limitation       | high       |
| dydexhandover_2025                 | 2025 | RL                          | 5            | R_dist, R_obj, R_contact, P_action, P_hand                      |                                                                                                      |                  | no            |                        |            |
| geometric_retargeting_2025         | 2025 | teleop-system               | 5            | Ldir, Lcover, Lflat, Lpinch, Lcol                               |                                                                                                      |                  | yes           |                        |            |
| groot_n16_2025                     | 2025 | VLA                         |              |                                                                 |                                                                                                      |                  | yes           | version-skew           | high       |
| groot_n1_2025                      | 2025 | VLA, flow                   |              |                                                                 |                                                                                                      |                  | yes           | version-skew           | high       |
| human2sim2robot_2025               | 2025 | RL                          | 1            | r_obj                                                           |                                                                                                      |                  | yes           |                        |            |

| method                            | yr   | paradigm           | terms stated | term names as the paper names them                                                                                                                              | where the reward was read                                                           | in reward matrix | code released | paper/code class | confidence |
|-----------------------------------|------|--------------------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|------------------|---------------|------------------|------------|
| humanoid_policy_human_policy_2025 | 2025 | BC                 | 2            |                                                                                                                                                                 |                                                                                     |                  | yes           |                  |            |
| humanoid_sim2real_recipe_2025     | 2025 | RL, distillation   | 2            |                                                                                                                                                                 |                                                                                     |                  | no            |                  |            |
| maniptrans_2025                   | 2025 | RL, BC, diffusion  | 5            | r_wrist, r_finger, r_smooth, r_object, r_contact                                                                                                                |                                                                                     |                  | yes           | parse-limitation | low        |
| metis_2025                        | 2025 | VLA                | 2            | Lar, Laction                                                                                                                                                    |                                                                                     |                  | no            |                  |            |
| pi05_2025                         | 2025 | VLA, flow          |              |                                                                                                                                                                 |                                                                                     |                  | yes           | code-absent      | high       |
| pistar06_2025                     | 2025 | VLA, RL, flow      |              |                                                                                                                                                                 |                                                                                     |                  | yes           | code-absent      | high       |
| wm_dex_human_videos_2025          | 2025 | world-model, MPC   | 2            |                                                                                                                                                                 |                                                                                     |                  | no            |                  |            |
| being_h05_2026                    | 2026 | VLA, flow          |              |                                                                                                                                                                 |                                                                                     |                  | yes           | code-absent      | high       |
| dexteleop0_2026                   | 2026 | teleop-system, MPC | 3            |                                                                                                                                                                 |                                                                                     |                  | no            |                  |            |
| force_grasp_sim2real_2026         | 2026 | RL                 | 7            | R_torque, R_Force, R_diff, R_outer, terminal-state penalty, R_action, R_vel                                                                                     | Method, reward (Sec 3.5.1, Table 2); no weights survive the parse; no code released | yes              | no            |                  |            |
| poise_2026                        | 2026 | RL                 | 4            | pose reaching, goal completion, grasp maintenance, drop and actuation regularization                                                                            | Method, reward (Sec. IV-D); weights lost to the parse; no code released             | yes              | no            |                  |            |
| simtoolreal_2026                  | 2026 | RL                 | 5            | r_smooth, r_approach, r_lift, r_goal, b_succ                                                                                                                    |                                                                                     |                  | no            |                  |            |
| teledexter_2026                   | 2026 | RL, teleop-system  |              |                                                                                                                                                                 | Reward block (Sec. 3.1); term formulas lost to the parse; no code released          | yes              | no            |                  |            |
| toporetarget_2026                 | 2026 | trajopt, RL        | 4            | object, link-position, joint-position, action-smoothness                                                                                                        |                                                                                     |                  | no            |                  |            |
| viserdex_2026                     | 2026 | RL, distillation   | 10           | orientation tracking, success bonus, object dropped, object distance, object velocity, joint velocity, action magnitude, action rate, joint work, joint torques | Method, reward (App. Table IX); no code released                                    | yes              | no            |                  |            |

77 rows. 253 of 770 cells (32 percent) are values no source stated. A blank `terms stated` with a filled `term names` column is a paper that names its terms without numbering them. `in reward matrix` marks the rows that are also in the reward-term matrix, which covers in-hand reorientation only.

## C.2 Paper against released code, in full

Each entry below is the disagreement text stored in the row, unedited. The class is what Sec. 5.8 and Sec. 8.1 count. `contradiction` means the paper states one value and the shipped code demonstrably states another. `parse-limitation` means this survey's own parse could not settle it and the accusation is withdrawn. `code-absent` means the described component is not in the released repository. `version-skew` means the repository is a later generation than the paper. `internal-inconsistency` means the paper disagrees with itself and no code is implicated. Every `contradiction` entry carries an `Artefact` line: the repository, the commit `corpus/code_manifest.json` records, the file inside it and where in that file the value sits, so the entry can be checked without asking anyone. All of those authors were written to on 19 September 2026, before this survey was posted; section 5.6 says so beside the finding, gives the date the reply window closes, and states the route by which a disputed entry is corrected. The letters are in `outreach/` in the corpus, which is available from the author at the address in the byline.

### contradiction, 8 rows.

- `pddm_2019` (high). Table 2 states obs-dim 46 for In-hand Reorientation while the released `cube_env.py` code sums to 39; the Baoding reward code includes an extra `-10*wrist_too_high` term absent from Table 2's printed formula. Artefact: `google-research/pddm` at `06b88cdbaf`, `pddm/envs/cube/cube_env.py` (`_get_obs`).
- `dextreme_2022` (high). Action Delta Penalty weight is -0.25 in paper Table 2 but -0.2 in the ADR yaml and -0.01 in the ManualDR yaml; the Joint Velocity Penalty in code normalises velocity by `(max_velocity-vel_tolerance)` unlike the paper's stated formula; code has a `timeout_rew` term absent from the paper's reward table; Appendix Table 12 states critic learning rate  $5e-4$  and KL threshold 0.16, vs body text/code values of  $5e-5$  and 0.016. Artefact: `isaac-sim/IsaacGymEnvs` at `aeed298638`, `isaacgymenvs/cfg/task/AllegroHandDextremeADR.yaml` (`actionDeltaPenaltyScale`).
- `visual_dexterity_2022` (high). Table S1 states 32000 teacher training environments, but the released config sets `alg.num_envs` to 8000 (parent config 16384); `fallDistance` differs across two shipped configs (0.24 vs 0.15, only the latter matching Table S1's threshold); the paper's Eq 8 penultimate-joint penalty (`c7=-2`) does not appear anywhere in the released reward code; the config carries a `dead distRewardScale=-10.0` key never used in `compute_reward`; and the paper's table-friction lower bound (0.05) differs by a factor of 10 from the code's randomized lower bound (0.005). Artefact: `Improbable-AI/dexenv` at `ad9634e9d2`, `dexenv/conf/dclaw.yaml` (`alg.num_envs`).
- `dexpbt_2023` (high). Paper presents the reward as 4 mutually exclusive stage terms (`r_reach`, `r_pick`, `r_targ`, `-r_vel`), but code's `compute_kuka_reward` sums 8 named components, one of which (`hand_delta_penalty`) is multiplied by 0 and disabled; there is no single `r_vel` term in code, instead separate kuka/allegro action penalties whose exact formula is not shown. Also, the paper reports zero experiments with domain randomization, yet the shipped `AllegroKuka.yaml` already carries a fully specified DR schedule (disabled via `randomize: False`). Artefact: `NVIDIA-Omniverse/IsaacGymEnvs` at `aeed298638`, `isaacgymenvs/tasks/allegro_kuka/allegro_kuka_base.py` (`compute_kuka_reward`). Review: R3 adversarial review: the disabled-randomisation half is withdrawn, since the note finds it consistent with the paper; the zeroed reward term stands
- `physhoi_2023` (high). The released `compute_humanoid_reward` sets the body position-velocity error and the object rotation and rotation-velocity errors to zeros\_like, unconditionally, with the computation that would produce them commented out on the same lines, while Table 4 lists nonzero  $\lambda^{\text{or}}=0.1/\lambda^{\text{orv}}=0.01$  weights for GRAB: in that file the object's orientation error is the constant zero and those weights cannot change the reward. Artefact: `wyhuai/PhysH0I` at `6095c605e2`, `physhoi/env/tasks/physhoi.py` (`compute_humanoid_reward`).
- `unidexgrasp_2023` (high). The paper describes a four-term weighted reward (`r_goal + r_reach + r_lift + r_move` via Table 7's omega weights) but the released `compute_hand_reward` implements a different threshold-gated torch.where cascade with distinct hardcoded coefficients that do not map one-to-one onto the paper's weights. Artefact: `PKU-`

EPIC/UniDexGrasp at 36c9bfcf7c, dexgrasp\_policy/dexgrasp/tasks/shadow\_hand\_grasp.py (compute\_hand\_reward, goal\_cond branch).

- penspin\_2024 (medium). The appendix states a randomised disturbance force, and the released configs/task/AllegroHandHora.yaml ships forceScale: 0.0, so no shipped configuration applies it. Artefact: HaozhiQi/penspin at 5035c52dc9, configs/task/AllegroHandHora.yaml (forceScale). Review: Narrowed while the letter to its authors was being drafted, before that letter went out on 19 September 2026. The original comparison also said the released code disables the paper's tactile channel, and that half is withdrawn: the config read has numObservations 96 and enable\_tactile False, which is consistent with the proprioception-only student rather than the oracle, and the paper never claims the student has tactile input. The disturbance-force half is unaffected.
- pianomime\_2024 (high). Paper's Table 3 states 2 weighted reward terms (Key Press 2/3, Mimic 1/3), but the released code sums roughly 5 unweighted terms (key press doubled, sustain, energy and fingering hardcoded to return 0, forearm-collision) plus a separately-added mimic wrapper term. Artefact: sNiper-Qian/pianomime at c4abefac8d, single\_task/piano\_with\_shadow\_hands\_res.py (\_set\_rewards).

#### internal-inconsistency, 4 rows.

- aloha\_act\_2023 (high). Algorithm 1 pseudocode states  $L_{\text{reconst}} = \text{MSE}$ , but Sec.IV.C's prose explicitly states L1 loss is used instead; the algorithm box and the implementation text disagree; code/md captures only function signatures, not bodies, for policy.py's loss implementation
- dp3\_2024 (medium). the paper's prose states the network predicts the noise added to the data, but the shipped default config trains with prediction\_type: sample (predicting the denoised action  $a^0$  directly), not epsilon; the paper only qualifies this later in the same section (Fig. 7 ablates both).
- omni2o\_2024 (medium). stumble weight -0.00125 (paper) vs -1250 (code); max-feet-height sign/magnitude differ (+1000 paper vs -2500 code, a penalty not a bonus); paper's  $\exp(-c*///)$  form vs code's  $\exp(-\text{err}^2/\sigma)$ ; curriculum level-down threshold 40 (paper) vs 50 (code) Review: Withdrawn as a contradiction at the point of drafting the letter to its authors, which was never sent. Four sibling weights match the paper to the digit under a systematic x1.25 curriculum factor and only the stumble weight differs, by a factor of about a million, which is a typo signature in the paper's own table rather than evidence of a different trained objective. The hands in this work are driven open-loop from VR pose, outside the policy and outside the reward, so it is a weak fit for a dexterous-manipulation reward census in the first place.
- dextramachina\_2025 (low). paper describes a plain weighted sum  $\lambda_{\text{task\_task}} + \lambda_{\text{imir\_imi}} + \lambda_{\text{bcr\_bc}} + \lambda_{\text{conr\_con}}$  with unspecified weights; code implements a multiplicative task term with per-component beta decay, an unmentioned 0.1 force-penalty term, and curriculum-driven decay of auxiliary weights not described as such in the paper Review: R3 adversarial review: the multiplicative form credited to the code is printed in the paper itself

#### version-skew, 4 rows.

- hora\_2022 (high). The parsed code commit's joint-noise range, disabled default disturbance force, and default cube object type differ from the paper's stated values; the code README itself says to use tag v0.0.1, not the parsed commit, to reproduce paper numbers.
- pi0\_2024 (high). The released repo (openpi, commit 215abfb2) has evolved past this paper: it also documents pi0-FAST (autoregressive) and pi0.5 checkpoints/configs not described in this paper, which describes only the flow-matching pi0 and the non-VLM pi0-small ablation.
- groot\_n16\_2025 (high). code/md is the current Isaac-GR00T main branch (N1.7 generation, commit 51d4c89f), not the N1.6 checkpoint; the repo names a separate n1d6 branch for N1.6 that was not fetched. The page states N1.6's backbone is "an internal NVIDIA Cosmos-2B VLM variant," but the repo's own changelog states the Eagle backbone (nvidia/Eagle-Block2A-2B-v2) was used through N1.6 and only replaced by Cosmos-Reason2-2B in N1.7 -- whether the page's Cosmos-2B variant is a third distinct model or the page is describing N1.7 under an N1.6 headline is not resolvable from what was fetched. Review: The page attributes a Cosmos backbone to N1.6 while the repository changelog says N1.6 used Eagle and N1.7 made the switch, so the repository is a later generation than the page describes.

- `groot_n1_2025` (high). The parsed code repo (Isaac-GR00T, commit 51d4c89f) is a later N1.7 generation, not the paper's GR00T-N1-2B: VLM backbone changed from the paper's Eagle-2 to Cosmos-Reason2-2B (via Qwen3-VL); action horizon expanded from the paper's stated  $H=16$  to 40; state/action dimensions expanded from 29 to 132; DiT layers changed from 32 to 16; embodiment scope widened to include Unitree G1, AgiBot G1, and a 'YAM' arm not described in the paper.

#### code-absent, 8 rows.

- `asymdex_2024` (medium). Code contains a GraspAndPlace-task reward variant (AllegroHandDualArmGraspAndPlaceAsymDex.py) with exponential hand-distance and block-alignment terms not one of the four BiDexHands tasks described in the paper body.
- `dexmimicgen_2024` (high). The environment code includes an unused 'reward\_shaping' branch (if self.reward\_shaping: pass) that is never described or used in the paper text; only the sparse binary completion reward is used, as the paper states.
- `dreureka_2024` (high). The released DrEureka repo (code/md) contains only forward\_locomotion/ and globe\_walking/ trees; the LEAP-hand cube-rotation training/deployment code and the quoted Prompt-13 reward are the LEAP-hand authors' own code, not included in the released repo, so the cube-rotation reward cannot be cross-checked against a code file.
- `open_television_2024` (low). The paper's prose states 25k training iterations, lr 5e-5, batch size 45, but the code repo's example training command instead shows 50000 epochs and an explicit kl\_weight=10 not mentioned anywhere in the paper text. Review: R3 adversarial review: a behaviour-cloning paper with no reward; it compared paper iterations to a README example's epochs
- `being_h0_2025` (high). The actual downstream action-head module referenced in §4.3 (learnable query tokens, proprioceptive projector f\_p, regression head f\_r) is not present in the parsed code; the repo's own TODO list marks 'Training code and scripts' as unreleased, so the action-chunk-length=16 value and exact proprioception-vector fields come only from inference/eval CLI examples, not a verifiable implementation.
- `pi05_2025` (high). The paper describes a hybrid discrete-FAST-token-pretrain-then-flow-matching-post-train recipe (Eq. 1, alpha: 0 to 10), but the released repo states plainly it 'currently only support[s] the flow matching head for both pi0.5 training and inference.'
- `pistar06_2025` (high). The general openpi framework (pi0/pi0.5/FAST, flow-matching action head, ALOHA/DROID/LIBERO adapters) is released at commit 215abfb2, but the RECAP-specific pieces described in the paper -- the advantage-conditioning input, the distributional value function, the RECAP training loop (Algorithm 1), and the espresso/laundry/box-assembly checkpoints and datasets -- are not present in the parsed repo.
- `being_h05_2026` (high). Released code (commit e12ac44f) exposes only benchmark/inference/config scaffolding (policy.py, beingh\_policy.py, dataset-transform configs). no MoT/MoF model definition, no rectified-flow action-expert source, and no config for any dexterous-hand embodiment; the only post-train configs present (libero\_robotcasa.yaml, libero\_all.yaml, robocasa\_human.yaml, so101\_example.yaml) are all gripper-only, so the dexterous-hand claims cannot be cross-checked against released code.

#### parse-limitation, 14 rows.

- `dexpoint_2022` (low). The released reward reshapes the paper's reach term into an inverse-distance form,  $1.0/(0.06+\text{finger\_object\_dist})$  in place of the plain distance of Eq. 5, and adds a 2 cm lift-threshold bonus, a target-distance term and an IK controller-tracking penalty that Eq. 5 does not list. Artefact: `yzqin/dexpoint-release` at `17f1e238bb`, `dexpoint/env/rl_env/relocate_env.py` (`AllegroRelocateREnv.get_reward`). Review: Reclassified from contradiction to parse-limitation after the corresponding author replied, 2026-09-20. Three parts of the original claim do not stand. The lift term is not a departure: `object_lift` is already `h_current` minus `h_init`, set against a resting height fixed at reset, so  $10*\text{clip}(\text{object\_lift}, 0, 0.2)$  is the paper's formula with the stated weight plus a saturation, and this survey's parse captured the reward body but not the line defining `object_lift`, so the claim rested on material the snapshot did not contain. The rotation bonus is inactive at the paper's settings, since `rotation_reward_weight` defaults to

zero and the term is multiplied by it, so it contributed nothing to the reported numbers. The reach-term difference is real but is version skew rather than a contradiction: the inverse form is a later, lower-variance revision and the paper's reported variance is that of the plain-distance formula the paper states. The remaining extra terms have stated purposes, the 2 cm bonus blocking a height reward hack on meshes with no flat resting surface and the IK penalty correcting a SAPIEN-specific controller mismatch.

- **artigrasp\_2023** (low). paper's two regulariser weights ( $w_{rh}=0.5$ ,  $w_{ro}=0.2$ ) correspond to five separate velocity penalties in code  $(-0.5,-0.2,-0.5,-0.5,-0.3)$ ; the fingertip weight 12.0 and lambda 5.0 in Table 6 do not appear in the yaml; reward weights differ between curriculum phase-1 and phase-2 configs though Table 6 reports only one set Review: R3 adversarial review: the reward body is C++ and absent from the parse; the weight split is the paper's documented curriculum
- **diffusion\_policy\_2023** (high). the paper's core loss equations (DDPM training loss, EBM/InfoNCE, score-matching) are unrecovered images in the parsed text, and code/md is a signature-only API map, so compute\_loss's body could not be verified from code; only the exact shipped noise-scheduler config (DDPMScheduler, 100 train steps) is confirmed.
- **eureka\_2023** (low). The released repo's default eureka/cfg/config.yaml ships iteration:1, sample:3 (a fast-test default), not the paper's reported experimental configuration of  $N=5$  iterations  $\times$   $K=16$  samples  $\times$  5 independent runs, and LLM-generated reward code itself is never checked into the repo (written at runtime to a gitignored outputs/ directory). Review: R3 adversarial review: the repo README documents the paper's defaults; the cited config block is a fast-test preset
- **unidexgrasp\_pp\_2023** (low). The paper's four named reward weights (0.5, 0.1, 2, 10) numerically match literals in the released compute\_hand\_reward, but the code never labels them with the paper's variable names, and the paper's own display equations were dropped from the PDF, so the correspondence is inferred rather than confirmed by either source.
- **bidexhd\_2024** (high). Table 5 weights ( $w_r$ ,  $w_t$ ,  $w1-w4$ ,  $\lambda_w=0.12$ ,  $\lambda_{ft}=0.48$ ) cannot be matched to code reward lines because the code md truncates the reward function before stage-2/final sum;  $\lambda_w$  and  $\lambda_{ft}$  numerically match the code's grasp-condition thresholds (0.12, 0.12x4 fingers) rather than confirmed reward weights
- **graspxl\_2024** (low). The code splits the paper's single  $r_{reg} = -w_h/u_h//^2 - w_o/u_o//^2$  into four separately-weighted terms (wrist vel, wrist qvel, object vel, object qvel), and the floating-phase object-velocity coefficient (-1.5) does not match the paper's stated  $w_o=0.1$ , so the regularization weighting cannot be reconciled 1:1 with Table 9. Review: R3 adversarial review: the environment source is not in the parse and the note says the reward cannot be verified
- **humanplus\_2024** (medium). code implements a superset of legged\_gym reward terms (lin\_vel\_z, ang\_vel\_xy, orientation, base\_height, torques, dof\_vel, dof\_acc, action\_rate, collision, termination, dof/torque limits, tracking\_lin/ang\_vel, feet\_air\_time, stumble, stand\_still, feet\_contact\_forces, target\_jt) beyond the 8 terms documented in the paper's Table 1
- **omnigrasp\_2024** (medium). The code's compute\_pregrasp\_reward\_time hard-codes  $w_{pos}$ ,  $w_{rot} = 0.9, 0.1$ , silently overriding the rwd\_specs weight arguments passed into the function; separately, the paper's pre-grasp reward weights ( $w_{hp}$ ,  $w_{hr}$ ) are not resolvable from either the paper text or Appendix Table 7.
- **resdex\_2024** (medium). The base-policy and residual-policy task files implement different variants of compute\_hand\_reward (whether finger/hand-distance terms are included alongside goal\_hand\_rew/hand\_up/bonus differs), a difference the paper's Sec 4.1/4.4 text does not describe, and the released configs' env count (4096 default) and iteration counts (2500/5000) do not match the paper's stated 11,000 envs and 5,000/20,000 iterations.
- **dexgrasvla\_2025** (high). Hardware-interfacing code is explicitly withheld ("Due to intellectual property constraints, we are unable to open-source the hardware-related code"); only function signatures for the controller loss (compute\_loss, noise\_assignment) are present in code/md, not their bodies, so the MSE/Immiscible-Diffusion loss cannot be verified against code.
- **dextrack\_2025** (low). The paper gives one canonical reward-weight table (Table 3), but the released repo's task-config YAMLS contain several divergent, unreconciled reward-coefficient sets across AllegroHandTracking/... TrackingGeneralist/other variants, and it is not identifiable from the parsed source which config produced the paper's headline Table 1 numbers.

- `dexvla_2025` (high). code/md only carries class/function signatures, not bodies, for the diffusion head (modeling\_scaledp.py) and no config file with the  $\alpha$  weight was captured; the paper's diffusion loss and denoising process cannot be verified against code bodies. only the multi-head ScaleDP signature and a DDIM fp32 patch are available, so none of the paper's tables can be reproduced from what was parsed.
- `maniptrans_2025` (low). paper's PPO learning rate (5e-4) and env count (4096) differ from the shipped README/yaml defaults (2e-4, 8192) Review: R3 adversarial review: the cited values are a README override and an unused fallback; the repo's own config matches the paper

## Appendix D. The existing surveys, and what each covers

Fourteen corpus entries are themselves surveys or engine-comparison studies. Table 10 sets them on one set of columns. The columns record what each work covers, not how well. Section 1 states what this survey adds over them and does not repeat the comparison here.

`an_dexil_survey_2025` is the closest in subject, covering imitation learning for multi-fingered hands by learning family, end-effector class and demonstration source. It gives reinforcement learning no taxonomy, treats bimanual work as a single "multi-agent" subsection, compares no simulator, and defines no evaluation metric. `welte_iil_survey_2025` finds only seven dexterous works that use interactive imitation learning and carries a fifteen-hand commercial table, with no bimanual section, no benchmark table and no contact modelling. `zhao_sim2real_survey_2020` supplies the standard sim-to-real split and predates GPU-parallel simulation. `firoozi_foundation_models_2023` has zero occurrences of bimanual, tactile or in-hand.

`bai_unified_manip_survey_2025` spans all of manipulation across 212 pages. Its Sec. 4.3 on dexterous manipulation runs about 720 words, the third longest of its ten task subsections behind grasping and quadrupedal manipulation, which is a real treatment and not a passing mention. The difference is elsewhere. Its Sec. 1.2 lists dexterous manipulation among the topics that "existing surveys" cover from "narrower perspectives" and defers it to two of them.

`zhao_dexhand_survey_2026` is the most recent hand-centred survey, with a 29-hand anatomy table and a task-by-paradigm taxonomy. Its Sec. IV-C names the reference-versus-rollout split that Section 1 credits it with, and its Sec. III-F is the one bimanual subsection, 16 cited works with no coordination analysis.

`nine_physics_engines_review_2024` is the predecessor closest to the engine comparison, and it reviews Brax, Chrono, Gazebo, MuJoCo, ODE, PhysX, PyBullet, Unity and Webots for reinforcement learning research. It scores each on documentation, model and environment creation, URDF and MJCF support, and readiness for multi-agent work. It runs no benchmark of its own and says so in its Sec. V, that implementing the same scenarios across nine engines "goes beyond the scope of this paper". Its running bodies are ant-and-humanoid RL benchmarks rather than hands, and it discusses no timestep, no friction model, no contact formulation and no penetration.

### Table 10. Existing surveys and what each covers

| survey                           | yr   | scope                                                                                                                                      | taxonomy used                                                                                                           | bimanual covered                                                               | hardware covered                                                               | evaluation covered                                                                                                               | gaps it names                                                                                                                           |
|----------------------------------|------|--------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|--------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| isaac_sim_2026                   | 2026 | one simulator's ecosystem and application domains, reviewed rather than measured                                                           | qualitative capability matrix, Table 1, over simulators                                                                 | one cited GR00T task called bimanual, no hand, DoF or number                   | no hand named anywhere in paper or code parse                                  | none. The paper runs no experiment of its own                                                                                    | computational cost, configuration complexity, learning curve                                                                            |
| zhao_dexhand_survey_2026         | 2026 | dexterous hands end to end: hardware anatomy, methods, datasets, directions                                                                | five task categories, each split by learning paradigm. Hardware is split by actuation, transmission and perception      | one subsection, III-F, 16 cited works, no coordination analysis                | Table I, 29 hands, 12 columns, secondary values                                | names two layers: physical plausibility including penetration before execution, success during it. No threshold, method or count | hardware feasibility, perception fusion, learning beyond benchmark-centric optimisation, industrialisation, absent evaluation standards |
| an_dexil_survey_2025             | 2025 | imitation learning for multi-fingered end-effectors                                                                                        | IL family by end-effector class by demonstration source. RL gets none                                                   | one subsection, II.E, framed as multi-agent                                    | hands named in prose, no hand table                                            | no metric defined and no trial count. Calls for protocols, proposes none                                                         | contact dynamics in engines, data-collection standards, cross-hand transfer, failure datasets, end-effector morphology                  |
| bai_unified_manip_survey_2025    | 2025 | all of robot manipulation. Dexterous manipulation is one of ten task subsections, about 720 words, and Sec. 1.2 defers it to other surveys | high-level planning, action modelling, actuation control, plus a bottleneck taxonomy of data and generalisation         | bimanual means two arms. One dual-hand mention in the paper                    | hands named, no DoF or actuation table                                         | success rate and checkpoint selection, six lines. Never says how success is judged for a dexterous task                          | no scaling law, sim-to-real for contact-rich tasks, fragmented datasets, reliability as important as success                            |
| welte_iil_survey_2025            | 2025 | interactive imitation learning, seven dexterous works found                                                                                | IIL feedback type, plus a keyword bibliometric of 326 papers                                                            | three mentions in 687 lines, no section                                        | Table 1, 15 commercial hands, manufacturer figures                             | no metric defined, no benchmark table                                                                                            | tactile feedback, long-horizon tasks, generalisation, human-feedback interface                                                          |
| nine_physics_engines_review_2024 | 2024 | nine physics engines for RL research, scored on documentation and usability                                                                | 13-axis feature and usability matrix, Table II, plus citation-count popularity                                          | none. MARL readiness is the multi-agent axis                                   | none. Ant and humanoid RL bodies are the running examples                      | no benchmark of its own. Throughput claims are second-hand                                                                       | no cross-engine MARL performance comparison exists in the literature                                                                    |
| contact_models_comparison_2023   | 2023 | LCP, CCP, RaiSim and NCP contact models re-implemented in one framework and ranked                                                         | contact model by solver, with the physical property each one violates                                                   | none                                                                           | one Allegro hand as a benchmark system, a ball dropped into it                 | NCP criterion, self-consistency against a 1e-5 s reference, iteration cost                                                       | no fully satisfactory contact model, and gradients through simulation artifacts are unexplored                                          |
| firoozi_foundation_models_2023   | 2023 | foundation models in robot decision-making, perception and embodied AI                                                                     | background, robotics, and robotics-adjacent papers, then by application                                                 | none, zero occurrences                                                         | none. Parallel-jaw end effectors throughout                                    | none defined. Benchmarking appears as a reproducibility problem                                                                  | data scarcity, variability, uncertainty, safety, real-time inference, and simulators that neglect contact physics                       |
| zhao_sim2real_survey_2020        | 2020 | sim-to-real transfer in deep RL, eight pages, 21 works tabulated                                                                           | zero-shot, system identification, domain randomisation, domain adaptation, learning with disturbances, simulator choice | none, zero occurrences                                                         | two cited hand works, no hand table                                            | no metric defined, no trial count, no success rate                                                                               | domain randomisation has no formal account, and domain adaptation assumes matched feature spaces                                        |
| piazza_century_2019              | 2019 | <i>source not obtained</i>                                                                                                                 | <i>source not obtained</i>                                                                                              | <i>source not obtained</i>                                                     | <i>source not obtained</i>                                                     | <i>source not obtained</i>                                                                                                       | <i>source not obtained</i>                                                                                                              |
| physics_engine_comparison_2015   | 2015 | five engines on one shared model: speed, self-consistency, conservation, grasp stability                                                   | none. Four test systems, one comparison per test                                                                        | none                                                                           | one 35-DoF rig modelled on the Shadow Hand                                     | largest timestep that holds a grasp, and a speed-accuracy Pareto curve                                                           | restricted feature subset by design, and the authors are MuJoCo's developers                                                            |
| roa_suarez_grasp_quality_2015    | 2015 | <i>source not obtained</i>                                                                                                                 | <i>source not obtained</i>                                                                                              | <i>source not obtained</i>                                                     | <i>source not obtained</i>                                                     | <i>source not obtained</i>                                                                                                       | <i>source not obtained</i>                                                                                                              |
| ma_dollar_dexterity_2011         | 2011 | <i>PDF fetched 2026-09-18 after the note was written, no note read from it</i>                                                             | <i>PDF fetched 2026-09-18 after the note was written, no note read from it</i>                                          | <i>PDF fetched 2026-09-18 after the note was written, no note read from it</i> | <i>PDF fetched 2026-09-18 after the note was written, no note read from it</i> | <i>PDF fetched 2026-09-18 after the note was written, no note read from it</i>                                                   | <i>PDF fetched 2026-09-18 after the note was written, no note read from it</i>                                                          |
| okamura_overview_2000            | 2000 | <i>source not obtained</i>                                                                                                                 | <i>source not obtained</i>                                                                                              | <i>source not obtained</i>                                                     | <i>source not obtained</i>                                                     | <i>source not obtained</i>                                                                                                       | <i>source not obtained</i>                                                                                                              |

14 rows, one per corpus entry of class `survey`, 3 of which could not be obtained and are entered as such, and 1 of which was fetched too late to be read into a note. Every filled cell is read from `papers/notes/<key>.md` and is checked against a quotation from that note by `tools/make_survey_table.py`. A cell whose evidence has gone from the note prints empty rather than printing an unchecked claim. The columns record what each work covers, not how well, and a blank cell in `bimanual covered` or `hardware covered` is a scope decision by its authors rather than a failure.

Table 10's last rows carry the cost of the corpus. `okamura_overview_2000`, `piazza_century_2019` and `roa_suarez_grasp_quality_2015` are behind publisher paywalls with no author-hosted copy found on 2026-09-18, and `bicchi_hands_2000` is in the same position with no row at all. `ma_dollar_dexterity_2011` is a different case. The fetch that failed when its note was written succeeded afterwards, so a seven-page PDF is on disk with a recorded hash, and no note has been read from it. All five are cited by metadata only and nothing in this survey describes their contents. The open chapter `bicchi_grasping_chapter_2001` overlaps the paywalled Bicchi paper without being identical to it, so it is quoted in its own right.

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## Appendix E. Where the protocol's counts come from

Every count in Table 8 is derived below, and each axis is derived for the statistic that axis actually reports rather than by one convention applied to all of them: a single rate takes a Wilson half-width, a matched comparison takes McNemar, a ratio takes the standard error of the log ratio, and a correlation takes the Fisher-z interval. Section 7.3 states what these derivations conclude. This survey re-ran no method, so every count here rests on an interval width, a power calculation or another paper's measurement, and never on a measurement of our own. The counts an earlier draft quoted and this one withdrew are kept, so that a reader can see which test was the wrong one and why.

Fix the width first, then read off the count. Take a 95 percent Wilson interval on a single reported rate, at the worst case of  $p = 0.5$ . A half-width of 20 points needs 21 trials, 15 points needs 39, 10 points needs 93 and 5 points needs 381. Ten points is the coarsest width at which a single rate is worth printing, so the absolute-rate minimum is 93, rounded to 100. At 100 trials a reported 80 percent has an interval of 71 to 87 percent, and a reported 50 percent has 40 to 60. A comparison is a different question and a harder one: two rates each carrying  $\pm 10$  points do not resolve a 10-point difference between them, because the difference's standard error is larger by a factor of  $\sqrt{2}$ , so the width argument sets a floor on what is worth reporting and not on what can be compared.

For the A/B comparison the relevant calculation is power, and the design is paired. Table 8 matches initial conditions by image overlay and interleaves the two policies in one session, so the unit is a matched pair and the count follows McNemar, which depends on the discordance rate. The share of initial conditions on which the two policies disagree, and not on the two rates alone. To separate 50 from 70 percent at  $\alpha = 0.05$  with 80 percent power: 37 pairs per arm at a discordance of 0.2, 57 at 0.3, 77 at 0.4 and 96 at 0.5. The protocol assumes 0.3 and asks for 57, and states the sensitivity rather than hiding it, because 0.5 is the discordance the same two rates produce when the pairing buys nothing, and at that value the paired count returns to the 93 per arm an unpaired test would need. The saving from pairing is real but smaller than the pair counts suggest, since a pair costs two rollouts: 57 pairs is 114 rollouts against 186. An earlier version of this section quoted 93, 169 and 387 per arm for gaps of 20, 15 and 10 points, which are correct for independent arms and are the wrong test for this protocol. That 93 was also the same integer as the half-width calculation in the paragraph above, which is a coincidence of the worst-case arithmetic and not a second derivation of the same number.

One hundred is a cap and not a bill, because on a graded score a sequential test reached its decision in 12 to 36 paired hardware trials in `beyond_binary_success_2026`. At 30 rollouts a continuous score already carries a half-width of  $\pm 0.36$  standard deviations, which is why a graded score can stop where a binary one cannot. A cell that stops early does not report a Wilson interval. Optional stopping breaks the coverage of a fixed-n interval, which is the reason `beyond_binary_success_2026` and `suresim_2025` use anytime-valid betting intervals rather than Wilson, so Table 8 asks a cell run to a fixed 100 for a Wilson interval and a cell stopped early for a confidence sequence, and never for both. Intervals are also marginal rather than simultaneous. For example, evaluating twelve methods on seven axes produces 84 intervals. At 84 independent 95 percent intervals, four excursions are expected by construction, so a paper comparing  $k$  policies on  $m$  tasks

corrects its  $k(k-1)/2$  pairwise tests to a global 95 percent level, as `lbm_careful_examination_2025` does, or says its intervals are not simultaneous.

Simulation is cheap, so simulated cells take 200 episodes, giving a 6.9-point half-width. Perturbation axes are screened rather than certified, and 40 per axis buys a 15-point half-width on each axis's own absolute rate, which is enough to rank the axes and pick the two worst for hardware. It is not enough for the ratio to the anchor that an earlier draft asked each cell to report. At 40 trials in each arm, a fall from a 0.50 anchor to 0.30 is a ratio of 0.60 with a 95 percent interval of 0.34 to 1.06, which contains 1: at the screening count you cannot establish that the perturbation hurt at all. Certifying that same drop takes 101 per arm at 80 percent power, or about 50 for an interval that merely excludes 1, so Table 8 now asks for absolute rates with their own intervals at 40, reports the ratio without an interval, and prescribes 101 before any claim that a named axis hurt.

For unseen objects the resampling unit is the object and not the trial, so 20 objects at 5 trials each gives 100 trials and an object-level half-width near 20 points. That 20 points is the Wilson width at  $n = 20$  and it treats each object's outcome as a single Bernoulli draw, which the five within-object trials are not. It is the right order of magnitude and the assumption belongs in the cell. A 10-point claim about an object distribution needs about 93 objects. Seven of the 39 rows that state an unseen count reach that: 225 in `bimangrasp_2024`, 241 in `resdex_2024` and `unidexgrasp_2023`, 360 in `dexgraspvla_2025`, 500 in `dexmv_2021`, 2029 in `clutterdexgrasp_2025` and 503409 in `graspxl_2024`. For a continuous score the half-width is 1.96 standard deviations over the square root of the count, so 100 rollouts give  $\pm 0.20$  standard deviations, and the unit is the rollout because frames within one are correlated.

The transfer axis is the one where 100 is least defensible. On 100 matched pairs a measured correlation of 0.70 carries a Fisher-z interval of 0.58 to 0.79, a half-width of about 0.10 that 130 pairs would be needed to guarantee. That is enough to establish that a simulator tracks reality at all, and it is not enough to separate `suresim_2025`'s useful regime from its marginal one, since those differ by about 0.11 in correlation. Both limits come within 0.05 of the estimate only at about 457 pairs. Table 8 states which of the two decisions each count supports rather than leaving a reader to assume the larger one.